# TECHNICAL QUALIFICATION.
# PRODUCT INFORMATION.

## ENGINE MECHANICS – PETROL ENGINE.




BMW GROUP

# General information

### Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:



Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

### Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as the result of the equipment specification in specific markets or countries.

### Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

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The information contained in this document forms an integral part of the BMW Group Technical Qualification and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

Information status: **March 2017**
**Technical qualification.**

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# ENGINE MECHANICS – PETROL ENGINE.
# 1. INTRODUCTION.

## 1.1. Introduction

This product information is part of the “BMW Group Automotive Senior Technician” qualification and is a reference work for engine mechanics - petrol engine. The purpose of this product information is to provide system-wide background knowledge about the engine mechanics of the current BMW and MINI vehicles.

Target group of the qualification:

- BMW and MINI employees, who have already successfully completed a “BMW Group Automotive Technician” qualification.
- New BMW and MINI employees, who already have basic technical knowledge about motor vehicles and are familiar with the basic knowledge about BMW and MINI vehicles.



These documents make information available that will serve as an overview and give you a better understanding of the qualification content. **As a rule, the instructions and other information in the workshop information system must be used for all maintenance and repair work.**

## 1.2. Required competencies

Several areas of competency are required for the "BMW Group Automotive Senior Technician" qualification; these must either exist already or must have been obtained through the "BMW Group Automotive Technician" qualification.

**In general the participants for the qualification should be able to perform general and simple repair work professionally and according to the specifications of the current repair instructions.**

In addition, the following basic technical knowledge is required about the engine mechanics in the petrol engine of BMW and MINI vehicles:

| Training module | Drive I | |
|---|---|---|
| Engine mechanics | • Engine designation and engine identification<br>• Designs, direction of rotation and installation of the engines<br>• Design and variants of engine housings<br>• Equipment specifications and functions of the valve gears, crankshaft drives and belt drives | • Functions, design and installation locations of the oil supply<br>• Functions and variants of the cooling<br>• Functions and designs of the vacuum system |

# ENGINE MECHANICS – PETROL ENGINE.
# 1. INTRODUCTION.

## 1.3. Further information

This document introduces you to the functions and special features of the engine mechanics of petrol engines.

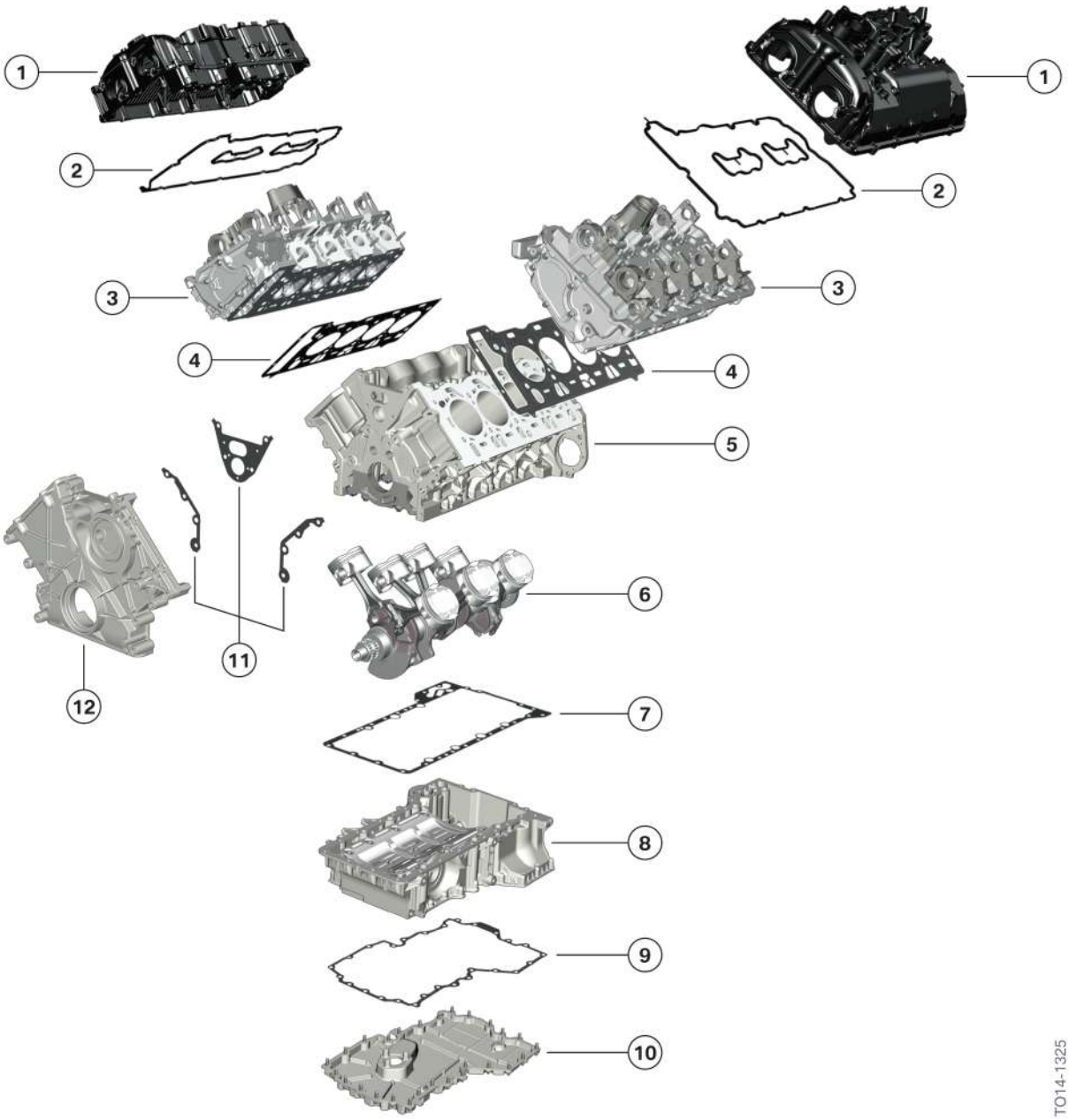
For more on information on the ICT petrol engine topic, please refer to the following documents:

| Document | Information about |
|---|---|
| ICT petrol engine I – engine electrical system | • Digital Motor Electronics DME<br>• Sensors<br>• Actuators<br>• Ignition system |
| ICT petrol engine II – fuel systems | • Fuel supply<br>• Fuel preparation |
| ICT petrol engine II – exhaust emission systems | • Exhaust turbocharger<br>• Exhaust system |

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.

## 2.1. Overview

Here, you can see the main components of the engine housing of V-engines.




N63TU2 engine, overview of engine housing

| Index | Explanation |
|---|---|
| 1 | Cylinder head cover |
| 2 | Cylinder head cover gasket |
| 3 | Cylinder head |
| 4 | Cylinder head gasket |
| 5 | Crankcase |
| 6 | Crankshaft drive |
| 7 | Oil sump gasket, top |
| 8 | Oil sump, top |
| 9 | Oil sump gasket, bottom |
| 10 | Oil sump, bottom |
| 11 | Timing case cover gasket |
| 12 | Timing case cover |

## 2.2. Crankcase venting components

When the engine is in operation, gases (referred to as "blow-by gases") from the combustion chamber pass through the cylinder walls and enter the crankcase. These blow-by gases contain unburned fuel and all elements of the exhaust gas. In the crankcase they are mixed with engine oil which is available there in the form of oil mist.

The volume of the blow-by gases is dependent on the engine speed and the load. Without crankcase ventilation excess pressure would arise in the crankcase. This excess pressure would be present in all cavities connected to the crankcase (e.g. oil return duct, chain shaft, etc.) and lead to oil leakage at the seals.

The crankcase ventilation prevents this. It routes the blow-by gases, which are largely oil-free, into the air intake system and the separated droplets of oil flow back to the oil sump via an oil return line. In addition, the crankcase ventilation, in combination with a pressure control valve, ensures a low vacuum in the crankcase during engine operation.

**Tasks of crankcase ventilation:**

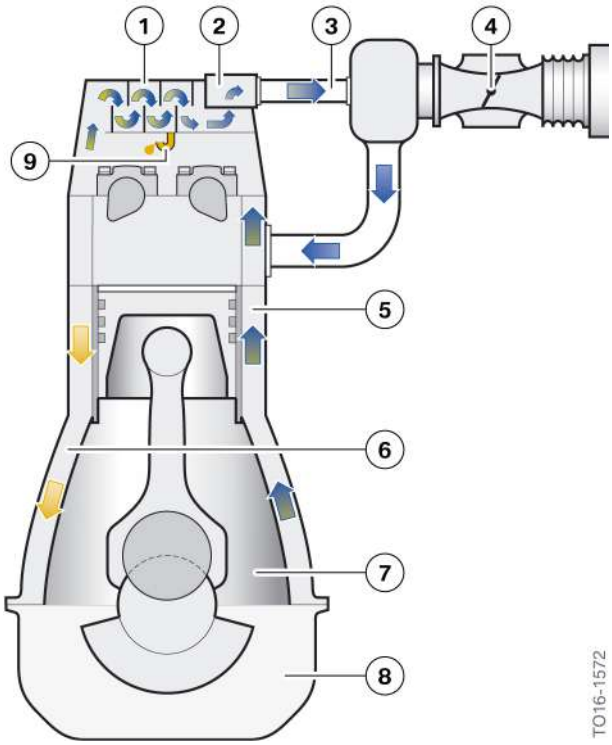
- Cleaning the blow-by gases to remove engine oil
- Recirculation of the cleaned blow-by gases in the intake area
- Regulation of the internal engine pressure.

On naturally aspirated engines the crankcase ventilation is designed to be single-stage and two-stage on turbo-charged engines. This means that, on turbo-charged engines the blow-by gases are routed via different channels depending on the load condition.

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.

Always ensure absolute cleanliness when filling the engine with engine oil. Clean the oil filler neck before you screw on the sealing cap. Engine oil residue at the sealing cap may lead to misdiagnosis at the crankcase ventilation.

## 2.2.1. Single-stage crankcase ventilation




N12 Engine, single-stage crankcase ventilation

| Index | Explanation |
|---|---|
| 1 | Oil separator (labyrinth) |
| 2 | Pressure control valve |
| 3 | Channel to the intake system |
| 4 | Throttle valve |
| 5 | Blow-by channel from crankcase |
| 6 | Oil return |
| 7 | Crank chamber |
| 8 | Oil sump |
| 9 | Siphon |

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.

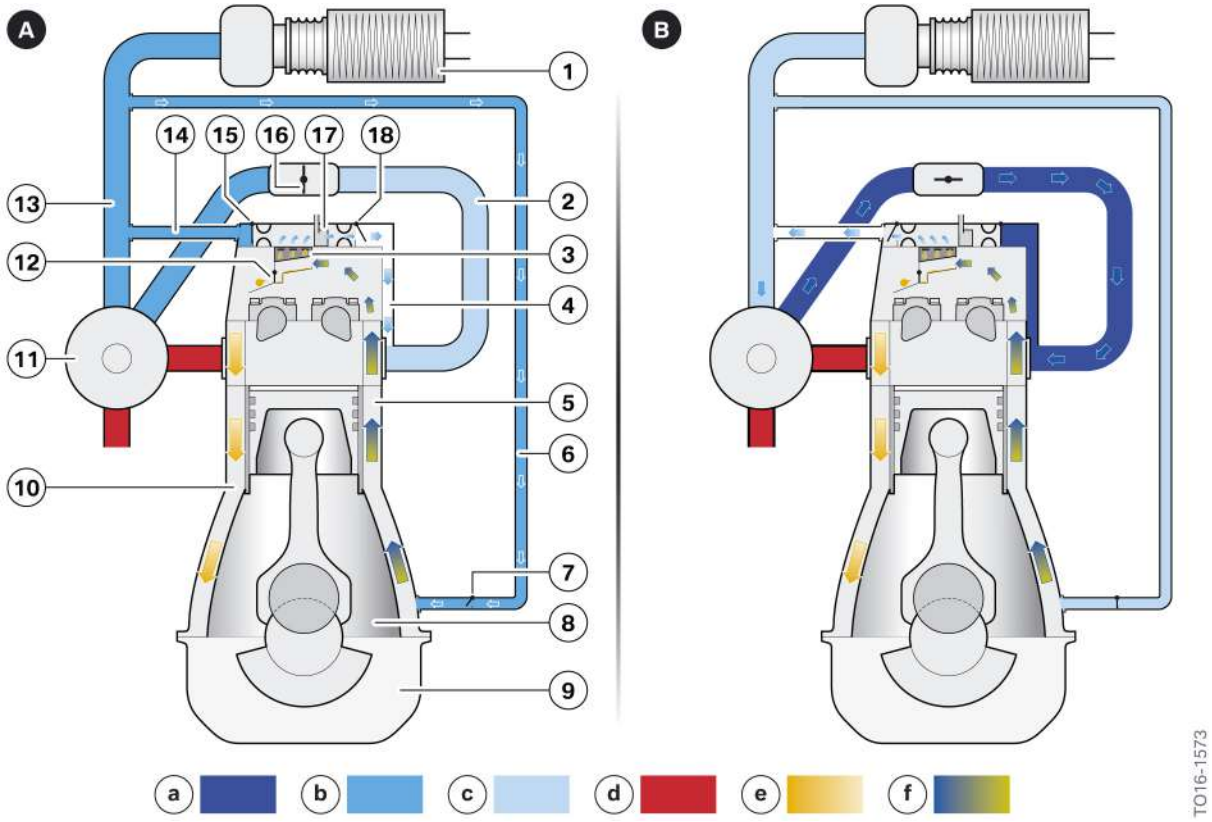
There is a relative vacuum in the intake pipe due to the flow resistance of the throttle valve and the air cleaner. Due to the pressure difference in relation to the crankcase, the blow-by-gas is drawn into the cylinder head via the blow-by channel and first reaches the settling chamber there.

Among other things, the purpose of the settling chamber is to stop any oil spray, e.g. from the camshafts, from getting into the crankcase ventilation. If the oil is subsequently separated via a labyrinth, the settling chamber additionally serves to reduce gas oscillations of the blow-by gas. This avoids excitation of the diaphragm in the pressure control valve.

On engines with cyclone filter (helix), these gas oscillations are definitely desired since they improve the oil separation efficiency. The gas then settles down in the cyclone filter. The settling chamber is therefore designed differently here than for an oil separation by means of a labyrinth.

The fine oil mist is separated from the blow-by gases with the assistance of a separator and the oil droplets formed are routed back to the oil sump via the oil return.

## 2.2.2. Two-stage crankcase ventilation




B48 Engine, two-stage crankcase ventilation

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.

| Index | Explanation |
|---|---|
| A | Naturally aspirated engine operation |
| B | Turbo-charging |
| a | Charging pressure |
| b | Ambient pressure |
| c | Vacuum |
| d | Exhaust gas |
| e | Oil |
| f | Blow-by gas |
| 1 | Air filter |
| 2 | Intake manifold |
| 3 | Separator |
| 4 | Blow-by channel (upstream of the intake valves) |
| 5 | Blow-by channel (from crankcase) |
| 6 | Purge air line |
| 7 | Non-return valve |
| 8 | Crank chamber |
| 9 | Oil sump |
| 10 | Oil return |
| 11 | Exhaust turbocharger |
| 12 | Non-return valve, oil return |
| 13 | Clean air pipe |
| 14 | Hose to clean air pipe |
| 15 | Non-return valve with restrictor |
| 16 | Throttle valve |
| 17 | Pressure control valve |
| 18 | Non-return valve with restrictor |

**Naturally aspirated engine operation**

The throttle valve is closed during naturally-aspirated engine operation. This produces a vacuum in the intake system. This vacuum facilitates a drawing in of the blow-by gases via blow-by channels in the intake system or directly upstream of the intake valves. On engines with Valvetronic, the throttle valve is also lightly shut in these operating conditions to guarantee the engine ventilation function.

As of the engines N18 and N20, purge air lines are used to feed fresh air directly into the crankcase via a non-return valve. The bigger the vacuum in the crankcase, the bigger the air mass introduced into the crankcase. By flushing the crankcase with fresh air, the chemical ageing of the lubricating oil and also the water content in the blow-by gases are reduced and icing up of the pressure control valve is prevented.

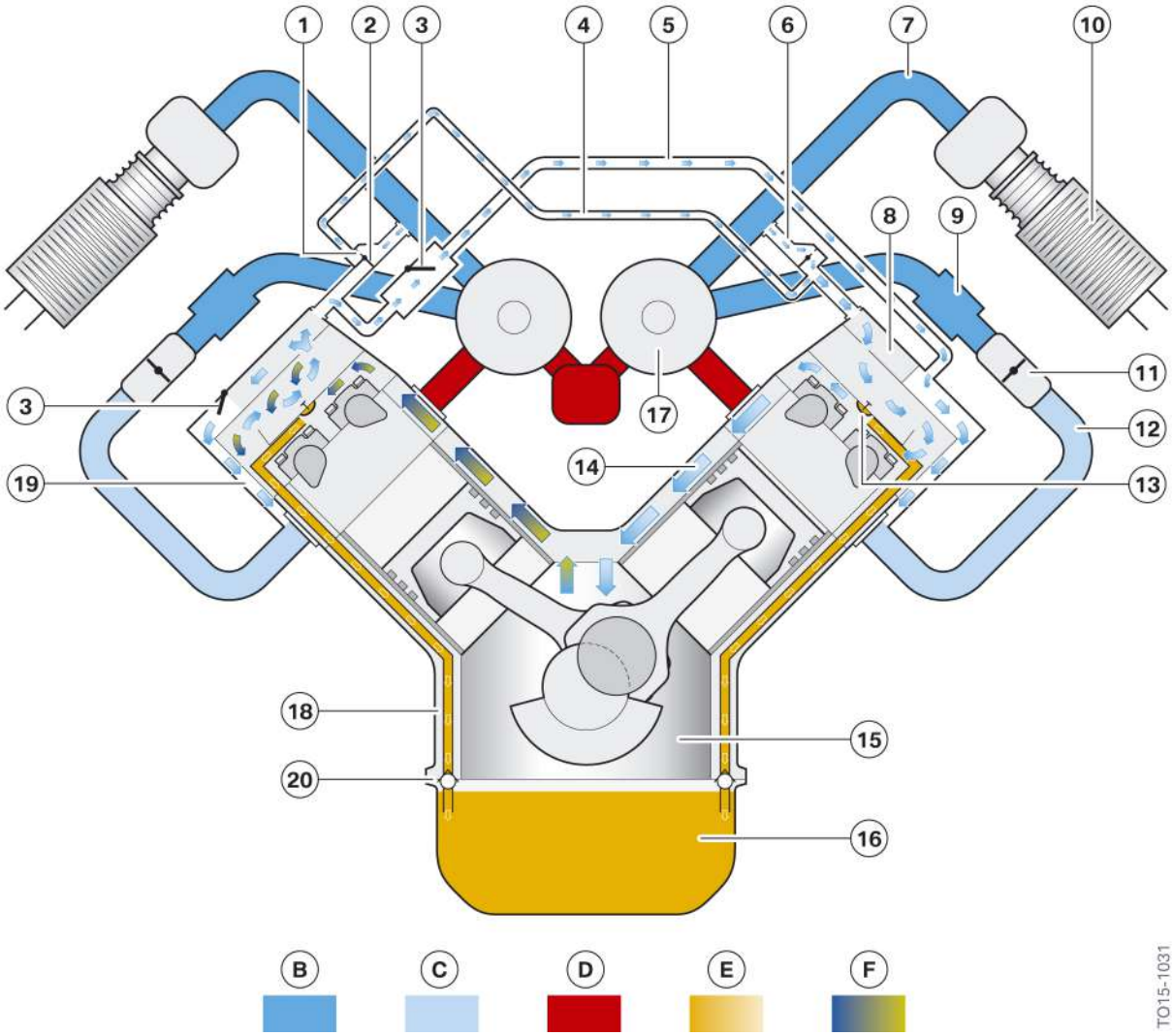
**Turbo-charging**

The cylinder head cover is connected to the clean air pipe by a hose. During turbo-charged operation, a vacuum is produced in the clean air pipe. This vacuum facilitates a drawing in of the blow-by gases via the clean air pipe.

## 2.2.3. Register ventilation

With the N74 engine, register ventilation is used for the V-engines. In naturally aspirated mode, the crankcase is ventilated via the oil separation in cylinder bank 2 only. In doing so, the efficiency of the oil separator is increased in partial load operation. The crankcase is ventilated in the counterflow via the separator in cylinder bank 1.

In turbo-charged mode, the crankcase is ventilated via the oil separation of both cylinder banks.

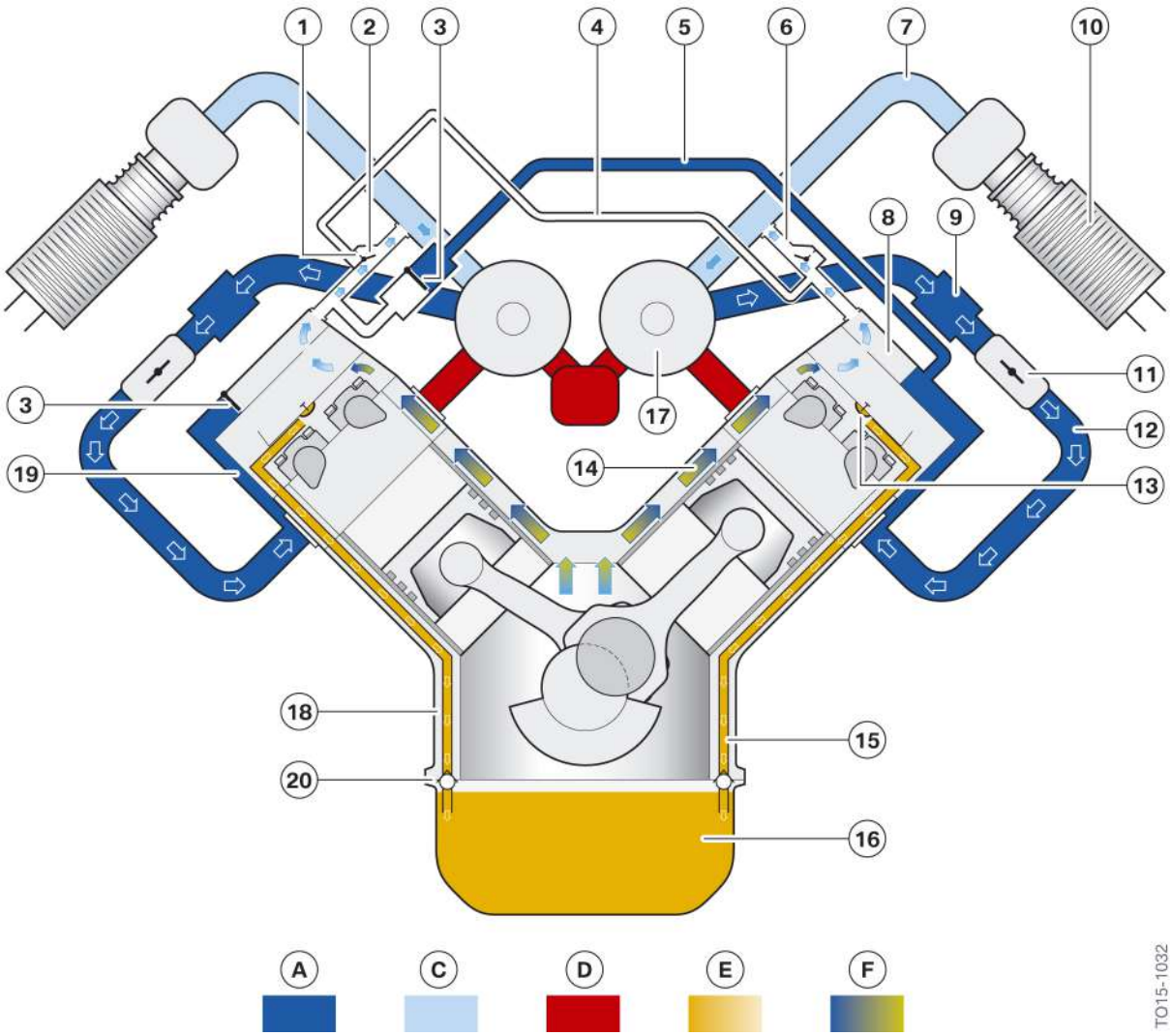



N63TU2 engine, crankcase ventilation, naturally aspirated mode

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.

| Index | Explanation |
|---|---|
| B | Ambient pressure |
| C | Vacuum |
| D | Exhaust gas |
| E | Oil |
| F | Blow-by gases |
| 1 | Leakage hole in the housing of the non-return valve |
| 2 | Non-return valve for the clean air pipe (closed) |
| 3 | Volume control valves for the intake system with throttle function (opened) |
| 4 | Purge air line |
| 5 | Intake pipe from bank 2 to bank 1 |
| 6 | Line to the clean air pipe (with closed non-return valve and leakage hole) |
| 7 | Clean air pipe |
| 8 | Oil separator |
| 9 | Charge air cooler |
| 10 | Intake silencer with hot film air mass meter |
| 11 | Throttle valve |
| 12 | Intake manifold |
| 13 | Oil return |
| 14 | Ventilation duct |
| 15 | Crank chamber |
| 16 | Oil sump |
| 17 | Exhaust turbocharger |
| 18 | Oil return duct |
| 19 | Duct in the cylinder head |
| 20 | Non-return valve |

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.




N63TU2 engine, crankcase ventilation, turbo-charged mode

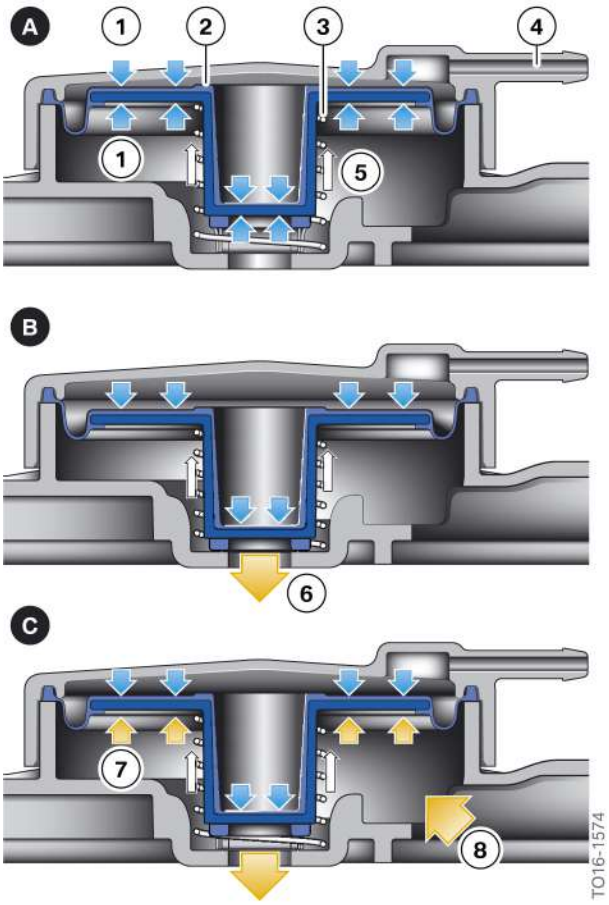
| Index | Explanation |
|---|---|
| A | Charging pressure |
| C | Vacuum |
| D | Exhaust gas |
| E | Oil |
| F | Blow-by gases |
| 1 | Leakage hole in the housing of the non-return valve |
| 2 | Non-return valve for the clean air pipe (open) |
| 3 | Volume control valves for the intake system with throttle function (closed) |
| 4 | Purge air line |
| 5 | Intake pipe from bank 2 to bank 1 |

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.

| Index | Explanation |
|---|---|
| 6 | Line to the clean air pipe (with opened non-return valve and leakage hole) |
| 7 | Clean air pipe |
| 8 | Oil separator |
| 9 | Charge air cooler |
| 10 | Intake silencer with hot film air mass meter |
| 11 | Throttle valve |
| 12 | Intake manifold |
| 13 | Oil return |
| 14 | Ventilation duct |
| 15 | Crank chamber |
| 16 | Oil sump |
| 17 | Exhaust turbocharger |
| 18 | Oil return duct |
| 19 | Duct in the cylinder head |
| 20 | Non-return valve |

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.

## 2.2.4. Pressure control valve




N14 engine, pressure control valve

| Index | Explanation |
|---|---|
| A | Pressure control valve opened (engine standstill) |
| B | Pressure control valve closed (idle or coasting overrun mode) |
| C | Pressure control valve in control operation (load) |
| 1 | Ambient pressure |
| 2 | Diaphragm |
| 3 | Compression spring |
| 4 | Connection ambient pressure |
| 5 | Spring force compression spring |
| 6 | Vacuum (intake system) |
| 7 | Vacuum (in the housing) |
| 8 | Blow-by gas (from the crankcase) |

On modern engines, the crankcase ventilation is regulated by a pressure control valve. Its job is to ensure that there is as constant a vacuum as possible in the crankcase (up to 70 mbar, depending on the engine version). Thus, the tightness of the engine is optimised.

At engine standstill there is ambient pressure on both sides of the diaphragm and the diaphragm is pressed upwards by the spring force of the compression springs – the pressure control valve is completely opened.

If the engine is started, the pressure reduces in the intake pipe and below the diaphragm. The ambient pressure presses this downwards, thus closing the pressure control valve. This state arises at idle and in coasting overrun mode.

Blow-by gas is produced by load and speed and increases the pressure under the diaphragm again. The pressure control valve opens and the blow-by gases are drawn in via the air intake system. The pressure control valve opens wide until an equilibrium of ambient pressure and pressure under the membrane plus spring force arises.

The more blow-by gases are produced, the wider the pressure control valve opens. Thus, a specified vacuum in the crankcase is maintained.

### 2.2.5. Oil separation

To remove the blow-by gas from the engine oil, different oil separator designs are used for BMW and MINI and partially combined:

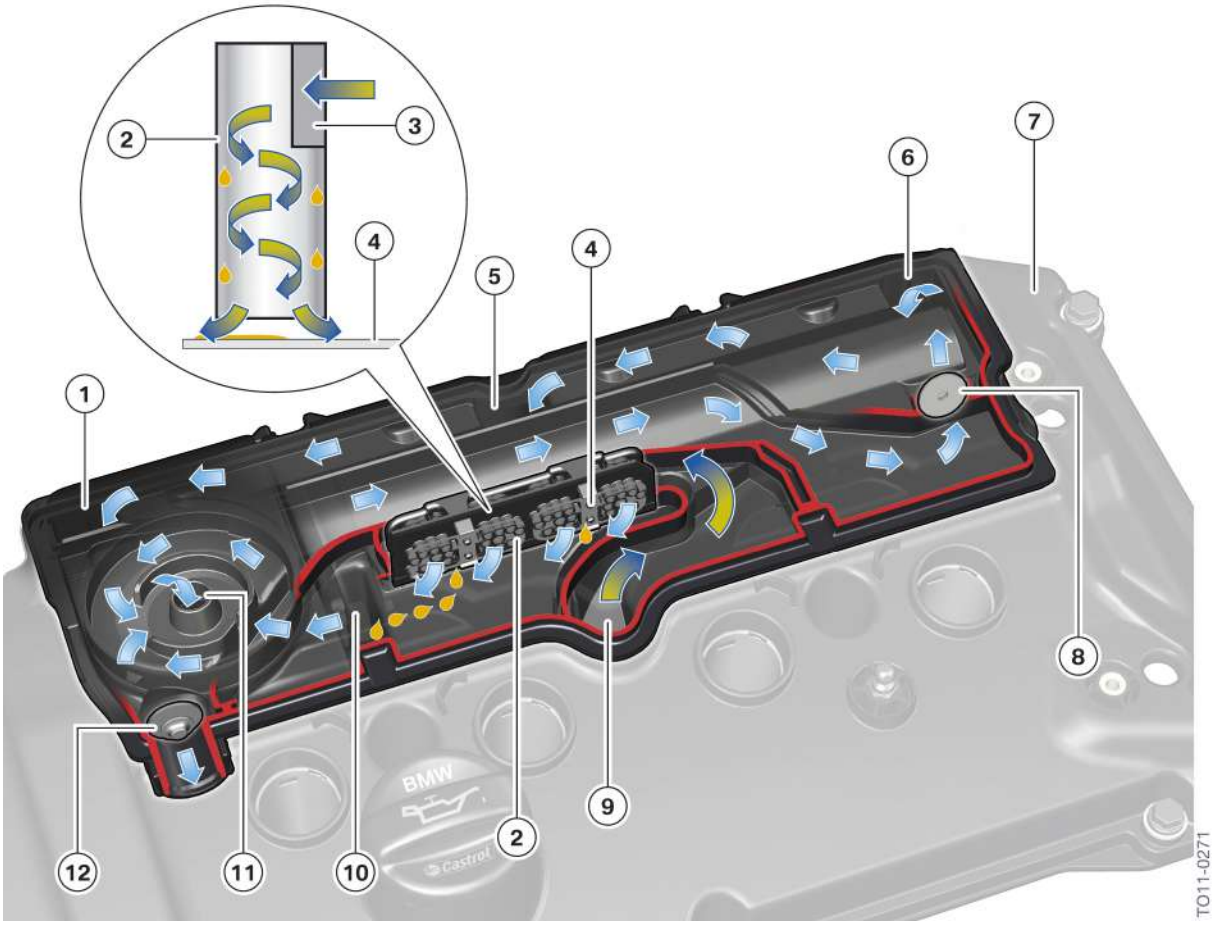
- Labyrinth oil separator
- Cyclone filter
- Cyclone flexible tongue separator
- Fine cobbing plates with small air vents
- Impact surfaces with upstream fleece.

The oil drops deposit on the obstructions and flow back into the cylinder head again.

On most BMW petrol engines, the oil separator is integrated into the cylinder head cover. However, it can also be located outside of the engine housing such as the cyclone filter of the N42 engine.

You can see examples of this in the two following graphics.

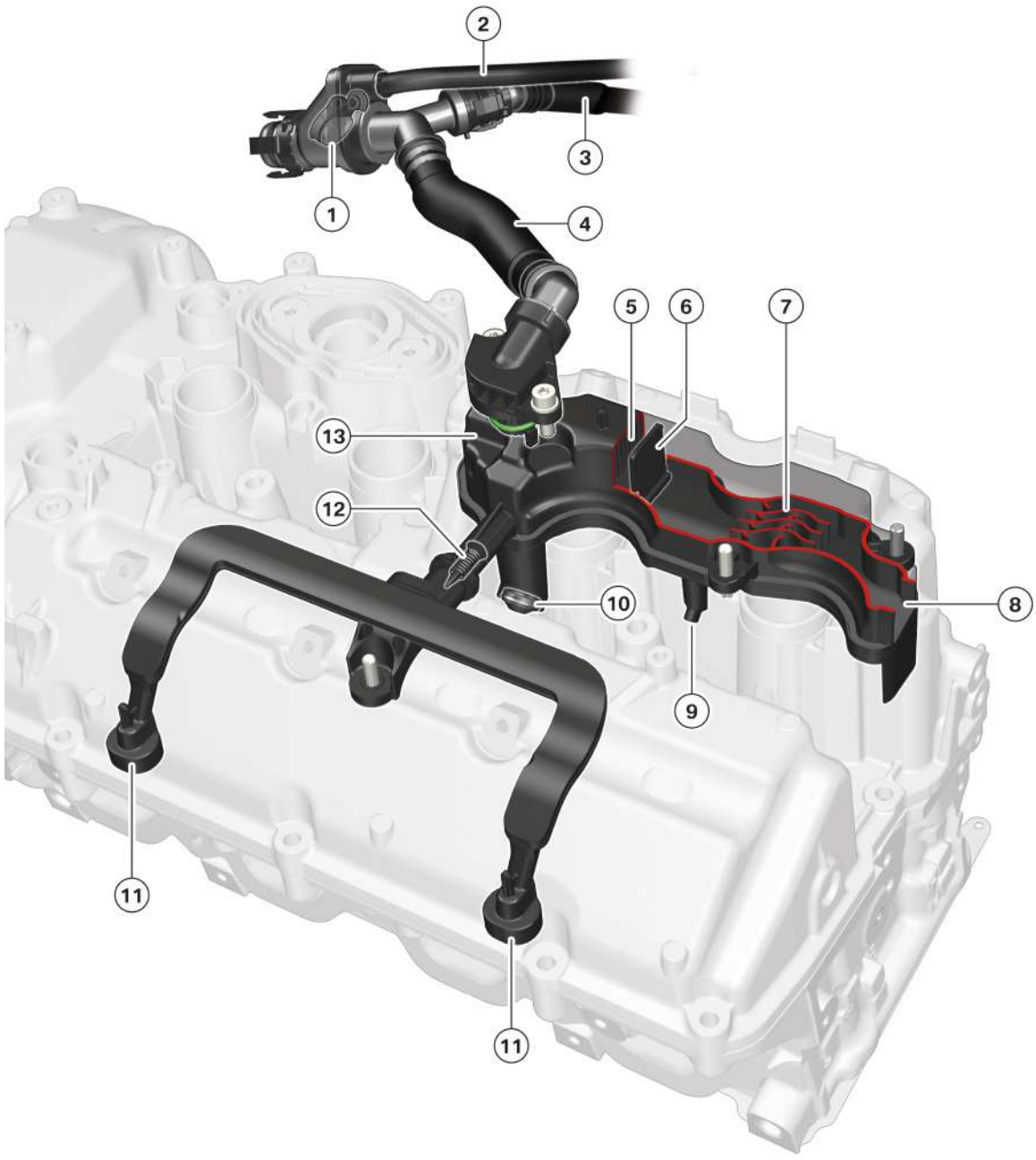
# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.




N13 engine, oil separation

| Index | Explanation |
|---|---|
| 1 | Duct to cylinder head into intake port, cylinder 1 |
| 2 | Cyclone flexible tongue separator |
| 3 | Side opening in cyclone separator |
| 4 | Flexible tongue |
| 5 | Duct to cylinder head into intake port, cylinder 2 and 3 |
| 6 | Duct to cylinder head into intake port, cylinder 4 |
| 7 | Cylinder head cover |
| 8 | Non-return valve in duct to intake ports |
| 9 | Opening to cylinder head |
| 10 | Oil return duct |
| 11 | Pressure control bore in pressure control valve |
| 12 | Non-return valve in duct to charge-air suction line |

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.




N63TU2 engine, cylinder head cover with crankcase ventilation

| Index | Explanation |
|---|---|
| 1 | Non-return valve for the clean air pipe with leak hole |
| 2 | Purge air line |
| 3 | Intake pipe with volume control valve |
| 4 | Line to clean air pipe |
| 5 | Impact surface with upstream fleece |

| Index | Explanation |
|---|---|
| 6 | Fine cobbing plate with small air vents |
| 7 | Pre-separator |
| 8 | Inlet for blow-by gases |
| 9 | Oil return |
| 10 | Oil separator |
| 11 | Volume control valve for the air intake system with throttle function |
| 12 | Connecting line via blow-by-gas channel for the intake port |

## 2.3. Crankcase

If the crankcase material is not capable of meeting the requirements of the cylinders, then cylinder liners made of suitable material are used or the cylinder liners are appropriately coated.

We differentiate between 2 types of cylinder liners:

- Dry cylinder liners: The cylinder liners do not come into contact with the water jacket.
- Wet cylinder liners: The cylinder liners come into direct contact with the water jacket.

### 2.3.1. Electric arc wire spraying procedure

As of the N20 engine, at BMW and MINI, the cylinder liners of the 3-, 4- and 6-cylinder engines are coated using the electric arc wire spraying procedure (LDS procedure).



Electric arc wire spraying procedure

In this procedure a conductive metal wire is heated using an electric arc until it melts. The melt is then sprayed onto the cylinder barrels at high pressure. This layer of ferrous material is roughly 0.3 mm thick, extremely wear-resistant and facilitates an efficient transfer of heat from the combustion chambers to the crankcase, and from there to the coolant ducts.

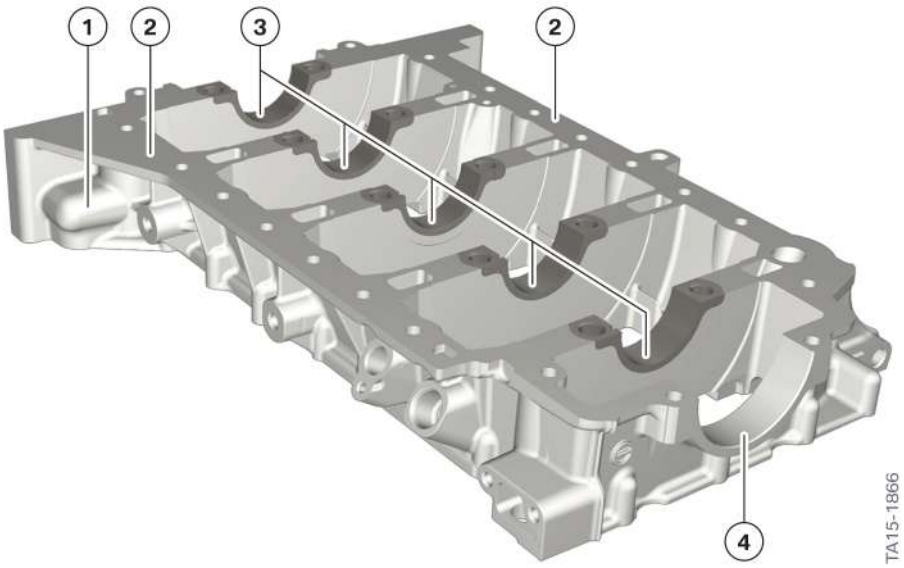
**Benefits:**

- Lower weight
- High wear resistance
- Good heat dissipation to the crankcase
- Lower internal engine friction thanks to excellent sliding properties.

Due to the thin material application during the electric arc wire-spraying procedure, subsequent processing of the cylinder barrels is not possible.

## 2.3.2. Bedplate

With exception of the B engine generations, all current 4- and 6-cylinder petrol engines have a split crankcase. The lower part of the crankcase is also referred to as "bedplate".




N20 engine, bedplate

| Index | Explanation |
|---|---|
| 1 | Bedplate |
| 2 | Sealing surface to upper crankcase housing section |
| 3 | Main bearing cap (without main bearing shells) |
| 4 | Sealing surface of crankshaft seal (belt side) |

# ENGINE MECHANICS – PETROL ENGINE.
# 2. ENGINE HOUSING.

In the bedplate, the main bearing caps are assembled in the lower housing section. This increases the rigidity of the entire crankcase. The bottom side of the bedplate forms the flange surface to the oil sump. The bedplate generally consist of the same material as the upper part of the crankcase.

The seal toward the upper crankcase housing section is created by means of a hardening liquid sealing compound since an exact positioning of the main bearing cap with a conventional seal is not possible.

Check the storage life of the liquid sealing compound. Never use liquid sealing compound that is past its storage date.

Depending on the engine type, the liquid sealing compound is applied in different ways:

**W1x, N1x and N20 petrol engines**

The liquid sealing compound is applied to the sealing surface of the upper part of the crankcase before the installation of the bedplate.

**N4x and N5x petrol engines**




N42 engine, seal bedplate

In the sealing surfaces, one grove each runs along the left and right bedplate flange. These lead to the crankshaft seals in each case. After screwing on the bedplate, the liquid sealing compound is pressed into these grooves using injector nozzles. The liquid sealing compound is introduced until it leaks out on both crankshaft seals on the left and right.

Using a primer, the sealing compound is hardened at the points of exit. The crankshaft seal itself has grooves on the points of exit, which must be positioned correctly during installation.

A special tool is available for pressing out the liquid sealing compound.

You must comply with the screwing sequence and the tightening torques during assembly. Leaks in the crankcase and engine damage may otherwise occur.

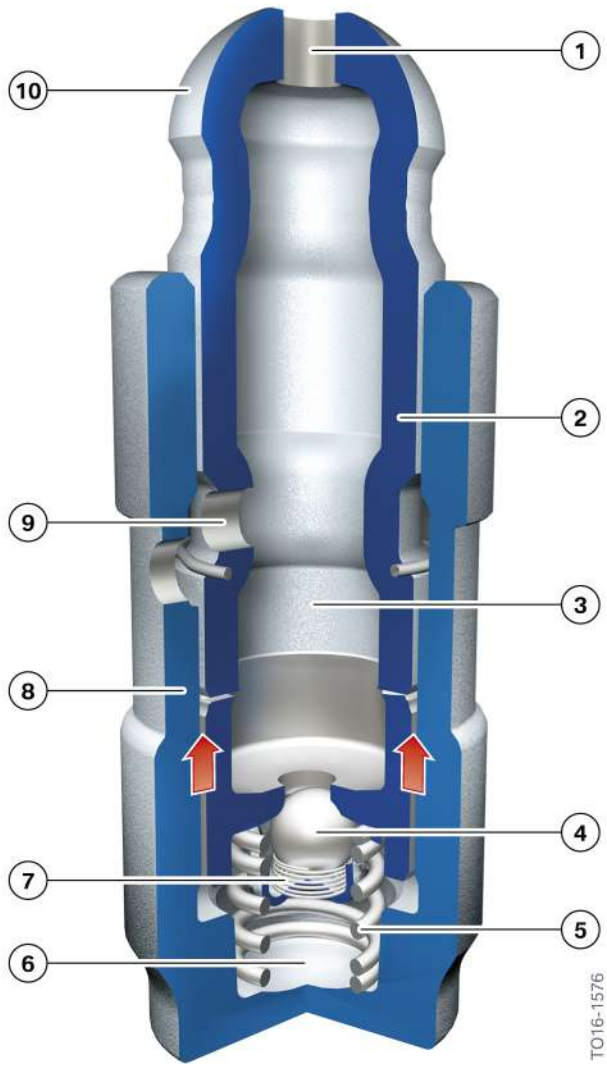
# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.

## 3.1. Hydraulic valve clearance compensation

The hydraulic valve clearance compensation (HVA) ensures that the intake and exhaust valves have a correct valve clearance under all operating conditions and thus close correctly. Even after longer service life of the engine, the valve clearance no longer needs to be adjusted.

On most of the current BMW and MINI engines, the roller cam follower rests on the hydraulic valve clearance compensating element. Exceptions are the W11 and W12 engine on which the hydraulic valve clearance compensating elements are integrated into the rocker arms.

Another exception are some of the engines from M GmbH. These rely on box tappets.




N12 engine, hydraulic valve clearance compensating element

| Index | Explanation |
|---|---|
| 1 | Vent hole |
| 2 | Piston |
| 3 | Oil supply chamber |
| 4 | Sealing cone |
| 5 | Spring for plunger |
| 6 | Pressure chamber |
| 7 | Spring for sealing cone |
| 8 | Pressure cylinder |
| 9 | Oil supply hole |
| 10 | Ball head |

If the roller cam follower presses on the ball head, the piston supports itself via the oil present on the pressure cylinder in the pressure chamber. A small amount of leakage oil between piston and pressure cylinder escapes upwards.

Therefore, a spring presses the piston upwards after the actuation, whereby the ball head always sits closely on the roller cam follower. In the process, the oil from the oil sump gets to the pressure chamber.

The locking ball seals the pressure chamber again when the latter is filled.

On the W10 and W11 engine, the hydraulic valve clearance compensating elements are integrated into the rocker arm and operate the valve directly.

If an hydraulic valve clearance compensating element is faulty, the oil is no longer able to be held in the pressure chamber. As a result, the affected valve will have a large valve clearance which is usually noticeable due to rattling noises.

## 3.2. VANOS

The variable camshaft timing control (VANOS) facilitates a movement of the camshaft in relation to the crankshaft. In the process, the timing is adapted to optimise the efficiency of the engine.

It makes a high torque possible in the low and medium engine speed range and a high maximum power in the higher engine speed ranges. A further benefit of the VANOS is the option of internal EGR. This reduces the emission of harmful nitrogen oxides NOx, particularly in the partial load range. The following is also achieved:

- Faster heating up of catalytic converter
- Lower pollutant emissions during cold start
- Reduction in consumption.

For the maximum engine performance, the closing time of the intake valve is decisive. At higher speeds, it is shifted towards "retarded" so that the cylinder filling is as optimal as possible and a higher performance is achieved.

# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.

A flow back of the gas from the combustion chamber to the intake port can be avoided by a moderate speed adjustment of the closing time of the intake valve.

With the variable camshaft timing control, the residual gas fraction in the cylinder can be controlled. The retention of the residual gas in the cylinder limits the temperature level of the combustion. Thus, the $NO_x$ emissions are also reduced.

The adjustment of the intake camshaft is thus used in the lower and middle engine speed range primarily for the torque increase and the internal exhaust-gas recirculation. At higher engine speeds the performance is to the fore.

All current BMW and MINI engines have a double VANOS. This means that in addition to the intake camshaft, the exhaust camshaft is also adjusted. The adjustment of the exhaust camshaft is used for optimal idling quality or achieving a maximum exhaust-gas recirculation rate.



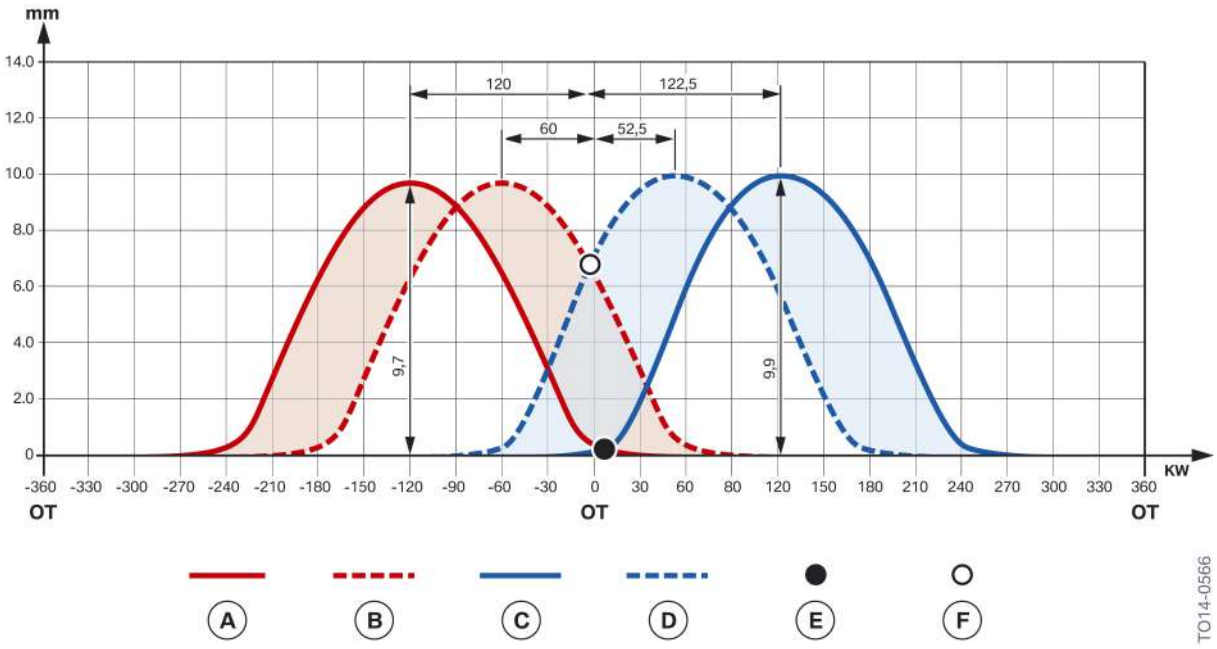
The VANOS adjusters for the intake and exhaust side differ; there is a risk of risk of mixing up (similar parts)!

Please note the identification on the VANOS adjusters, otherwise there is a risk of engine damage:

- The VANOS adjuster for the exhaust side is marked with “EX” or “AUS”.
- The VANOS adjuster for the intake side is marked with “IN” or “EIN”.

The driver will notice a malfunction of the VANOS by a loss of power and increased fuel consumption.

## 3.2.1. Timing diagram

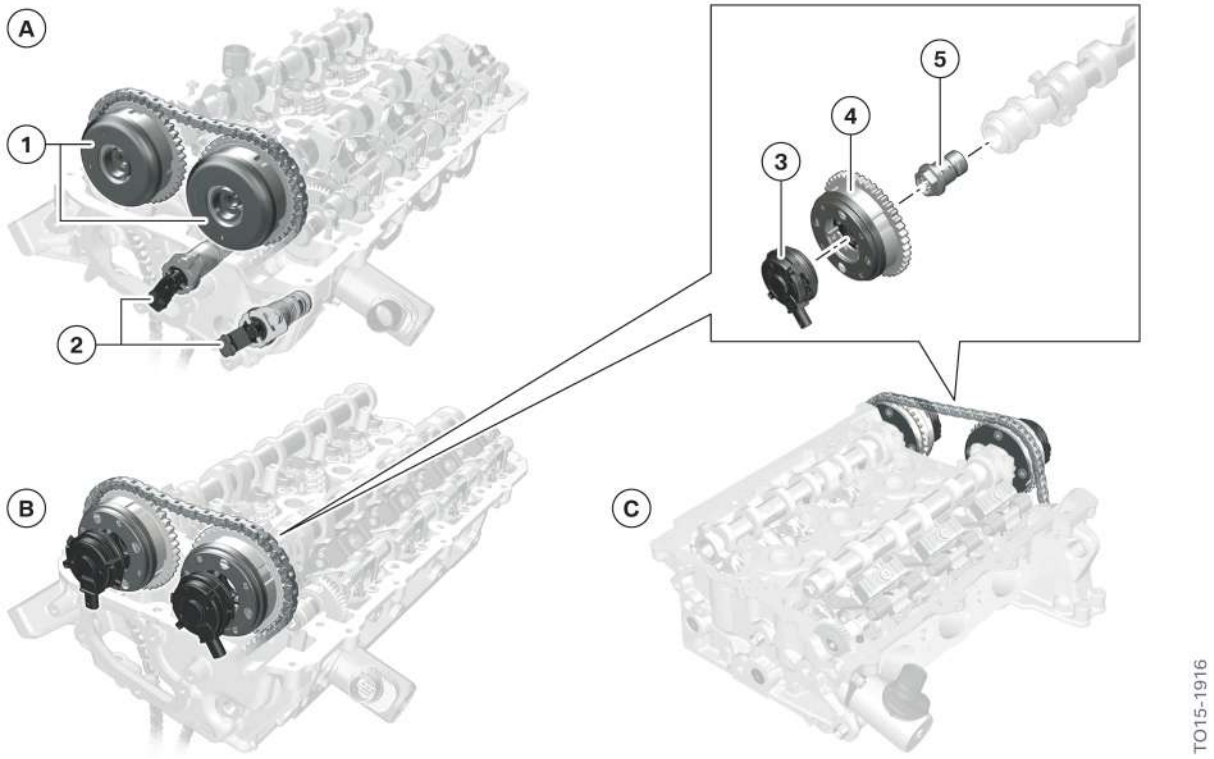



Variable valve overlap, Bx8 engine

# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.

| Index | Explanation |
|---|---|
| A | VANOS solenoid valve, exhaust, de-energized |
| B | VANOS solenoid valve, exhaust, duty cycle 100 % |
| C | VANOS solenoid valve, inlet, de-energized |
| D | VANOS solenoid valve, inlet, duty cycle 100 % |
| E | Smallest valve overlap in idle |
| F | Maximum valve overlap in partial load |

## 3.2.2. Component overview




VANOS systems component overview

| Index | Explanation |
|---|---|
| A | N63TU |
| B | N63TU2 |
| C | B38 engine |
| 1 | VANOS adjuster unit |

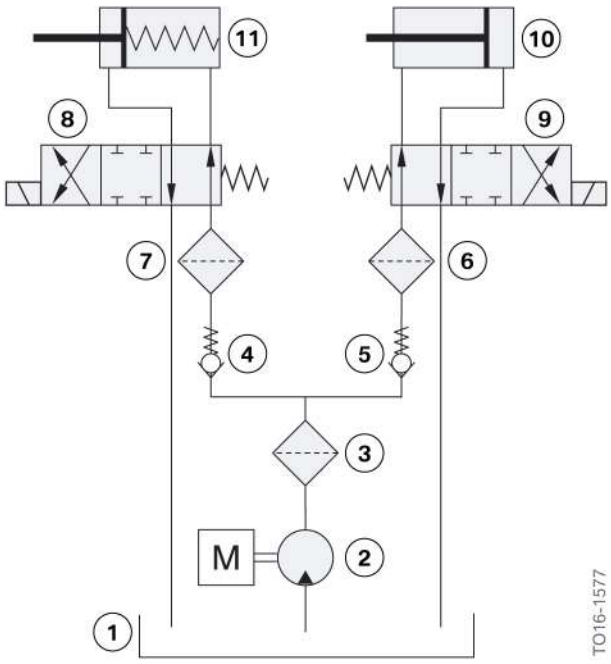
| Index | Explanation |
|---|---|
| 2 | VANOS solenoid valve |
| 3 | VANOS solenoid actuator |
| 4 | VANOS adjuster unit |
| 5 | VANOS central valve |

### 3.2.3. Function

The Digital Motor Electronics (DME) uses sensors to detect the position of the crankshaft and camshafts. There are characteristic maps stored in the DME for the positioning of the camshafts to the crankshaft. These characteristic maps essentially take the following parameters into consideration:

- Engine speed
- Load requirement
- Coolant temperature.

The DME thereupon controls the VANOS solenoid valves resp. the VANOS central valve via the VANOS solenoid actuators. These in turn control an oil pressure with which the VANOS adjusters and thus the camshafts are adjusted.




N46 engine, hydraulic circuit diagram VANOS

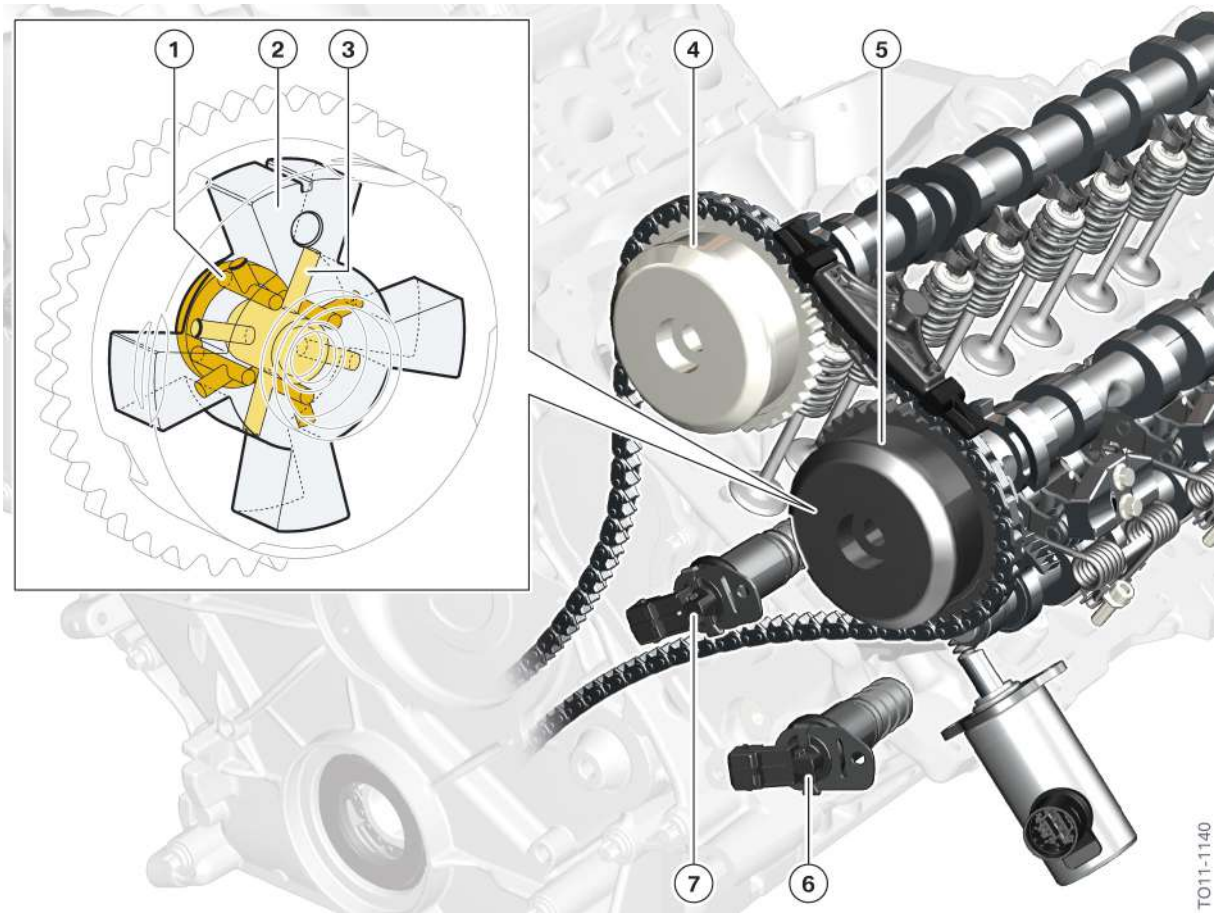
# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.

| Index | Explanation |
|---|---|
| 1 | Oil sump |
| 2 | Oil pump |
| 3 | Oil filter |
| 4 | Non-return valve |
| 5 | Non-return valve |
| 6 | Strainer |
| 7 | Strainer |
| 8 | VANOS solenoid valve |
| 9 | VANOS solenoid valve |
| 10 | VANOS adjuster unit |
| 11 | VANOS adjuster unit |

**VANOS adjuster with VANOS solenoid valve**

With the exception of the N20 and N63TU2 engines, as well as the engines of the new B engine generation, the VANOS adjusters are controlled by separate VANOS solenoid valves on current BMW and MINI engines. These are located in oil ducts in the cylinder head. The VANOS adjusters are each attached to the camshaft by a mounting bolt.

# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.



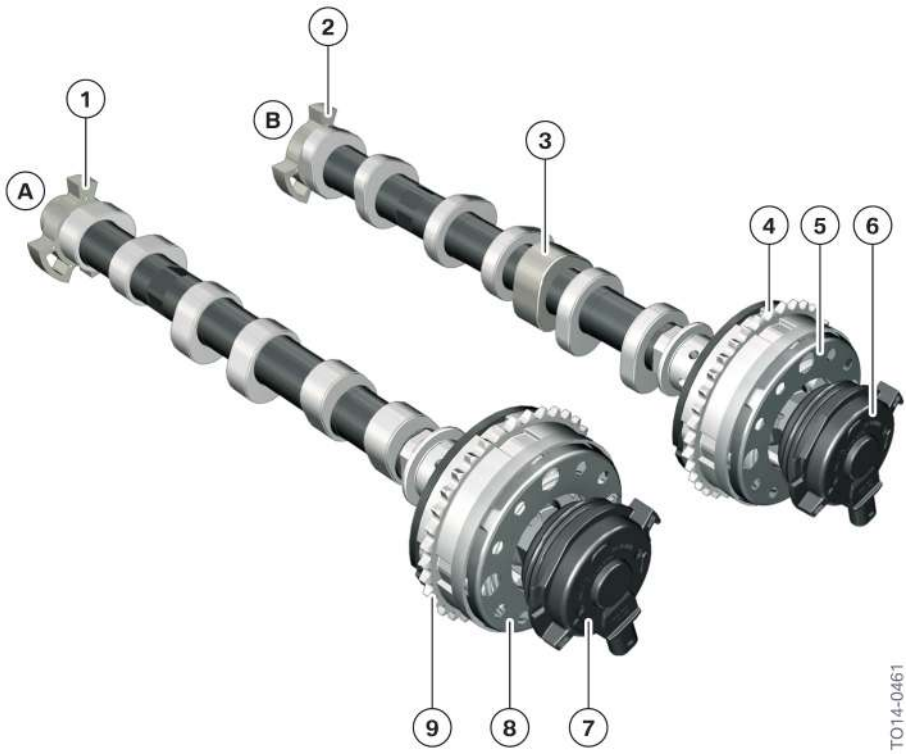

N63TU engine, VANOS adjuster with VANOS solenoid valve

| Index | Explanation |
|---|---|
| 1 | Oil duct for timing advance |
| 2 | Rotor |
| 3 | Oil duct for timing retardation |
| 4 | VANOS, exhaust side |
| 5 | VANOS, intake side |
| 6 | VANOS solenoid valve, intake side |
| 7 | VANOS solenoid valve, exhaust side |

**VANOS adjuster with VANOS solenoid actuator and VANOS central valve**

In the N20 and N63TU2 engines, as well as the engines of the new B engine generation, VANOS solenoid actuators and VANOS central valves are used. The VANOS central valve is operated by the VANOS solenoid actuator and takes over the mounting of the VANOS adjuster with the camshaft. The oil flow into the VANOS unit is simultaneously controlled by the VANOS central valve. As a result, the oil ducts in the cylinder head are reduced and the adjustment speed is increased.

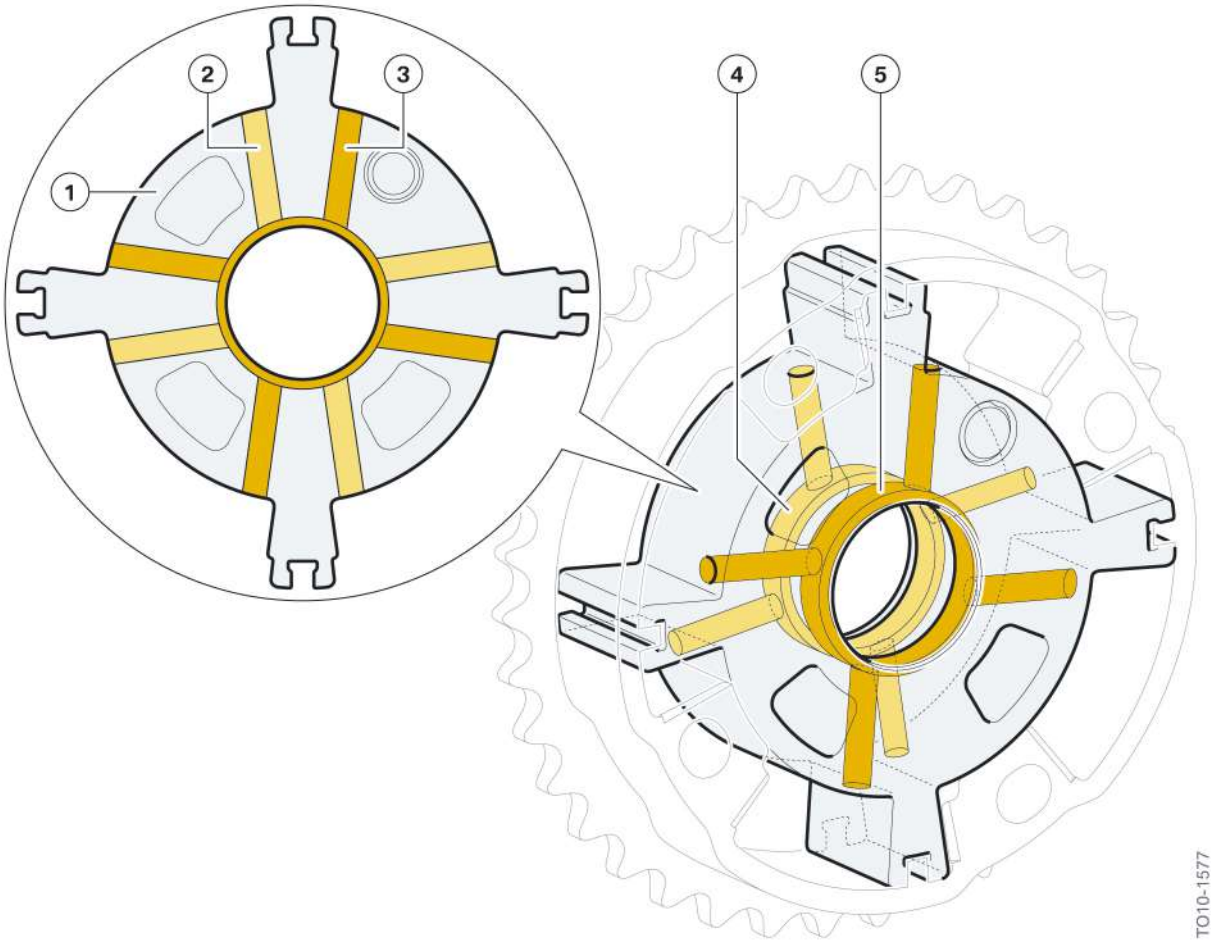
# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.




B38 engine, double VANOS

| Index | Explanation |
|---|---|
| A | Intake camshaft |
| B | Exhaust camshaft |
| 1 | Increment wheel, intake camshaft |
| 2 | Increment wheel, exhaust camshaft |
| 3 | Triple cam for high pressure pump drive system |
| 4 | Exhaust camshaft sprocket |
| 5 | VANOS unit, exhaust side |
| 6 | VANOS solenoid valve actuator, exhaust |
| 7 | VANOS solenoid valve actuator, intake |
| 8 | VANOS unit, intake side |
| 9 | Intake camshaft sprocket |

# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.

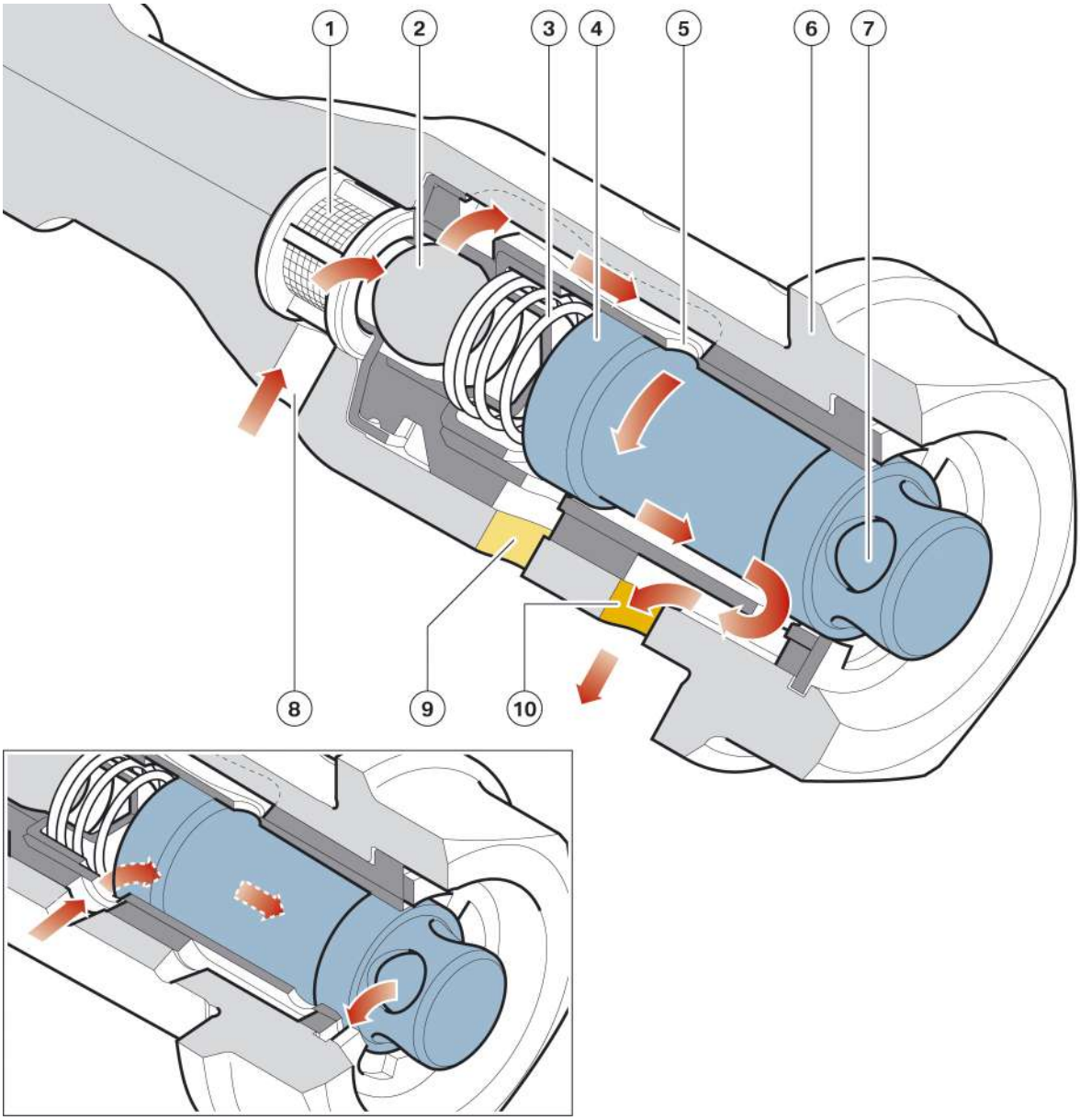



Bx8 engine, oil ducts of the VANOS adjuster intake camshaft

| Index | Explanation |
|---|---|
| 1 | Rotor |
| 2 | Oil duct for timing advance |
| 3 | Oil duct for timing retardation |
| 4 | Oil duct for timing advance |
| 5 | Oil duct for timing retardation |

The following graphic shows the oil flow rate in the VANOS central valve which is used to control the camshaft adjustment:

# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.

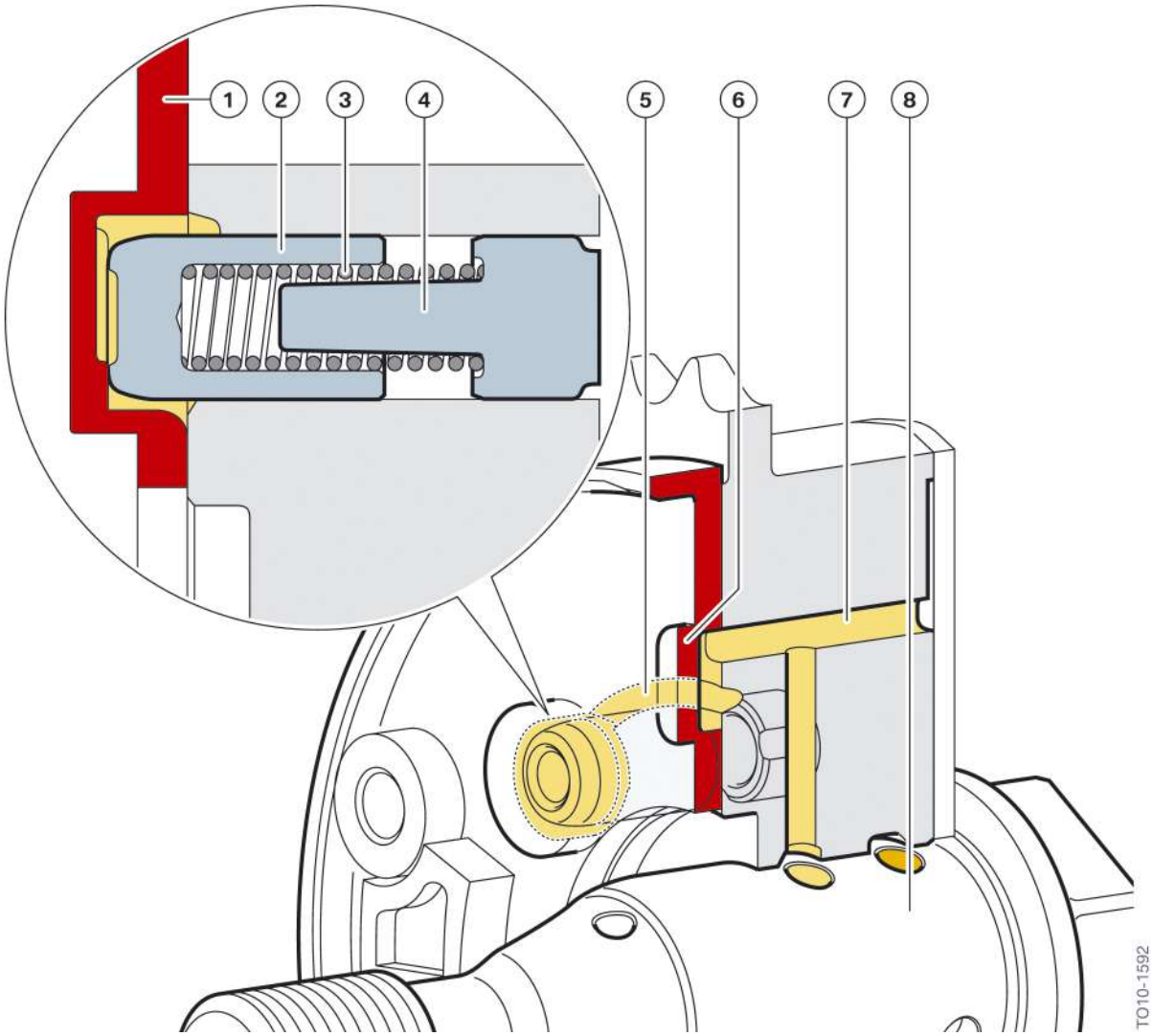



VANOS central valve, intake camshaft, Bx8 engine

| Index | Explanation |
|---|---|
| 1 | Filter |
| 2 | Ball |
| 3 | Spring |
| 4 | Piston |
| 5 | Sleeve |
| 6 | Housing |

| Index | Explanation |
|---|---|
| 7 | Opening in plunger |
| 8 | Oil supply from main oil duct |
| 9 | Bore to oil duct in VANOS (timing advance) |
| 10 | Bore to oil duct in VANOS (timing retardation) |

**Setting the timing**




Bx8 engine, locking pin

| Index | Explanation |
|---|---|
| 1 | Locking cover |
| 2 | Locking pin |
| 3 | Locking spring |
| 4 | Cartridge |
| 5 | Oil duct |
| 6 | Locking cover |
| 7 | Oil duct |
| 8 | VANOS central valve |

On both VANOS variants, a lock ensures that the VANOS adjuster assumes a base position in the depressurised state. For this purpose, a locking pin is pressed into the locking position by a compression spring thus blocking the VANOS adjuster. In this state, the timing can be checked and adjusted.

The oil pressure that exists for the adjustment also unlocks the locking pin. Here, the acting oil pressure presses the locking pin against the locking spring and enables the adjustment of the VANOS.

## 3.3. Valvetronic

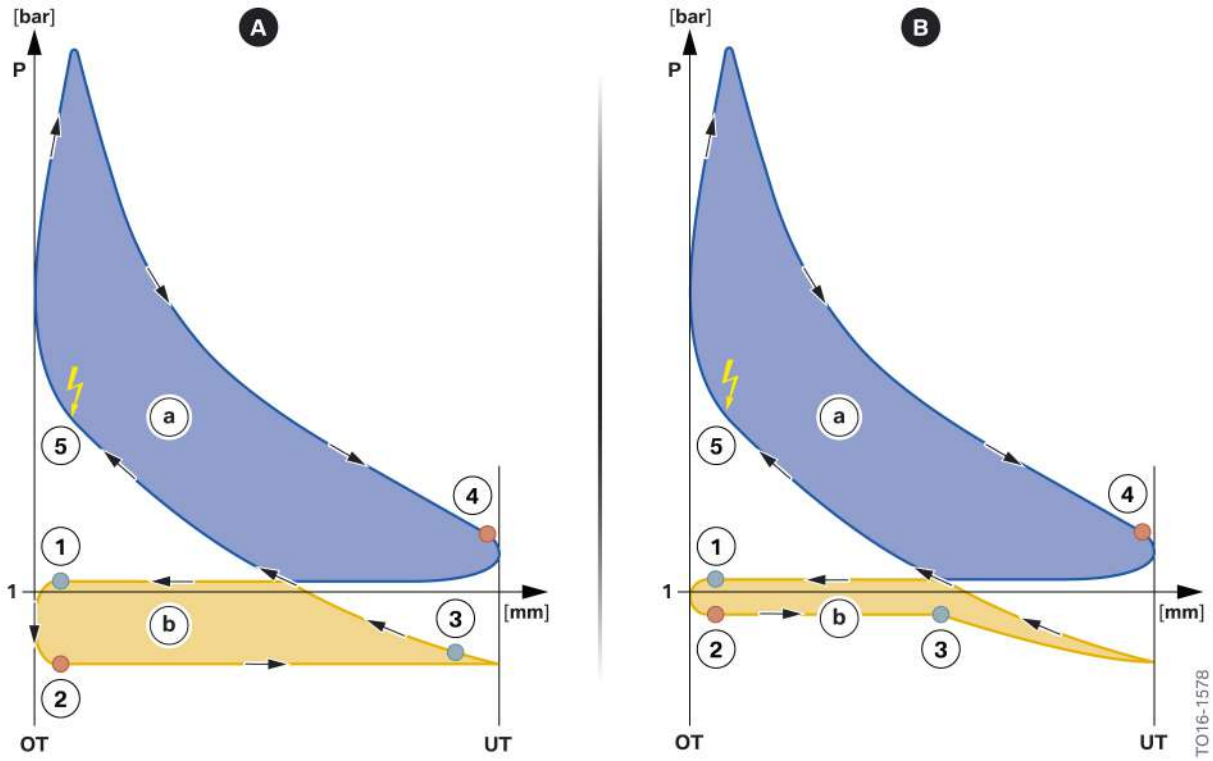
The Valvetronic comprises fully variable valve lift control **and** variable camshaft timing control (double VANOS), which makes the closing time of the intake valve freely selectable. The valve lift control is performed on the intake side only, while camshaft control is performed on both the intake and exhaust sides.

### 3.3.1. Throttle-free load control

The performance at the crankshaft on conventional engines is controlled via the throttling of the intake air by means of a throttle valve. Depending on the load point, a more or less higher vacuum is generated in the intake port. The vacuum generation is the reason that the proportion of gas exchange work increases considerably at lower loads.

With the realisation of throttle-free load control , a drawing in at ambient pressure is possible, whereby the gas exchange losses are drastically reduced. Consumption reduces significantly in the process. This is made possible by a fully variable valve gear, with which the closing time of the intake valves is freely selectable.

# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.




Valvetronic, efficiency comparison

| Index | Explanation |
|---|---|
| A | Throttle-valve-controlled engine |
| B | Valvetronic engine |
| P | Pressure |
| TDC | Top dead centre |
| BDC | Bottom dead centre |
| a | Gain |
| b | Loss |
| 1 | Intake opens |
| 2 | Exhaust closes |
| 3 | Intake closes |
| 4 | Exhaust opens |
| 5 | Firing point |

In the left graphic, the conventional procedure is shown with the slightly larger loss. In the right graphic, the reduced loss is visible. The upper area represents the power acquired in the combustion process of the petrol engine. The lower area illustrates the loss of this process.

The loss area can be compared to the gas exchange work. It is the energy that has to be used in order to eject the burnt gases from the cylinder and then draw in fresh gas to the cylinder. The intake of fresh gas to a throttle-valve-controlled engine is always effected, except in the full load position, always

against the resistance which the throttle valve sets against the incoming gases. During the intake process of an engine with Valvetronic, the throttle valve is almost always fully opened. The load control is effected via the closing time of the intake valve.

In comparison to the conventional engine, which is controlled via the throttle valve based on the load, no vacuum arises in the intake manifold. This means the energy required for the vacuum generation in the entire intake manifold is no longer necessary. The work for generating the vacuum in the cylinder, which arises after the closing of the intake cylinder, is fed in again after exceeding the bottom dead centre. The improved efficiency is thus achieved because of the lower power loss during the intake process.

The Valvetronic is the first fully variable valve control on the market. It was only 9 years from the first granting of the patent on the 19th August 1993, to the first series introduction in 2001.

**Throttle valve**

A throttle valve is still needed for engines with Valvetronic for the following tasks:

- Crankcase venting components
- Tank ventilation
- Emergency operation
- Catalytic converter heating.

By means of the throttle valve, a lower intake pipe pressure is adjusted. This vacuum is required in order to ensure the tank ventilation and crankcase ventilation. A pressure sensor in the intake manifold is used as a reference.

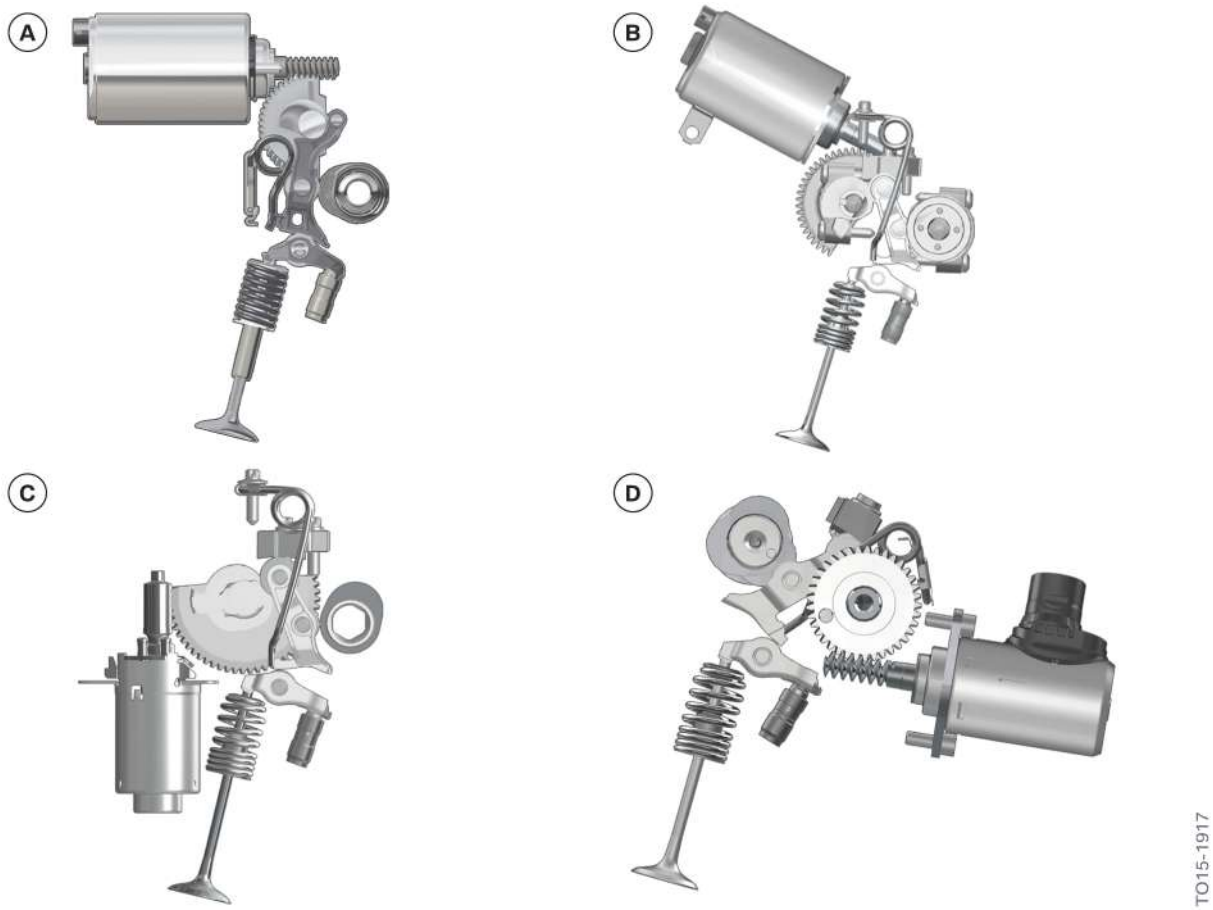
In the event of a fault in the Valvetronic, the valve lift is set to maximum where possible and the load control assumed by the throttle valve.

The idle speed control is map-controlled by the DME. During the starting operation the idle speed control can be effected via the throttle valve depending on the characteristic map. For an engine at operating temperature a switch is made to dethrottled operation after approx. 60 s (throttle valve is open fully). In cold weather conditions, however, the engine is started with a fully open throttle valve as this has a positive effect on the starting response.

## 3.3.2. Overview

Valvetronic has been used by BMW since 2001 and by MINI since 2006, and continuously developed ever since.

# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.




Valvetronic generation overview

| Index | Explanation |
|---|---|
| A | Valvetronic 1st generation |
| B | Valvetronic 2nd generation |
| C | Valvetronic 3rd generation |
| D | Valvetronic 4th generation |

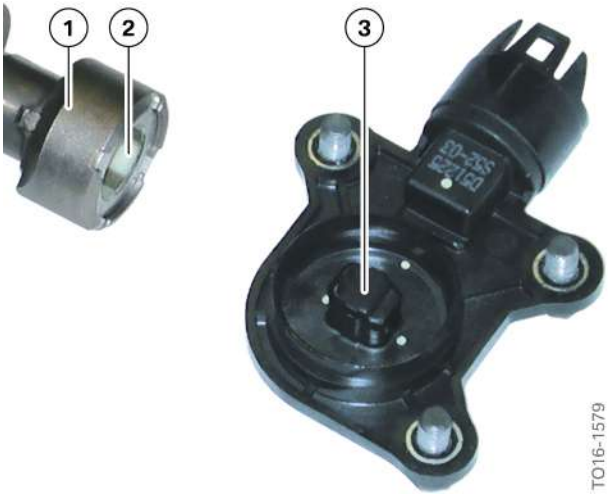
Today, almost all petrol engines by BMW and MINI have Valvetronic. The exceptions are the N74 engine and the B38A12U0 engine.

The following table shows which engine is equipped with which Valvetronic generation.

| 1st generation | 2nd generation | 3rd generation | 4th generation |
|---|---|---|---|
| • N42 | • N12 | • N13 | • B38 |
| • N62 | • N14 | • N18 | • B48 |
| • N62TU | • N16 | • N20 | • B58 |
| • N73 | • N46 | • N55 | • N63TU2 |
| | • N52 | • N63TU | |

### 3.3.3. Function

The actuation of the Valvetronic servomotor takes place by means of the Digital Motor Electronics DME or – with the 1st generation Valvetronic – by means of an additional valve lift control unit. In this case, the lift specification is sent via the CAN bus interface to the valve lift control unit and adjusted with help of the eccentric shaft sensor and the Valvetronic servomotor.




N52 Engine, eccentric shaft sensor and magnet wheel on eccentric shaft

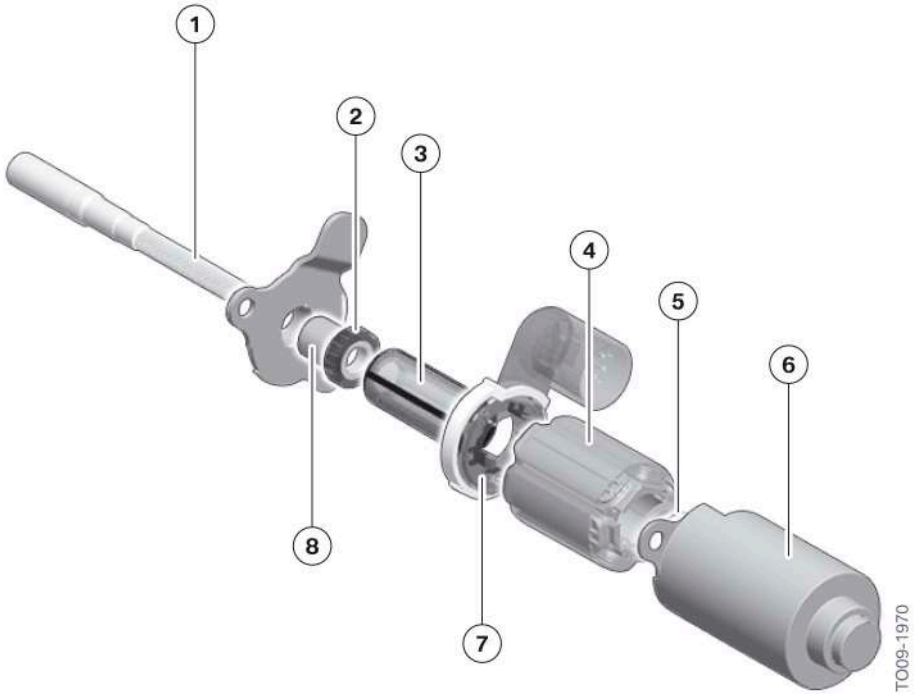
| Index | Explanation |
|---|---|
| 1 | Magnet wheel |
| 2 | Mounting bolt |
| 3 | Eccentric shaft sensor |

The eccentric shaft sensor records the angle of rotation of the eccentric shaft and supplies the actual value for the control.

The magnet wheel must be secured at the eccentric shaft with a non-magnetic mounting bolt as otherwise the eccentric shaft sensor will not work.

As of the Valvetronic of the 3rd generation, the eccentric shaft sensor no longer sits on the eccentric shaft, rather it is integrated into the Valvetronic servomotor:

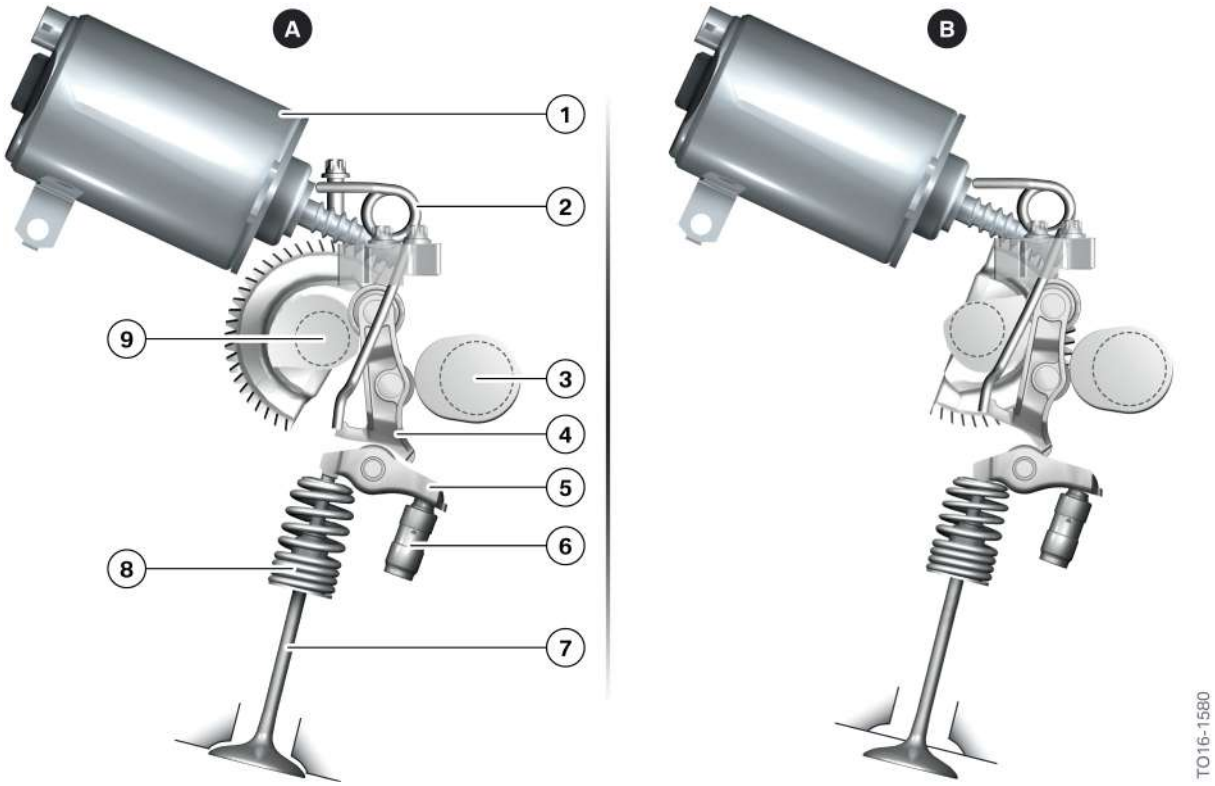
# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.




Valvetronic servomotor, Valvetronic of the 3rd generation

| Index | Explanation |
|---|---|
| 1 | Shaft |
| 2 | Magnet wheel for sensor system (increment wheel) |
| 3 | Rotor (permanent magnet) |
| 4 | Stator (excitation coil) |
| 5 | Ball bearing |
| 6 | Housing |
| 7 | Sensor system and connector |
| 8 | Needle bearing |

# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.




Valve stroke adjustment, Valvetronic of the 2nd generation

| Index | Explanation |
|---|---|
| A | Minimum stroke |
| B | Maximum stroke |
| 1 | Valvetronic servomotor |
| 2 | Spring |
| 3 | Intake camshaft |
| 4 | Intermediate lever |
| 5 | Roller cam follower |
| 6 | Hydraulic valve clearance compensating element |
| 7 | Intake valve |
| 8 | Valve spring on intake side |
| 9 | Eccentric shaft |

The intermediate lever rests with one side on the eccentric shaft and with the other side on the camshaft. If the Valvetronic servomotor twists the eccentric shaft, the intermediate lever moves in the direction of the camshaft. This allows the camshaft to move the linear increase of the intermediate lever further in the direction of the roller cam follower and thus opening the intake valve further. The intermediate lever thus modifies the gear ratio between camshaft and roller cam follower.

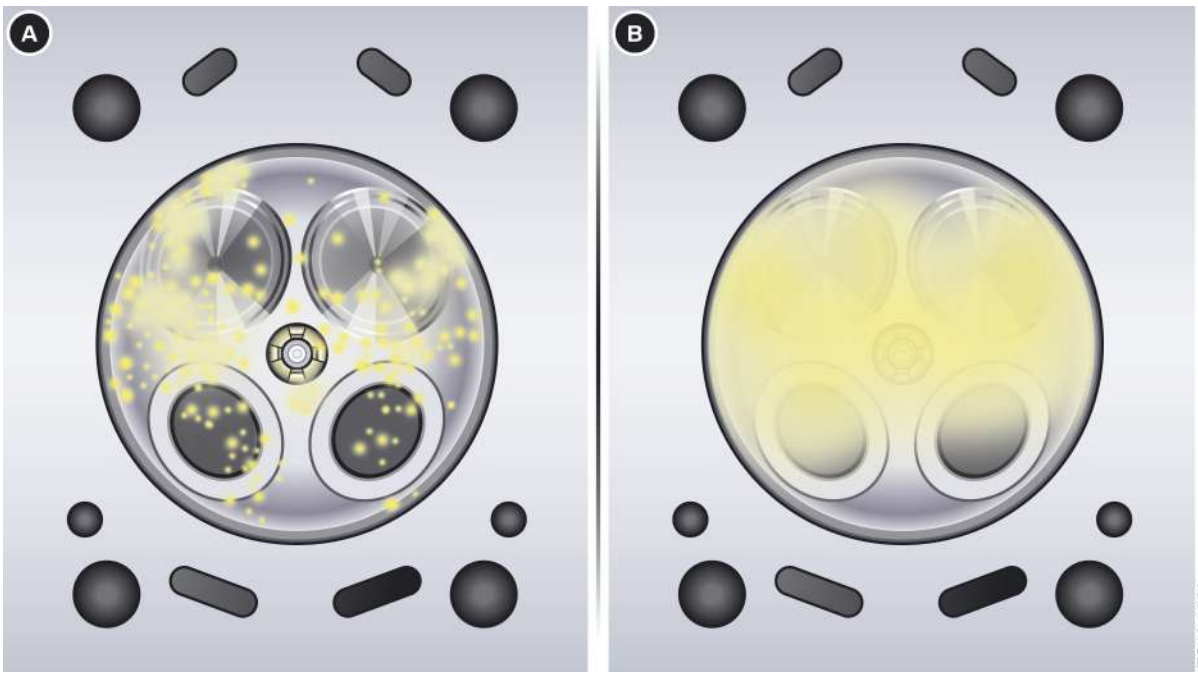
# ENGINE MECHANICS – PETROL ENGINE.
# 3. VALVE GEAR.

At full load operation, valve lift and opening period are at their greatest and at idle they are at their smallest. To ensure a smooth engine running, particularly at idle, all intake valves must be opened equally. For this reason, the deviations are measured at the factory and a specific intermediate lever is assigned to each roller cam follower.



When the valve gear is disassembled, you must ensure that all parts are installed again in the same position. Otherwise, it can lead to different cylinder charges thus causing an even engine running.

For small loads a very short opening period is necessary which is only possible with an overproportional reduction of the valve lift. Thus, the valve opening cross section is reduced so that the incoming flow velocity on the valve gap increases greatly and the entire flow around the valve is passed evenly. This effect facilitates the mixture preparation process.




Comparison of fuel distribution at idle

| Index | Explanation |
|---|---|
| A | Fuel distribution with full valve lift (conventionally throttled engine) |
| B | Fuel distribution with small valve lift (engine with Valvetronic) |

The left graphic shows a conventional engine at idle. The inhomogeneous fuel distribution with relatively large fuel drops does not suggest optimal combustion.

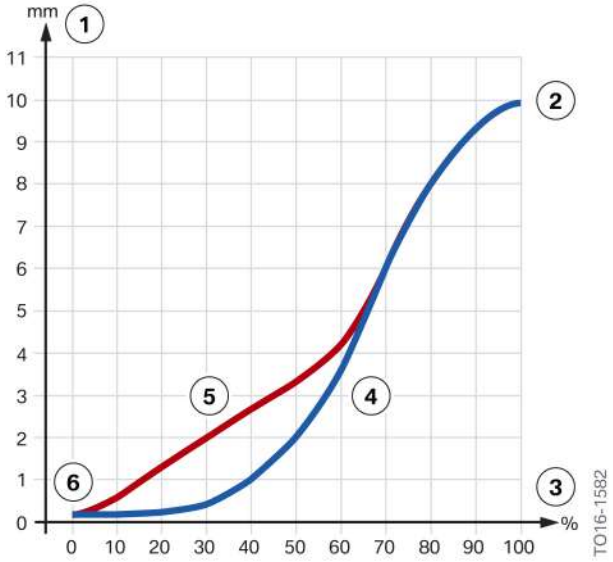
The right graphic shows the same situation on an engine with Valvetronic. As a result of the even flow of the mixture around the valve with a small valve lift, the mixture is distributed much better in the combustion chamber. Due to the very high incoming speed and the very large pressure difference in the valve gap, the drop size decreases. The advantages of this configuration are as follows:

- Very good mixture preparation
- Lower performance fluctuations
- Lower HC and $NO_x$ emissions.

Even at low temperatures, a cold start is possible with half the quantity of fuel of an engine controlled by throttle valve.

The response characteristics are also improved by the load control taking place directly at the cylinder. Unlike for the control with a throttle valve, the intake plenum and the intake pipes do not have to be filled first, but the Valvetronic has a direct effect on the cylinder charge without further deceleration.

### 3.3.4. Advance




Valve lift intake valve for advance

| Index | Explanation |
|---|---|
| 1 | Valve lift (in mm) |
| 2 | Maximum valve lift at full load |
| 3 | Load (in %) |
| 4 | Intake valve 2 |
| 5 | Intake valve 1 |
| 6 | Minimum valve lift for idle position |

As of 2nd generation Valvetronic, there is support for tuning the "advance" in the lower range of travel of the intake valves. The two intake valves of a cylinder are opened synchronously up to a stroke of 0.2 mm. As of this stroke, intake valve 1 starts to advance. Intake valve 2 opens later with a slight delay and catches up with intake valve 1 again at a stroke of approx. 6 mm. From there, they continue to open synchronously. The advance is facilitated by a different shaping of the two eccentrics of the eccentric shaft of a cylinder.

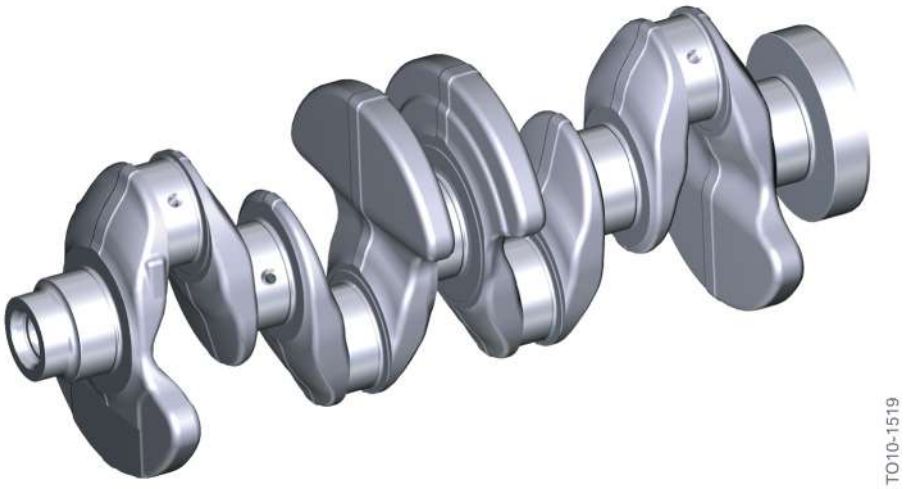
This opening behaviour favours the inflow of gas into the cylinder. As a result of the small opening cross-section of the intake valves, there is a significantly higher flow rate with a constant volume drawn in. This flow rate is used in conjunction with the geometry in the upper area of the combustion chamber for better mixing of the mixture drawn in.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

## 4.1. Activate

The crankshaft drive converts the straight-ahead motion of the piston into a rotational movement. This results in loads that are very complex and place high requirements on component parts and the performance of repair work.

## 4.2. Crankshaft




N20 engine, crankshaft

The connecting rod induces the force into the crankshaft as torque. The crankshaft rests on the main bearings.

The crankshaft also assumes the following tasks:

- Drive of the ancillary components via the belt drive
- Drive of the valve gear
- Frequently, the drive of the oil pump
- If necessary, the drive of counterbalance shafts.

The main bearing journals take over the bearing function in the crankcase. The connecting rod bearing journals are connected with the crankshaft via the crank webs. This area of the connecting rod bearing journals with crank webs is referred to as "crank throw".

The distance of the connecting rod bearing journals from the crankshaft axis determines the stroke of the engine.

The angle between the connecting rod bearing journals of the consecutive cylinders (firing order) is referred to as crank pin offset or crank throw angle. The firing interval is also determined by this, i.e. the angle of rotation of the crankshaft between 2 consecutive ignitions. During one working cycle, each cylinder has been activated one time. On a four-stroke engine, the working cycle (intake, compression, work, exhaust) extends over 2 full crankshaft revolutions, i.e. 720° angle of rotation.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

We usually say that for a four-stroke engine, the firing interval is the quotient of 720° and number of cylinders (firing interval = 720° : Number of cylinders):

- 3-cylinder engine: 240° crankshaft (crank angle)
- 4-cylinder engine: 180° (crank angle)
- 6-cylinder engine: 120° (crank angle)
- 8-cylinder engine: 90° (crank angle)
- 12-cylinder engine: 60° (crank angle).

The crankshaft has oil holes in it which provide the connecting rod bearings with oil. They lead from the main bearing journals to the connecting rod bearing journals and are tied to the oil circuit of the engine by the bearing seats.

Counterweights generate an overall symmetry around the crankshaft axis, thus enabling smooth engine running. However, the number of counterweights is quite varied:

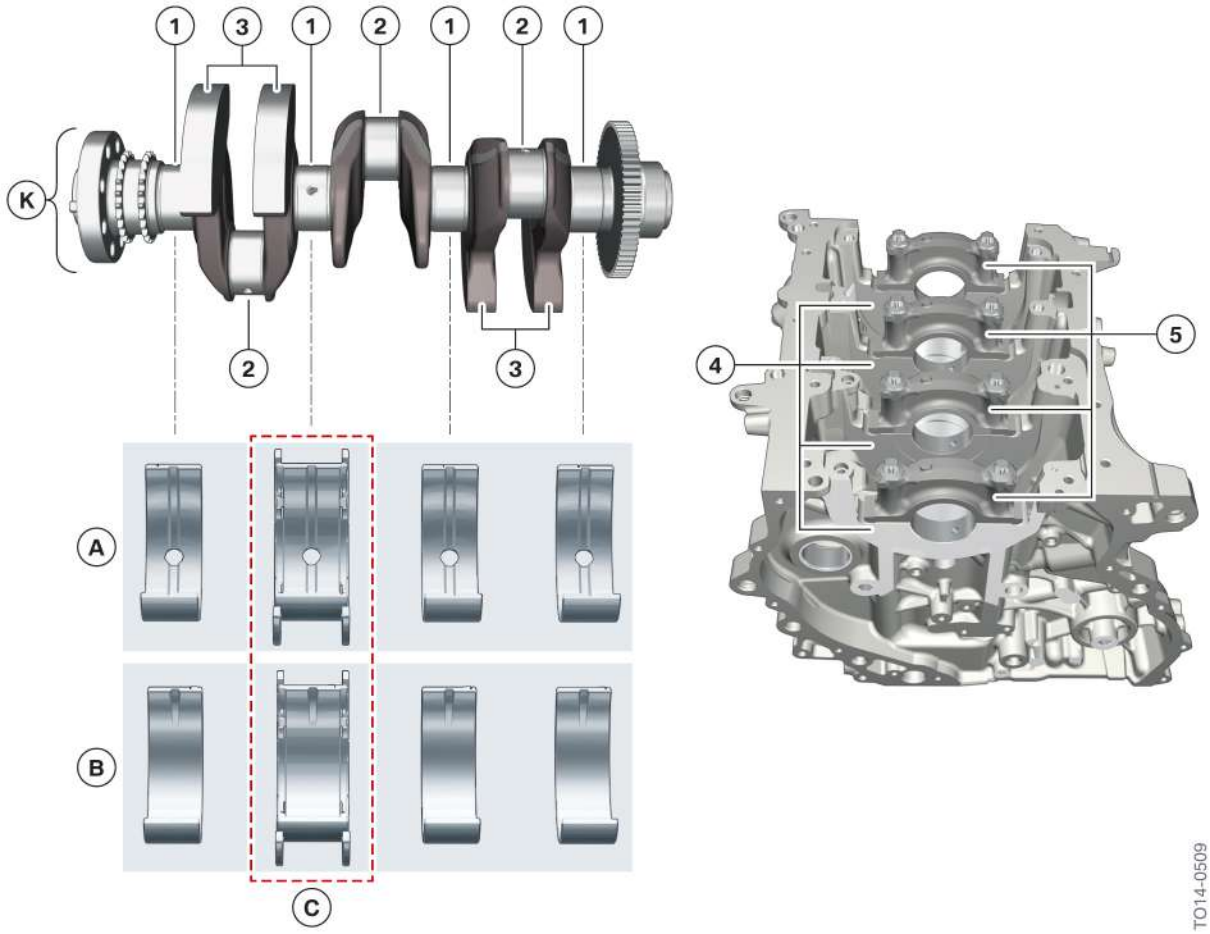
- N13 engine: 8 counterweights
- N20 engine: 4 counterweights
- N55 engine: 8 counterweights
- B58 engine: 12 counterweights

Depending on the load, the crankshaft is either cast or forged. Both procedures offer advantages:

| Advantages of cast crankshaft | Advantages of forged crankshaft |
|---|---|
| • Less expensive<br>• Reacts very well to surface treatment procedures for increasing the vibration resistance<br>• Lower weight<br>• Simpler processing<br>• Processing the crank webs can frequently be done without | • Stiffer<br>• Better vibration characteristics<br>• More favourable wear behaviour in the bearing journals |

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

## 4.2.1. Crankshaft bearings




B38 engine, overview of bearing shells

| Index | Explanation |
|---|---|
| A | Bearing shells in the bearing seat |
| B | Bearing shells in main bearing cap |
| C | Axial guide bearing |
| K | Force output side |
| 1 | Main bearing journal |
| 2 | Connecting rod bearing journal |
| 3 | Counterweights |
| 4 | Bearing seat |
| 5 | Main bearing cap |

The main bearings support the crankshaft against the crankcase. The high-load side is located in the crankshaft bearing cap. The force that is created by the combustion pressure is absorbed here.

The sliding surfaces of the bearing shells are provided with a special bearing material. The sliding surface is on the inside, i.e. the bearing shells do not rotate with the shaft; instead, they are fixed in the crankcase.

The crankshaft is mounted using different bearing shells. The bearing shells in the low-load zone of the bearing seat have oil holes and a peripheral groove for oil supply. The highly stressed bearing shells in the area of the main bearing cap have no oil holes or grooves.

The bearings are also cooled by the engine oil.



Due to the very low tolerances, special attention must be paid to cleanliness when handling bearing shells.

**Axial guidance**

The crankshaft has only one single axial guide bearing, that is frequently referred to as collar or thrust bearing. This bearing holds the crankshaft in the axial direction and must absorb forces in the longitudinal direction. In general, this arises for the following reasons:

- Gear with helical-cut splines for the oil pump drive
- Clutch control
- Acceleration.

Depending on the engine version, axial guide bearings can be designed as 180° or 360°bearings. This means some engines are only axially supported in the bearing seat and other engines are additionally supported in the main bearing cap.

Built-up bearings are made up of several separate components. With this technology a thrust washer is fitted on both sides. It facilitates a dimensionally stable, loose link with the crankshaft bearing. Hence, the thrust washers are movable, thereby making contact more uniform and reducing wear.

Three-part bearings consist of three separate parts. The thrust washers for the bearing are inserted loosely and are held in their position by the shaping of the bearing seat. The production is less expensive than with built-up bearings; the advantages are the same.

Overheating is usually the cause when a thrust bearing fails. A worn thrust bearing will result in the generation of noise, above all in the area of the torsional vibration damper.



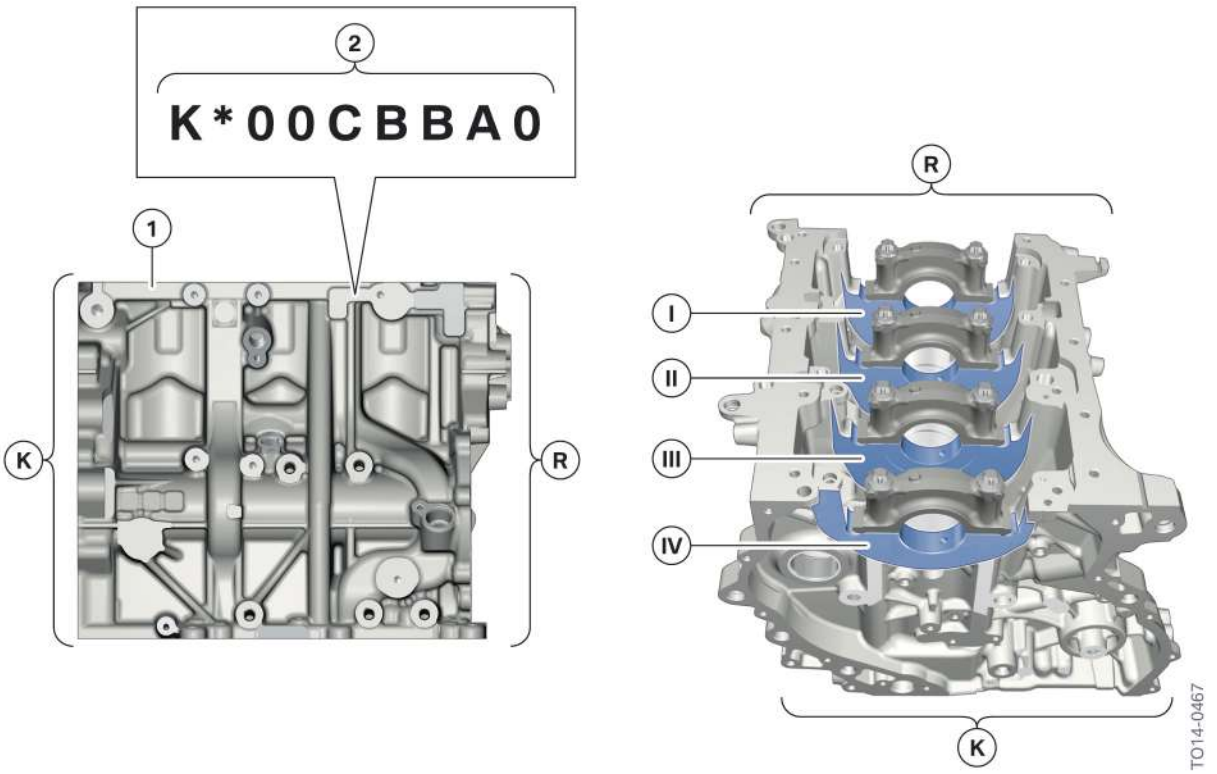
Before assembling the engine, the side clearance must be checked according to the repair instructions. Excessive side clearance may cause electrical faults or damage to the components.

**Determination of the bearing shells**

For the crankshaft to be able to carry out its main task effectively, namely the conversion of the stoke movement into a rotational movement, it must be positioned correctly. The dimension to be adjusted is created via varying strengths of crankshaft bearing shells.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

The stamped letters on the crankcase are used to determine the bearing shell sizes of the bearing seat (classification):



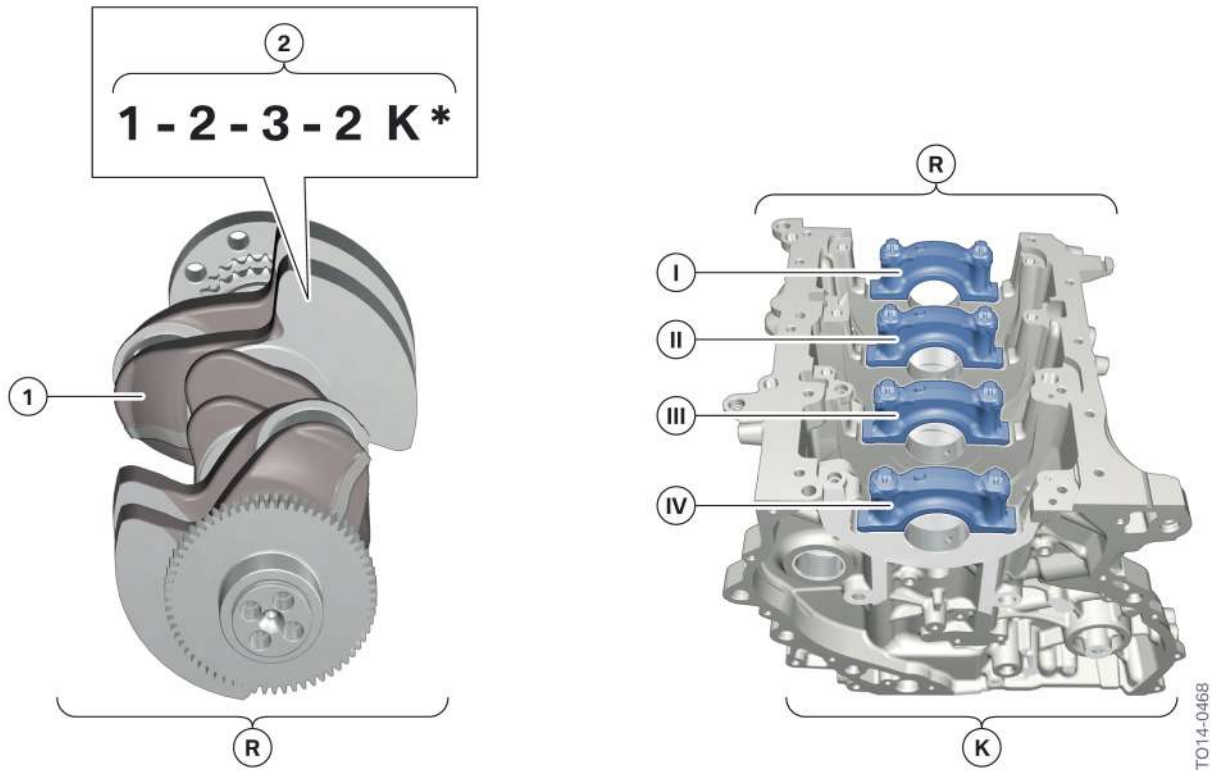

B38 engine, classification of the bearing shells in the bearing seat

| Index | Explanation |
|---|---|
| I | Bearing seat 1 |
| II | Bearing seat 2 |
| III | Bearing seat 3 |
| IV | Bearing seat 4 |
| K | Clutch side (side providing the force) |
| R | Belt side |
| 1 | Crankcase |
| 2 | Classification (stamped letters) |

The letter "K" stands for the clutch side of the crankcase. It also specifies the counting order for the assignment:

- If there is a "K" before the stamped letters, then the first letter of the code refers to the bearing shell of bearing seat 4, which is located on the clutch side. The subsequent letters, read from left to right, refer to the bearing seats 3, 2 and 1. The positions marked with "0" are placeholders for larger engines and are simply omitted.
- If there is **no** “K” before the stamped letters, then the first letter of the code refers to the bearing shell of the bearing seat 1, which is located opposite of the clutch side. The subsequent letters, read from left to right, refer to bearing seats 2, 3 and 4.

There are numbers stamped on the crankshaft for determining the bearing shell sizes in the main bearing cap (classification):



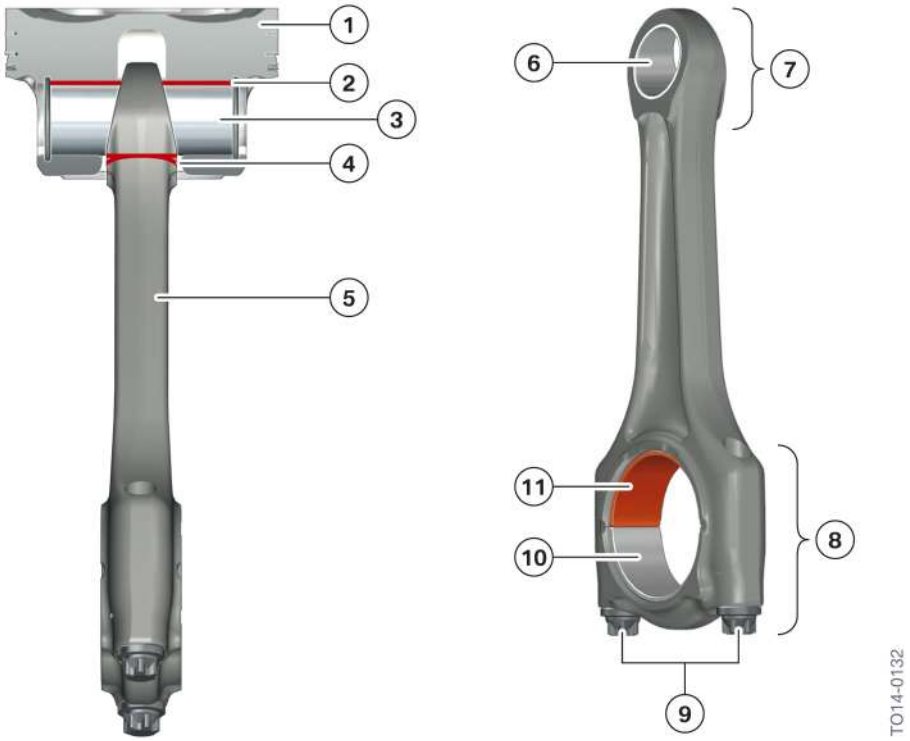

B38 engine, classification of the bearing shells in the main bearing cap

| Index | Explanation |
|---|---|
| I | Bearing cap 1 |
| II | Bearing cap 2 |
| III | Bearing cap 3 |
| IV | Bearing cap 4 |
| K | Clutch side (side providing the force) |
| R | Belt side |
| 1 | Crankshaft |
| 2 | Classification (stamped numbers) |

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

In the case of the crankshaft as well, the letter K defines the assignment of the bearing position. The digit next to the "K" is assigned to the main bearing cap 4, which is located on the clutch side. The subsequent digits, read from left to right, refer to the main bearing cap 3, 2 and 1.

## 4.3. Connecting rod




B38 engine, cracked connecting rod with even pitch

| Index | Explanation |
|---|---|
| 1 | Piston |
| 2 | Area transferring the force |
| 3 | Gudgeon pin |
| 4 | Connecting rod bearing bush with shaped bore hole |
| 5 | Connecting rod |
| 6 | Connecting rod bush |
| 7 | Small connecting rod eye (trapezoidal shape) |
| 8 | Large connecting rod eye (cracked) |
| 9 | Connecting rod bolts of the connecting rod bearing cap |
| 10 | Connecting rod bearing shell of the connecting rod bearing cap |
| 11 | Connecting rod bearing shell of the connecting rod (IROX-coated) |

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

The connecting rods in the crankshaft drive provide the connection between pistons and crankshaft. They covert the linear movement of the pistons into a rotational movement at the crankshaft. In addition, they transfer the forces from the pistons that arise during the combustion process to the crankshaft.

The connecting rod is subject to very strong acceleration and with its weight, it directly influences the performance and the smooth running of the engine.

The connecting rod is stressed by the gas forces in the combustion chamber and by the forces of inertia of the moving masses. It is subject to an alternating push and pull load. Particularly on reved-up petrol engines, the tensile forces are decisive. In addition, a centrifugal force arises due to the lateral deflection of the connecting rod, which generates bends.

The connecting rod has two connecting rod eyes. The small connecting rod eye creates the connection to the piston by means of a gudgeon pin. Due to the lateral deflection of the connecting rod during a crankshaft revolution, it must be attached to the piston so that it can rotate. This is done with the aid of a plain bearing. A bearing bush is pressed into the small connecting rod eye.

The split large connecting rod eye is located at the crankshaft end. The large connecting rod eye must be split to facilitate mounting onto the crankshaft. Plain bearings ensure the function. The plain bearing comprises two bearing shells. An oil hole in the crankshaft supplies the bearing with engine oil.



On some engines (e.g. W10 and W11 engine), the gudgeon pins are pressed into the connecting rod and may not be removed. In case of repair, both the piston and connecting rod must always be replaced together.

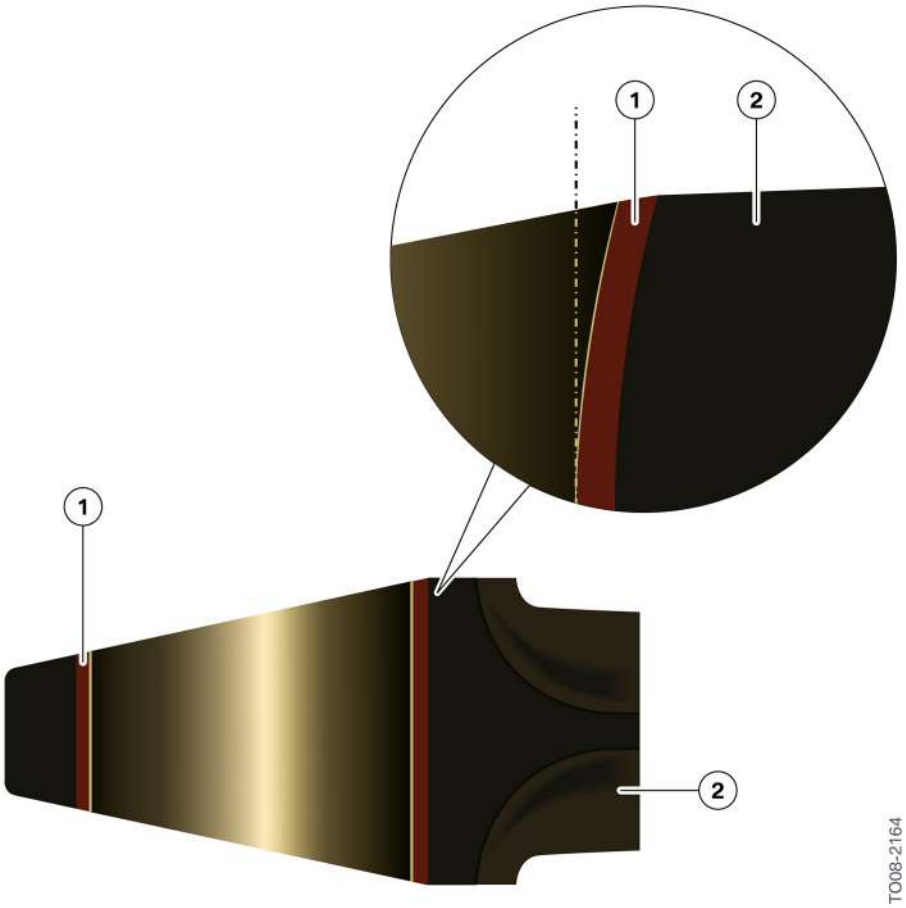
## 4.3.1. Trapezoidal connecting rod

In the case of a trapezoidal connecting rod the connecting rod eye is trapezoidal in shape in its cross-section. In other words, the connecting rod tapers at the small eye from the lug of the shank to the end.

This makes a further reduction in weight possible as material is saved on the "non-loaded" side while the full bearing width is maintained on the loaded side. In addition, it facilitates a smaller distance between the gudgeon pin bosses, which means a lower deflection of the gudgeon pin.

Another advantage lies in the fact that the oil hole in the small connecting rod eye can be dispensed with, as oil enters through the angled flank of the plain bearing. The omission of this oil hole also eliminates it negative influence on the strength of this bearing side. Once again this enables the connecting rod to be event narrow in design at this point. In this way both weight is saved and extra space gained in the piston.

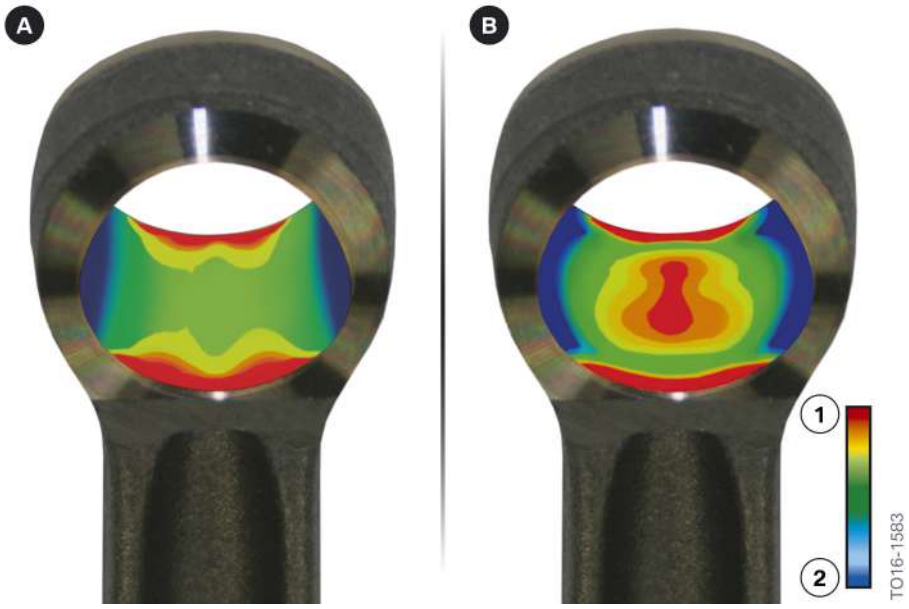
### 4.3.2. Shaped bore




N55 engine, profiled bore hole in small connecting rod eye

| Index | Explanation |
|---|---|
| 1 | Bushing |
| 2 | Connecting rod |

The force acting from the piston via the gudgeon pin is distributed evenly to the bushing surface by this shaped bore in the small connecting rod eye and the edge load reduced.




Comparison of connecting rod eye with/without shaped bore

| Index | Explanation |
|---|---|
| A | Small connecting rod eye without shaped bore (N54 engine) |
| B | Small connecting rod eye with shaped bore (N55 engine) |
| 1 | Hight surface load |
| 2 | Low surface load |

### 4.3.3. Bushless connecting rod




N63TU engine, connecting rod

| Index | Explanation |
|---|---|
| A | Connecting rod with bush |
| B | Bushless connecting rod |

In the N63TU2 and partially in the N63TU engine bushing-less connecting rods are used. Here, the small connecting rod eye is drilled into the head of the connecting rod, provided with a fine surface treatment and hardened.

### 4.3.4. Cracking

Cracked connecting rods are manufactured in one piece and then provided with score lines or laser scribing at the predetermined breaking point. They are then broken into 2 specific parts.

A fine fracture surface results from the cracking (breaking) process. This surface structure centres the connecting rod bearing cap to a precise fit when it is mounted on the connecting rod. Due to the individual fracture geometry both parts belong together and are not individually replaceable.

The advantage of cracking is that the contact surfaces do not need to be machined further. The two halves fit exactly on each other. Positioning with a fitting sleeve or screw is usually not necessary.



If a connecting rod bearing cap is mounted the wrong way round or on another connecting rod, the fracture structure of both parts is destroyed and the connecting rod bearing cap is not centred. In this event the entire connecting rod set must be replaced by new parts. In Service please observe the specified jointing torques and angle of rotation specifications in the repair instructions.

### 4.3.5. Classification of the connecting rods

To guarantee smooth engine running, the connecting rods are divided into weight classes. The connecting rods are weighed and divided into various classes according to their weight.



In Service only connecting rods of the same weight class may be used. This is why only a full set of connecting rods is available in the event of a replacement. This set comprises the corresponding number of connecting rods of a weight classification. Replacement of a single connecting rod is not permitted.

### 4.3.6. Determination of the bearing shells

On many petrol engines there are also various sizes of connecting rod bearing shells available. The connecting-rod bearing play must be determined with a special tool and the corresponding connecting rod bearing shell inserted.

On other petrol engines (for example, the new B engine generation) the connecting rod bearing shells are available in one standard size. Thus, a determination of the bearing shells is not necessary.

## 4.4. IROX coating

In order to comply with the increasingly stringent exhaust emission regulations, most combustion engines nowadays are equipped with an automatic engine start-stop function. This has led to a huge increase in starting cycles.

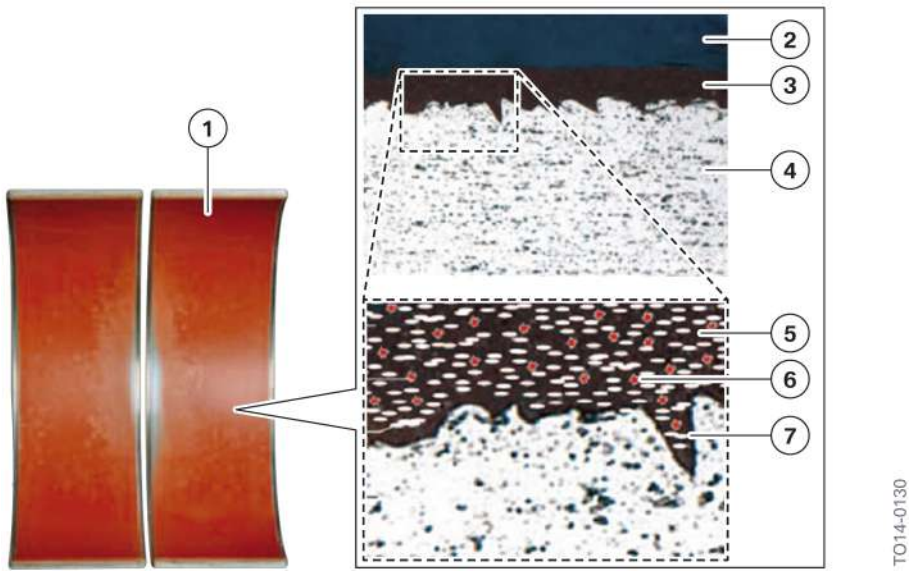
For proper engine running, the crankshaft must be supplied with sufficient lubricating oil at the bearing positions. If the oil supply is ensured, solid body contact will not occur between the bearing journal and bearing shell due to the thin lubricating film.

If the engine is now stopped, it will not be possible for the mechanically-driven oil pump to maintain the oil supply. The oil film between the bearing positions flows off. Solid body contact occurs between the connecting rod bearing journal and connecting rod bearing shell. Once the engine is restarted, it takes a certain amount of time for the lubricating film to fully re-establish itself. The connecting rod bearing shell may be subject to wear in this short period. The IROX coating technology reduces this wear to a minimum.

The IROX coating is currently used for the connecting rod bearing shells of the new B engine generation as well as for the connecting rod bearing shells and main bearing shells of the N74TU engine.

The IROX-coated bearing shells are only located on the stressed side of the connecting rod bearing resp. main bearing, i.e. on the side of the connecting rod resp. on the side of the main bearing cap.

The IROX bearings are red due to their special coating.

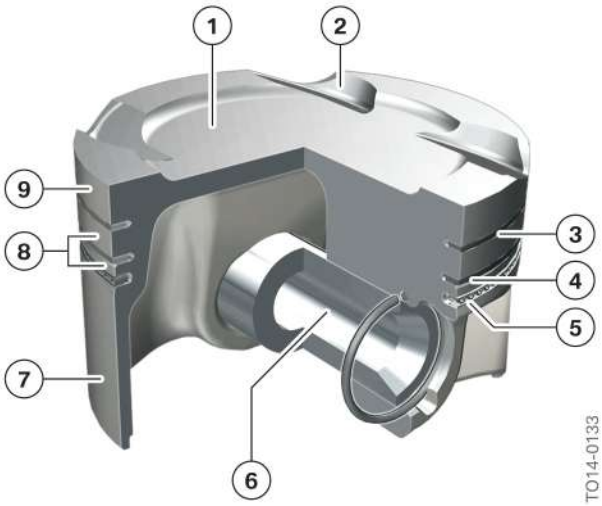



Detail enlargement of the IROX coating

| Index | Explanation |
|---|---|
| 1 | IROX-coated bearing shell |
| 2 | Oil film |
| 3 | IROX coating |
| 4 | Bearing shell |
| 5 | Binding resin |
| 6 | Hard particle |
| 7 | Solid lubricant |

The IROX coating is applied to a conventional bearing shell. It consists of a binding resin matrix made of polyamide-imide with embedded hard particles and solid lubricants. The polyamide-imide ensures, in combination with the hard particles, that the bearing shell surface is so hard that material abrasion is no longer possible. The solid lubricants reduce surface friction and replace the oil film which briefly no longer exists between the bearing shell and the bearing journals during the starting phase.

## 4.5. Piston with piston rings

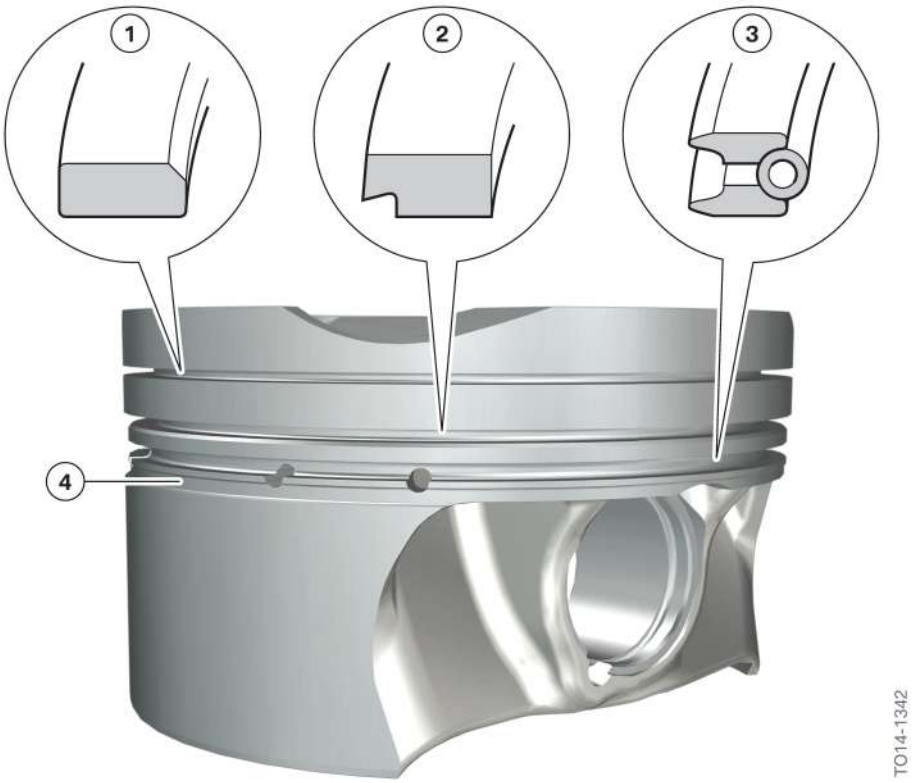



B38 engine, piston

| Index | Explanation |
|---|---|
| 1 | Piston crown |
| 2 | Valve relief |
| 3 | 1st piston ring |
| 4 | 2nd piston ring |
| 5 | 3rd piston ring |

| Index | Explanation |
|---|---|
| 6 | Gudgeon pin |
| 7 | Piston skirt |
| 8 | Ring bar |
| 9 | Fire land |

The piston is the first element in the chain of power-transmitting parts in the petrol engine. During combustion, it absorbs the resulting compressive forces and transfers these to the gudgeon pin on the connecting rod. In addition, it seals the combustion chamber, together with the piston rings, against the passage of gas and lubricating oil flow.




N63TU2 engine, cast piston with piston rings

| Index | Explanation |
|---|---|
| 1 | Plain compression ring |
| 2 | Stepped compression ring |
| 3 | Oil scraper ring with spiral expander |
| 4 | Additional oil groove |

To ensure a better oil drainage, some pistons are provided with an additional oil groove below the oil scraper ring groove (e.g N55 engine or N63TU2 engine). The additional oil groove works with the oil outlets in the piston skirt to help remove oil pushed down by the oil scraper ring during the piston's downward motion. This prevents the oil from being carried past the piston rings, in particular when the engine is in coasting overrun mode (during which a vacuum is generated in the combustion chamber).

The piston rings are metal seals with the following functions:

- Sealing the combustion chamber against the crankcase.
- Dissipating the heat from the piston to the cylinder wall.
- Regulating the oil supply to the liner

Due to the lateral force, which has an effect on the piston rings during the change of leaning direction of the piston, the piston rings rotate in their grooves and thus change the position of the impact. This process removes deposits from the ring grooves. It also prevents the piston ring gap from cutting into the cylinder barrel.

All BMW petrol engines are equipped with 2 compression rings and one oil scraper ring.



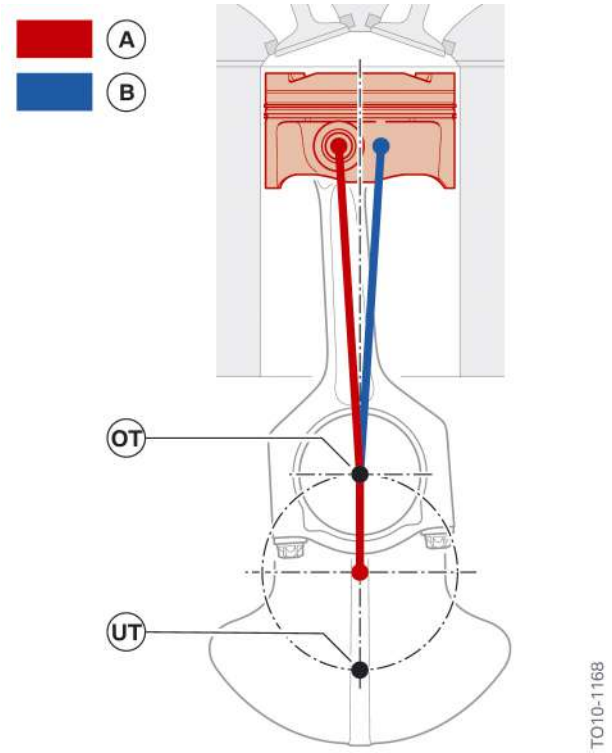
Many piston rings are equipped with a Top mark. When installing the piston rings, the Top mark must be facing in the direction of the piston crown.

Pistons and gudgeon pins are paired to each other and may not be mixed up.

## 4.6. Pin offset

The deflection of the connecting rod causes a lateral load of the piston perpendicular to the cylinder axis. Thereby and due to its articulation, the piston lays against one side of the cylinder wall to the other after the bottom and top dead centre in each case. This behaviour is called piston rocking or change of face and can causes a certain piston clacking sound. The greater the force that acts from above on the piston and the greater the running clearance, the greater too the piston knock.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.




Pin offset

| Index | Explanation |
|---|---|
| A | Major-thrust-face pin offset (positive) |
| B | Minor-thrust face pin offset (negative) |
| TDC | Top dead centre |
| BDC | Bottom dead centre |

Pin offset denotes the offset of the gudgeon pin axis from the cylinder centre line. A positive offset denotes offset to the major thrust face, a negative offset denotes offset to the minor thrust face. The major thrust face refers to that side of the piston on which the piston rests in the combustion cycle on its way to bottom dead centre.

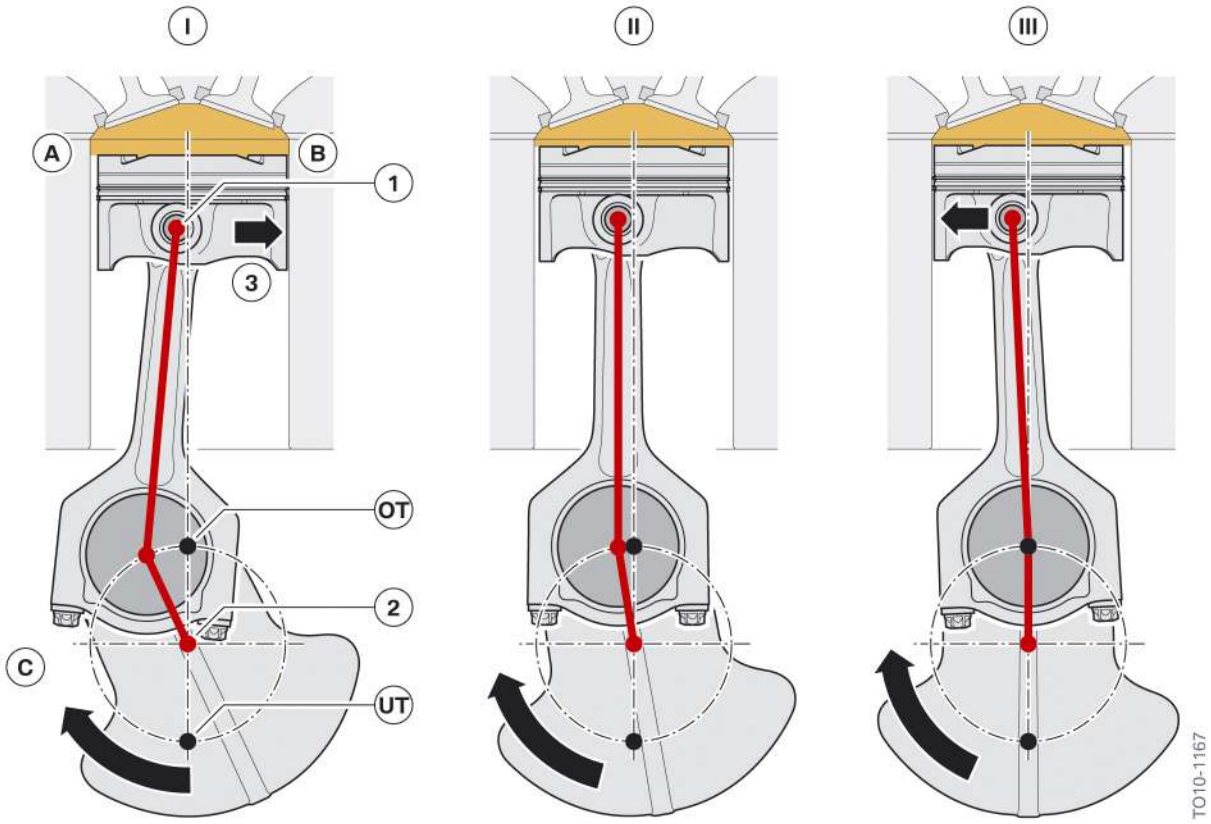
Pin offset can be effected towards the major thrust face (positive) and also towards the minor thrust side (negative).

- The major thrust face pin offset is also referred to as noise axial offset .
- The minor thrust face pin offset is also referred to as thermal pin offset. In this position the sealing effect of the piston rings is improved.

The offset amounts to approx. 0.3 – 0.8 mm and is thus nearly impossible to perceive with the eye. For this reason, there is a direction mark provided on the top side of the piston. Incorrect installation may result in extreme noise similar to that generated by piston damage.

To reduce the piston noises, the gudgeon pin is frequently offset to the major thrust face. This brings forward the time when the piston changes faces between the compression and power cycles to the range of lower pressure before top dead centre.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.




Piston rocking in an engine with pin offset

| Index | Explanation |
|---|---|
| I | Piston and crankshaft position before top dead centre |
| II | Piston and crankshaft position just before top dead centre, with connecting rod standing perpendicular |
| III | Piston and crankshaft position at top dead centre |
| A | Major thrust face |
| B | Minor thrust face |
| C | Direction of engine rotation |
| TDC | Top dead centre |
| BDC | Bottom dead centre |
| 1 | Gudgeon pin |
| 2 | Centre of crankshaft rotation |
| 3 | Thrust force |

The piston rests against the minor thrust face during the up-stroke. A neutral piston position is already achieved before TDC by offsetting the gudgeon pin. This is the case when the centre lines of the cylinder and of the big and small connecting rod eyes are parallel to each other. Already before TDC the piston thus changes face from the minor thrust face to the major thrust face. In this phase the force on the piston is still low. Due to the off-centre support of the piston, the force acting on the piston

from above has a higher lever arm on the one face than on the other. In this way, the piston is already tilted during the up-stroke, resulting in contract with the major thrust face at the upper edge. In its subsequent movement the piston again travels straight ahead, so that the piston rests completely on the major thrust face. The change of faces is much quieter than in a conventional crankshaft drive.

The disadvantage of the pin offset is the slightly increased friction on the major thrust face. This minor downside, however, is made up for by the reduced noise.

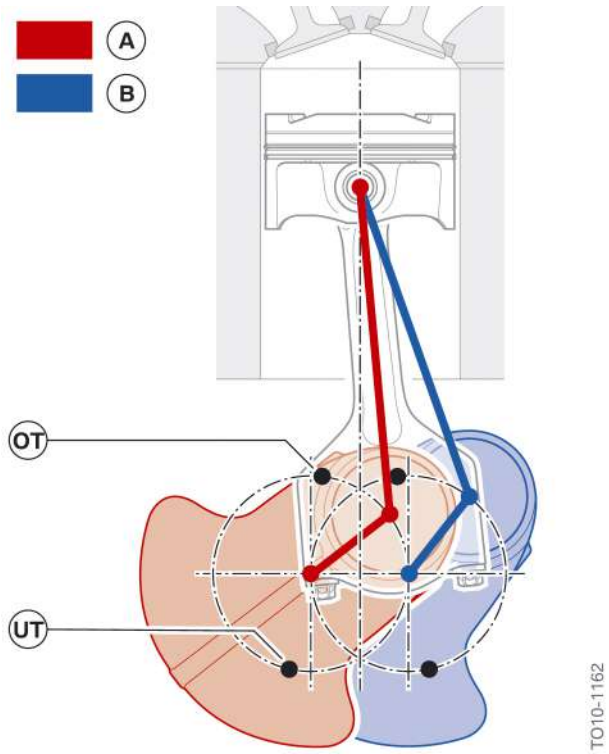
## 4.7. Crankshaft offset

In the N20 engine, BMW is using a crankcase with offset for the first time.

Crankshaft offset denotes the offset of the crankshaft axis from the cylinder centre line.

- Positive offset: Offset to the major thrust face
- Negative offset: Offset to the minor thrust face

Crankshaft offset can basically be effected in both directions, but up to now only the variation in the positive direction (A) has been used.



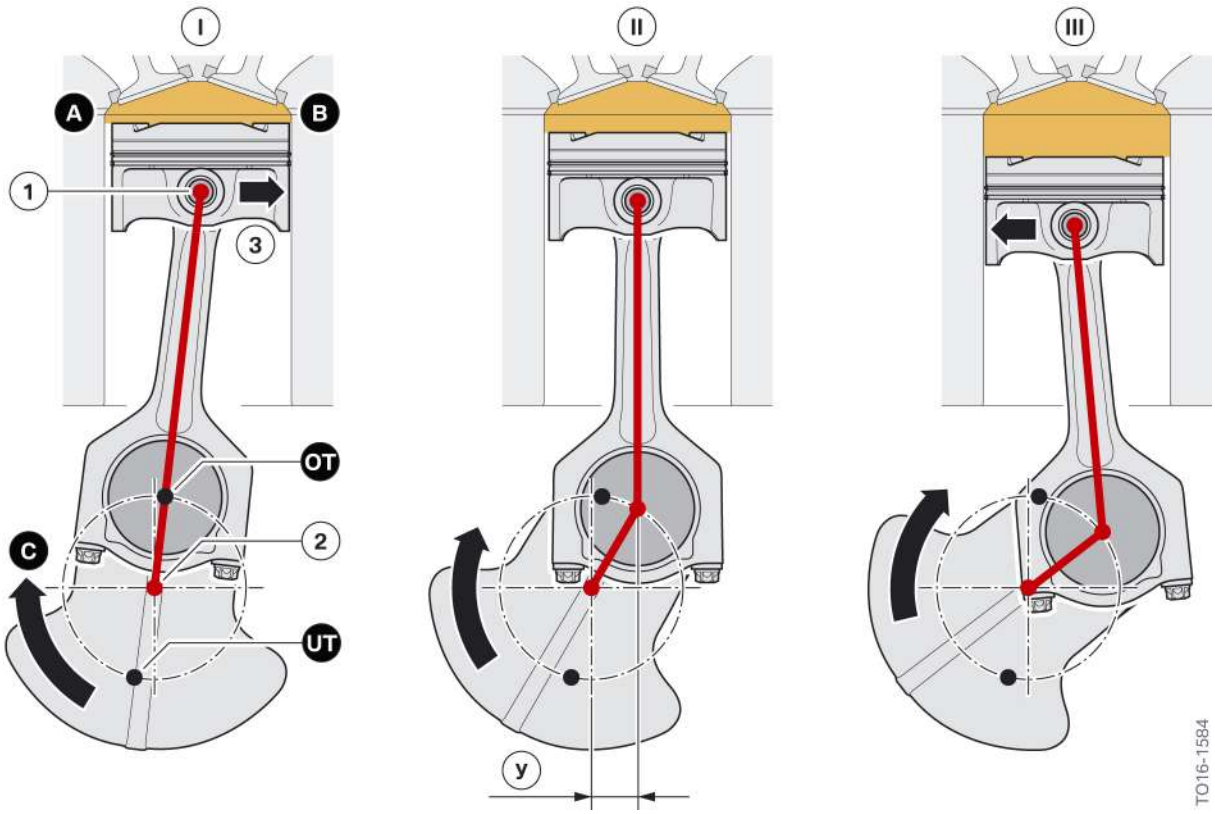

Crankshaft offset

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

| Index | Explanation |
|---|---|
| A | Positive offset |
| B | Negative offset |
| TDC | Top dead centre |
| BDC | Bottom dead centre |

On engines with positive offset, the connecting rod is in an almost perpendicular position during the power cycle. This measure significantly reduces the thrust force or the friction of the piston on the cylinder wall. The result is increased efficiency.

The following graphic clearly shows that positive crankshaft offset, when compared with positive pin offset, has an opposed effect on piston rocking. Thus piston rocking occurs much later and in the range of a high cylinder pressure.



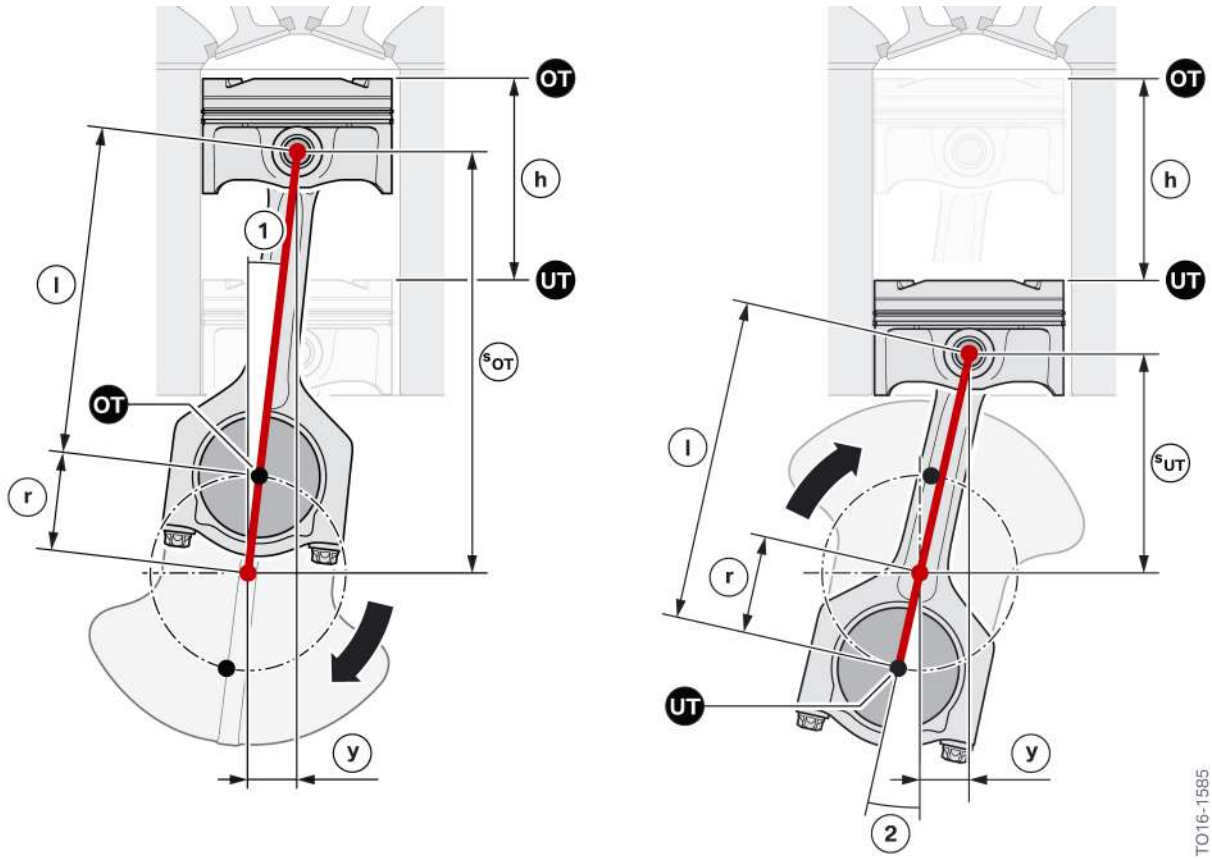

Piston rocking in an engine with crankshaft offset

| Index | Explanation |
|---|---|
| I | Piston and crankshaft position just after top dead centre |
| II | Piston and crankshaft position with perpendicular connecting rod |
| III | Piston and crankshaft position after piston rocking |
| A | Major thrust face |
| B | Minor thrust face |

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

| Index | Explanation |
|---|---|
| C | Direction of engine rotation |
| TDC | Top dead centre |
| BDC | Bottom dead centre |
| y | Crankshaft offset |
| 1 | Gudgeon pin |
| 2 | Crankshaft |
| 3 | Thrust force |

The top and bottom dead centres are shifted by the crankshaft offset. The top and bottom dead centres are achieved in the extended and overlap positions respectively. The connecting rod and the crankshaft point geometrically in the same direction.




TDC and BDC position on an engine with crankshaft offset

| Index | Explanation |
|---|---|
| TDC | Top dead centre |
| BDC | Bottom dead centre |
| l | Connecting rod length |
| r | Crank throw |

| Index | Explanation |
|---|---|
| y | Crankshaft offset |
| $s_{OT}$ | Line TDC (piston position at top dead centre to the crankshaft centre) |
| $s_{UT}$ | Line BDC (piston position at bottom dead centre to the crankshaft centre) |
| h | Piston stroke |
| 1 | Angle in TDC position $\alpha_{TDC}$ |
| 2 | Angle in BDC position $\alpha_{BDC}$ |

### 4.7.1. Calculations

At top dead centre (TDC) crankshaft offset (y), crank throw (r) + connecting rod length (l) and the line TDC ($s_{TDC}$) form a right triangle. Using the Pythagorean theorem, we can calculate the line ($s_{TDC}$):

$s_{TDC} = \sqrt{((l + r)^2 - y^2)}$

At bottom dead centre (BDC) crankshaft offset (y), connecting rod length (l) – crank throw (r) and the line BDC ($s_{BDC}$) form a right triangle. We can thus calculate the line ($s_{UT}$):

$s_{BDC} = \sqrt{((l - r)^2 - y^2)}$

The piston stroke (h) then results by subtraction of the line BDC ($s_{BDC}$) from the line TDC (sTDC):

$h = s_{TDC} - s_{BDC}$

The crankshaft angle at top and bottom dead centres can be calculated using the following formulas:

°Crankshaft $angle_{TDC} = \alpha_{TDC} = \text{arc sin}(y : (l + r))$

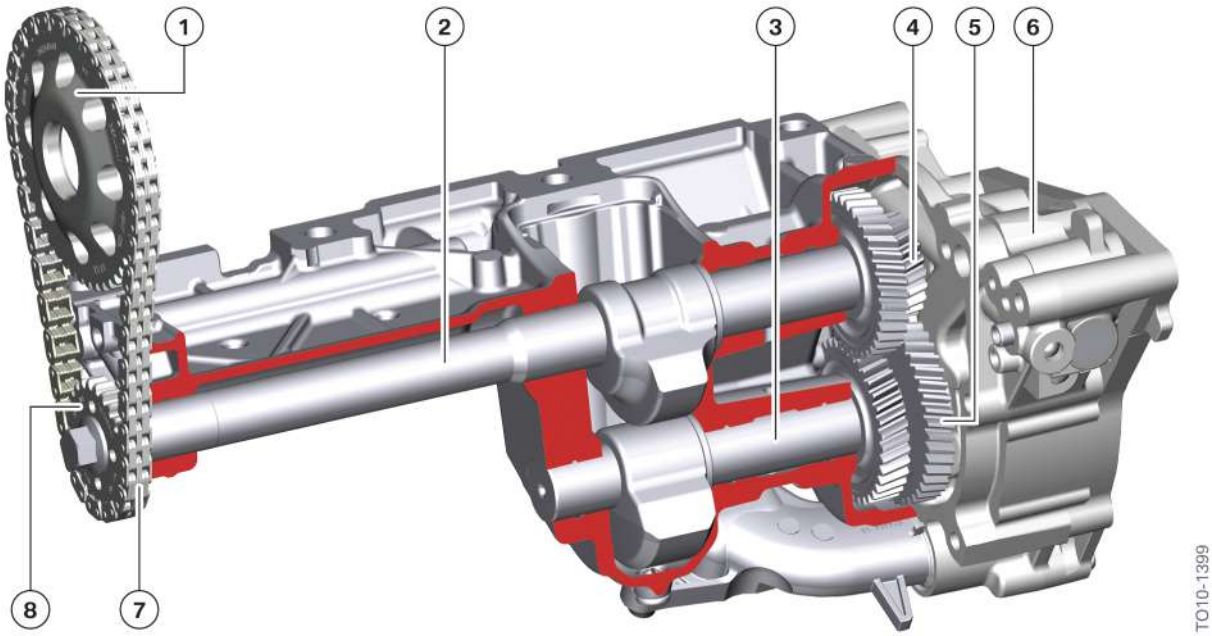
°Crankshaft $angle_{BDC} = 180° + \alpha_{TDC} = 180° + \text{arc sin}(y : (l - r))$

## 4.8. Counterbalance shafts




B38/B48 engine, counterbalance shaft system

| Index | Explanation |
|---|---|
| A | Crankshaft and counterbalance shaft in the B38 engine |
| B | Crankshaft and counterbalance shaft in the B48 engine |




N20 engine, counterbalance shafts

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

| Index | Explanation |
|---|---|
| 1 | Sprocket on crankshaft |
| 2 | Upper counterbalance shaft |
| 3 | Lower counterbalance shaft |
| 4 | Gear, upper counterbalance shaft |
| 5 | Gear, oil pump |
| 6 | Oil pump |
| 7 | Tooth-type chain, counterbalance shaft and oil pump drive |
| 8 | Counterbalance shaft sprocket |

Due to the operating principle of the piston engine, undesired oscillations occur at the engine housing when driving, which can be transmitted to the vehicle interior. To counter this BMW has been using counterbalance shafts for several engine generations. They compensate for imbalance caused due to the design and thus improve the smooth performance and the noise behaviour of the engine by:

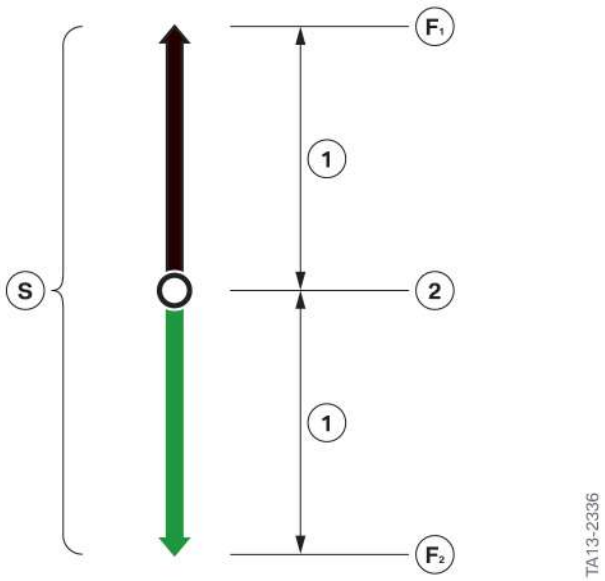
- Eradication of free inertial forces
- Eradication of free inertial torques.

Depending on the engine design and number of cylinders, varying degrees of free inertia forces and free moments of inertia occur.

Counterbalance shafts are used at BMW for the B38 engine as well as most of the 4-cylinder engines.

## 4.8.1. Forces and torques

### Forces



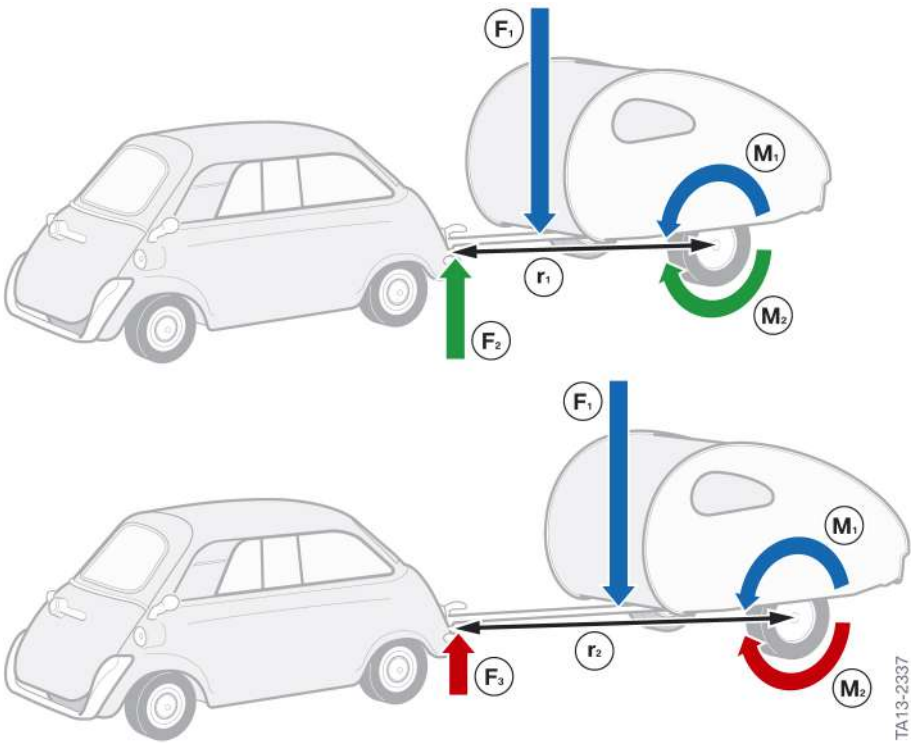

Static force

| Index | Explanation |
|---|---|
| $F_1$ | First force with upwards operating direction |
| $F_2$ | Second force with downwards operating direction |
| S | Static |
| 1 | Intensity of force |
| 2 | Point of action |

Forces are invisible and can only be noticed by their effects. They consist of a point of action and an operating direction. Their state can be dynamic or static. The forces are shown as an arrow (vector). The arrowhead provides information on the operating direction of the force. At the other end of the arrow (vector), opposite the arrowhead, is the point of action. The intensity of the force is determined by the arrow length (vector length). The physical unit of force is [N] and the symbol is [F].

The previous graphic shows 2 forces acting in exactly opposite directions with the same intensity. The system is balanced (static) as both forces cancel each other. When observed from the outside, a reaction cannot be determined.

**Torques**




Static torque

| Index | Explanation |
|---|---|
| $F_1$ | Attached trailer force |
| $F_2$ | Large counterforce |
| $F_3$ | Small counterforce |
| $M_1$ | Attached trailer torque |
| $M_2$ | Counter-torque |
| $r_1$ | Small lever arm |
| $r_2$ | Large lever arm |

Torques occur if a force is transmitted at its point of action via a lever arm. The physical unit of torque is [Nm] and its symbol is [M]. The lever arm and the force acting at the point of action are decisive for the size of the torque. The following mathematical context emerges:

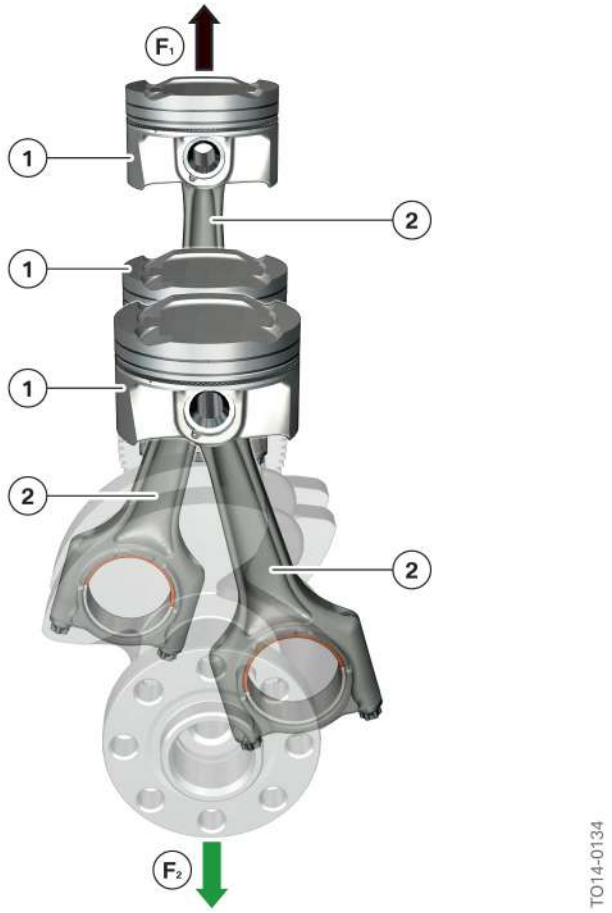
$M = F \cdot r$

The previous graphic shows 2 torques which cancel each other out. If the contact force is increased, without increasing the counterforce to the same degree, the two torques are no longer at equilibrium. The pendant would fall to the floor. To re-establish the static equilibrium of the system, either the counterforce has to be increased or the lever arm extended at which the counterforce acts.

### Oscillating and rotating forces

The upwards and downwards movement of the pistons, as well as the connecting rods, is called an oscillating force and the rotational movement of the crankshaft is called a rotating force. These two different forces also cause the undesired oscillations when the engine is running.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.



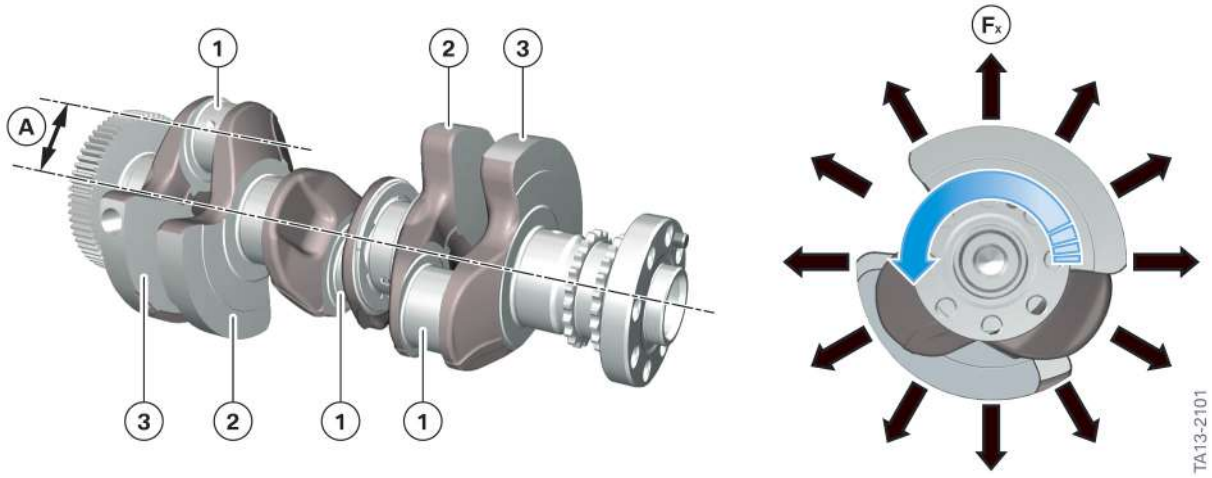

B38 engine, oscillating forces

| Index | Explanation |
|---|---|
| $F_1$ | Force directed upwards |
| $F_2$ | Force directed downwards |
| 1 | Piston |
| 2 | Connecting rod |

Oscillating forces occur in the piston engine when the direction changes at bottom and top dead centre. Due to the inertia of the piston and the connecting rod, the force is directed either upwards or downwards. The following therefore applies: the lower the mass of the piston and the connecting rod, the lower the corresponding oscillating forces will be.

To lower the fuel consumption further, the power output per litre is increased. Through the use of exhaust turbochargers more power is obtained from the same displacement. This means the pistons and the connecting rod are exposed to higher ignition pressures and must have a more solid design. The masses of the components subsequently increase, and thus the oscillating forces.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

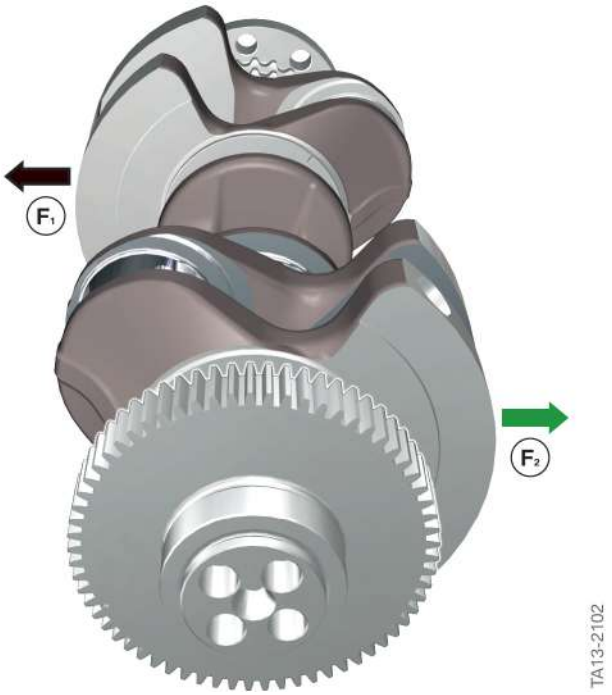



B38 engine, rotating forces

| Index | Explanation |
|---|---|
| A | Piston stroke (offset of connecting rod bearing journal) |
| $F_x$ | Rotating forces |
| 1 | Connecting rod bearing journal |
| 2 + 3 | Counterweights |

To produce a rotating movement of the crankshaft from the stroke of the piston, the connecting rod bearing journals are not located on the axis of rotation of the crankshaft. An imbalance when turning the crankshaft occurs by the offset. The arising imbalance is counteracted by the use of counterweights.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.




B38 engine, counterweights of crankshaft

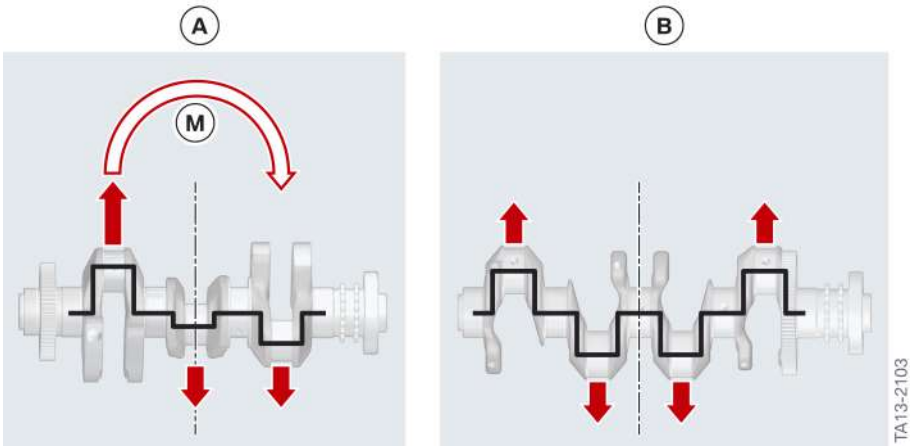
| Index | Explanation |
|---|---|
| $F_1$ | Force of the crankshaft weight directed to the left |
| $F_2$ | Force of the crankshaft weight directed to the right |

The rotating inertia forces can be completely neutralised with help of the counterweights at the crankshaft. Some of the weight at the crankshaft is also used to reduce the oscillating inertia forces (change of direction of the piston and connecting rod). If the entire oscillating inertia forces are reduced by counterweights at the crankshaft, the forces or torques (depending on the design and number of cylinders) in the top and bottom dead centre of the pistons are neutralised. However, this would also mean that in the piston intermediate position (piston between top dead centre and bottom dead centre) the inertia forces of the counterweights work in a lateral direction. The result would be vibrations which are transmitted via the crankshaft bearing to the crankcase and finally to the entire vehicle.

This conflict can be resolved through the use of counterbalance shafts, as well as clever weight classification of the counterweights at the crankshaft.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

### Oscillating torques




Oscillating torques

| Index | Explanation |
|---|---|
| A | 3-cylinder engine |
| B | 4-cylinder engine |
| M | Torque |

This example shows why the design plays a role in which forces or torques occur during engine operation.

The differences between the two engines become clear when the crankshaft is examined from the side. With the 3-cylinder engine, a free moment of inertia occurs at the crankshaft due to the positions of the connecting rod bearing journals when the engine is running. This torque is absorbed via the crankshaft bearing and transmitted at the crankcase. Unpleasant vibrations are caused at the vehicle.

In contrast, the forces which occur at the crankshaft of the 4-cylinder engine when the engine is running cancel each other out. Free moments of inertia therefore do not occur.

### Order

The inertia forces are divided into "orders". The more frequently an inertia force occurs during each crankshaft revolution (360°), the greater its order of magnitude will be.

The free inertial forces and torques categorised as follows:

- Inertia forces, 1st order
- Inertia forces, 2nd order
- Inertia torques, 1st order
- Inertia torques, 2nd order.

The following table provides an overview of the arising forces and torques depending on the design and number of cylinders

| | Inertia forces | | Inertia torques | |
|---|---|---|---|---|
| **In-line engine** | **1st degree** | **2nd degree** | **1st degree** | **2nd degree** |
| 3-cylinder | – | – | • | • |
| 4-cylinder | – | • | – | – |
| 6-cylinder engine | – | – | – | – |

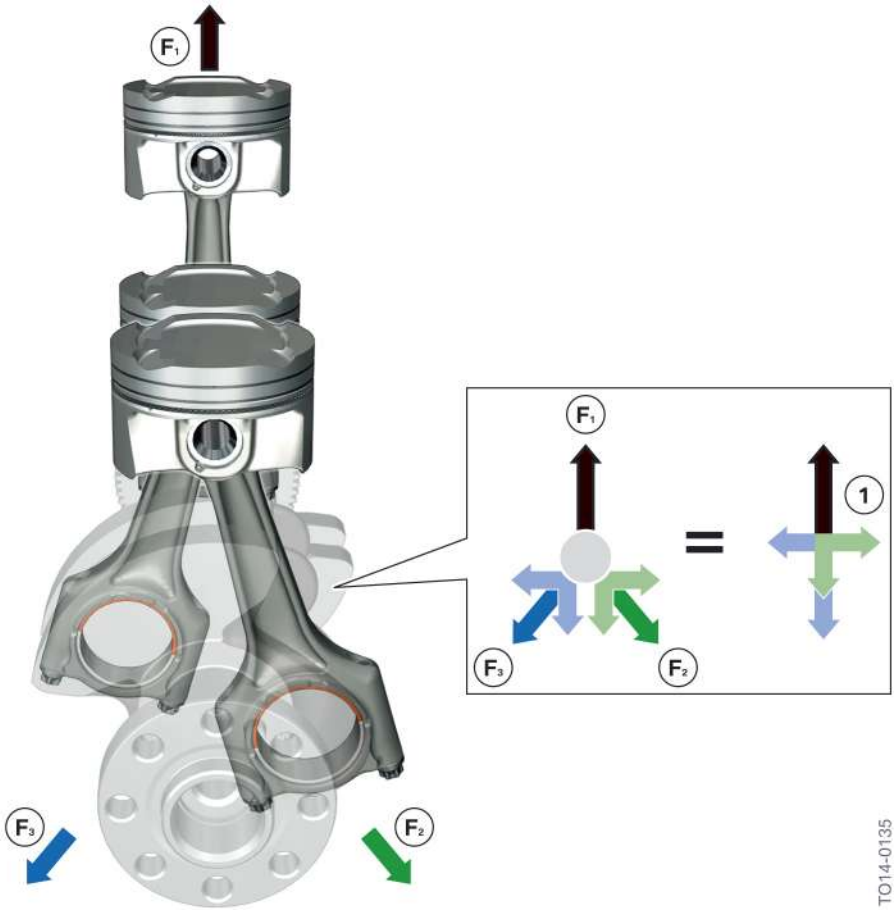
The table provides information on which forces and torques occur and their order of magnitude. Free inertial forces and moments of inertia of the first order are the most noticeable. Free inertia forces and moments of inertia of the second order are perceived is much less troublesome. The greater the order of magnitude is therefore, the more it can be disregarded for the purposes of mass balancing.

The table shows that the best engine design in terms of smooth running is the 6-cylinder in-line engine. All forces cancel each other out. Therefore, with this engine design no additional measures for neutralising the rotating or oscillating masses have to be implemented.

### 4.8.2. Counterbalance shaft 3-cylinder in-line engine

As the table shows, no free inertia forces occur in the 3-cylinder in-line engine. The following graphic demonstrates the reason for this

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.



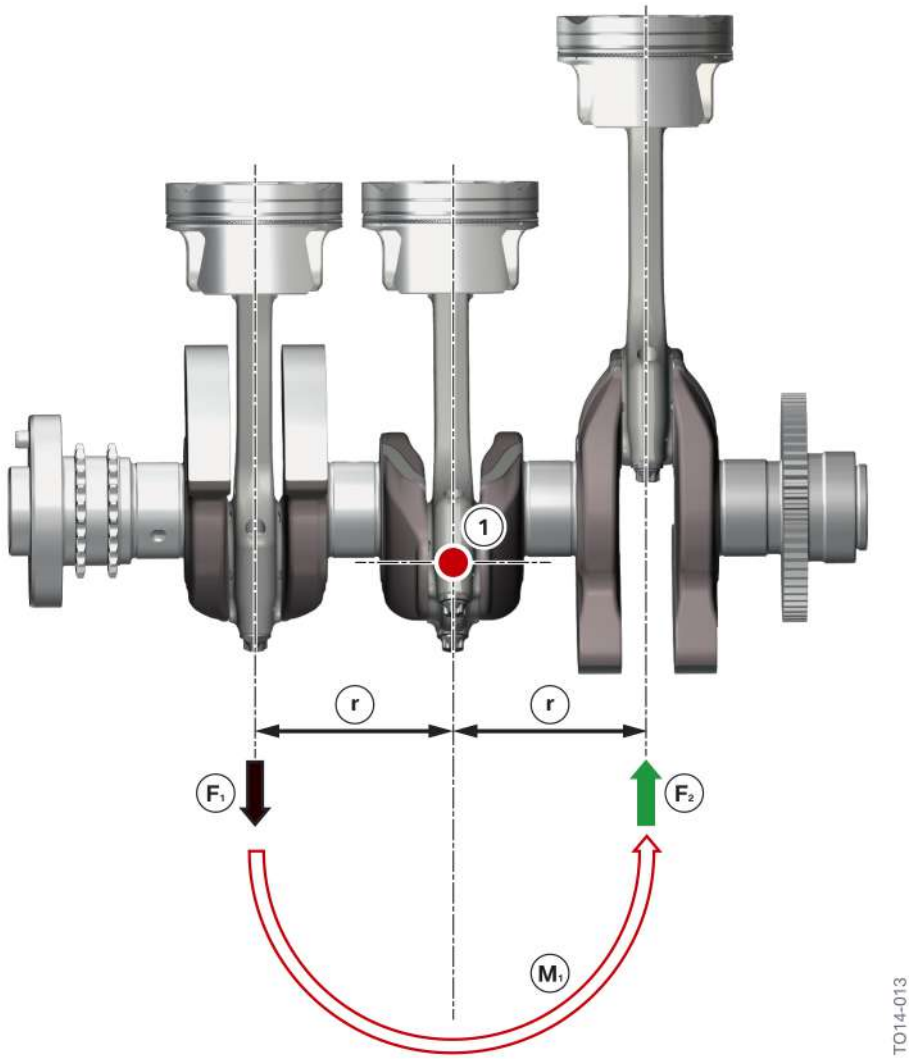

B38 engine, inertia forces

| Index | Explanation |
|---|---|
| $F_1$ | Oscillating and rotating inertia force with operating direction 0° crank angle |
| $F_2$ | Oscillating and rotating inertia force with operating direction 120° crank angle |
| $F_3$ | Oscillating and rotating inertia force with operating direction 240° crank angle |
| 1 | Inertia forces in equilibrium (static) |

Due to the uniform arrangement of the connecting rod bearing journals at an interval of 120° crankshaft angle, all forces occurring in the various operating directions at the crankshaft cancel each other out. The crankshaft is thus in equilibrium (static) when examined from the front.

If the crankshaft is examined from the side, the following schematic diagram of various forces arises:

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.



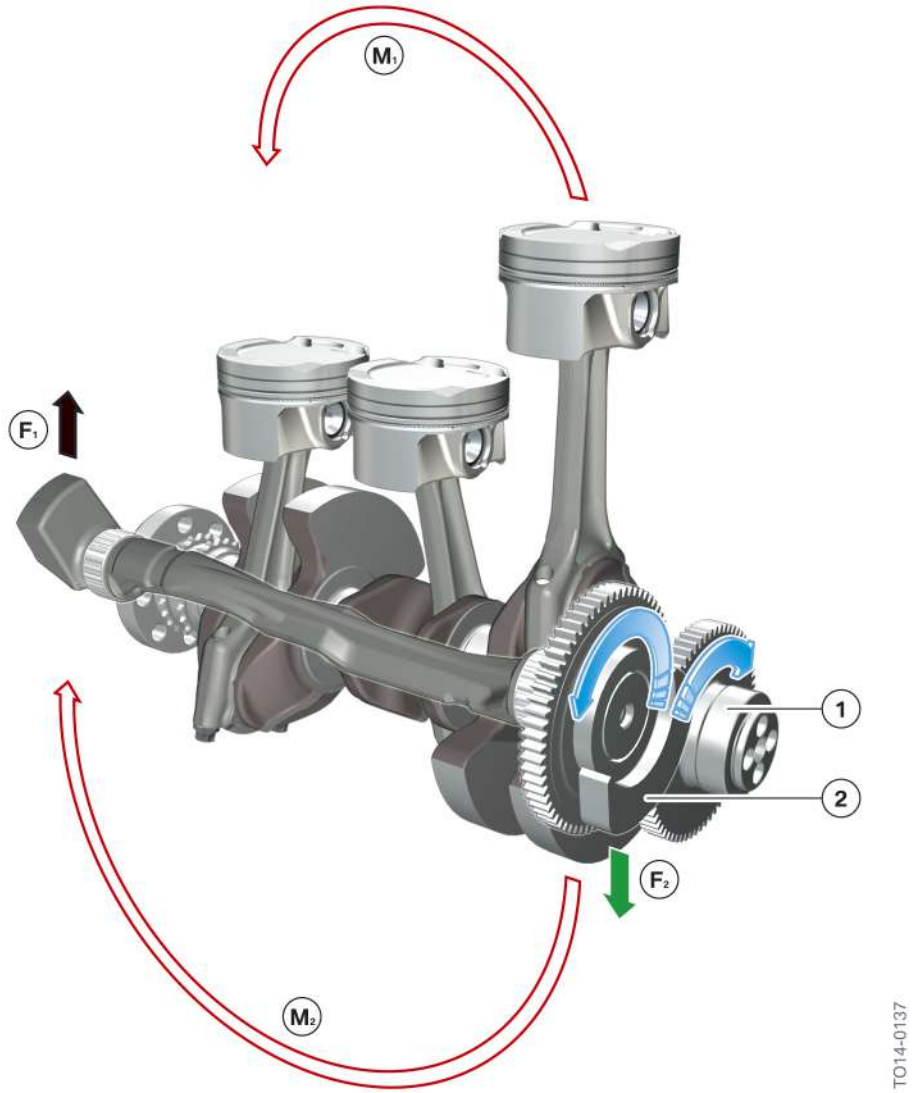

B38 engine, side view, free inertia torques

| Index | Explanation |
|---|---|
| $F_1$ | Force directed downwards |
| $F_2$ | Force directed upwards |
| $M_1$ | Torque |
| r | Lever length |
| 1 | Torque-neutral point |

Due to the positions of the connecting rod bearing journals in the 3-cylinder in-line engine, free moments of inertia of the 1st and 2nd order occur.

As the effects of the free moments of inertia of the 1st order are more intensive than the 2nd order, in the B38 engine the priority is to neutralise the free inertia torques of the 1st order.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.



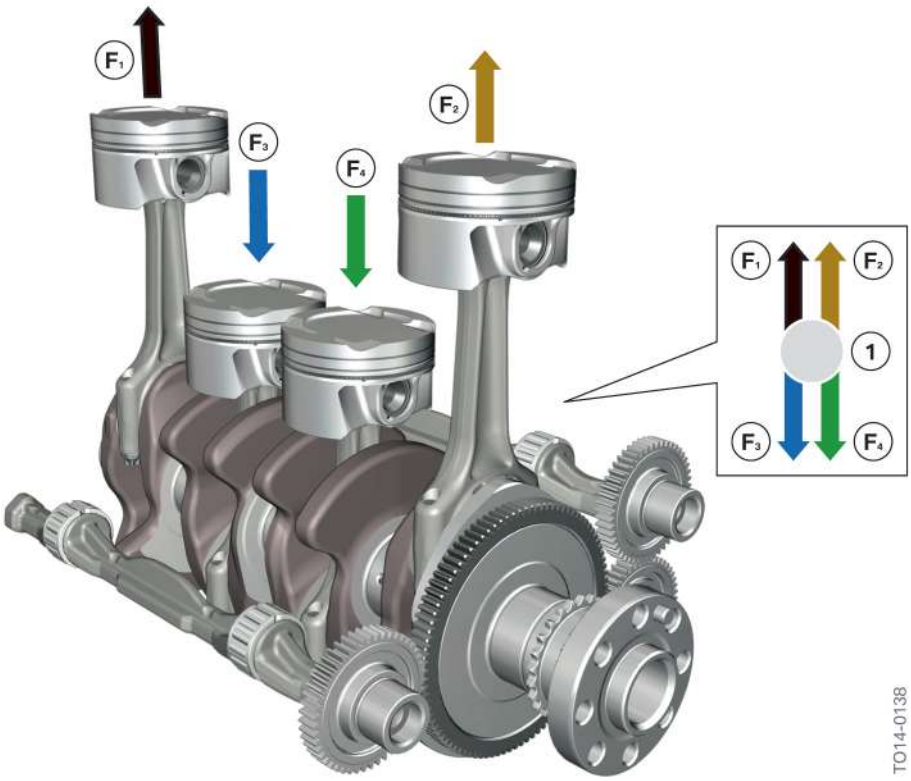

B38 engine, operating principle counterbalance shaft

| Index | Explanation |
|---|---|
| 1 | Crankshaft |
| 2 | Counterbalance shaft |
| $F_1$ | Force of counterbalance shaft directed upwards |
| $F_2$ | Force of counterbalance shaft directed downwards |
| $M_1$ | Torque at the crankshaft |
| $M_2$ | Counter-torque at the counterbalance shaft |

A counterbalance shaft, which rotates at crankshaft speed, is used in the B38 engine. The two weights of the counterbalance shaft are located at the outer sides of the counterbalance shaft in order to generate a high counter-torque with the least possible weight. The radial support is provided by 2 needle bearings. Lubrication is provided by the oil spray from the crankcase.

### 4.8.3. Counterbalance shaft 4-cylinder in-line engine

Free inertia forces of the 2nd order occur in the 4-cylinder in-line engine. The following graphic shows how the forces of the 1st order cancel each other.

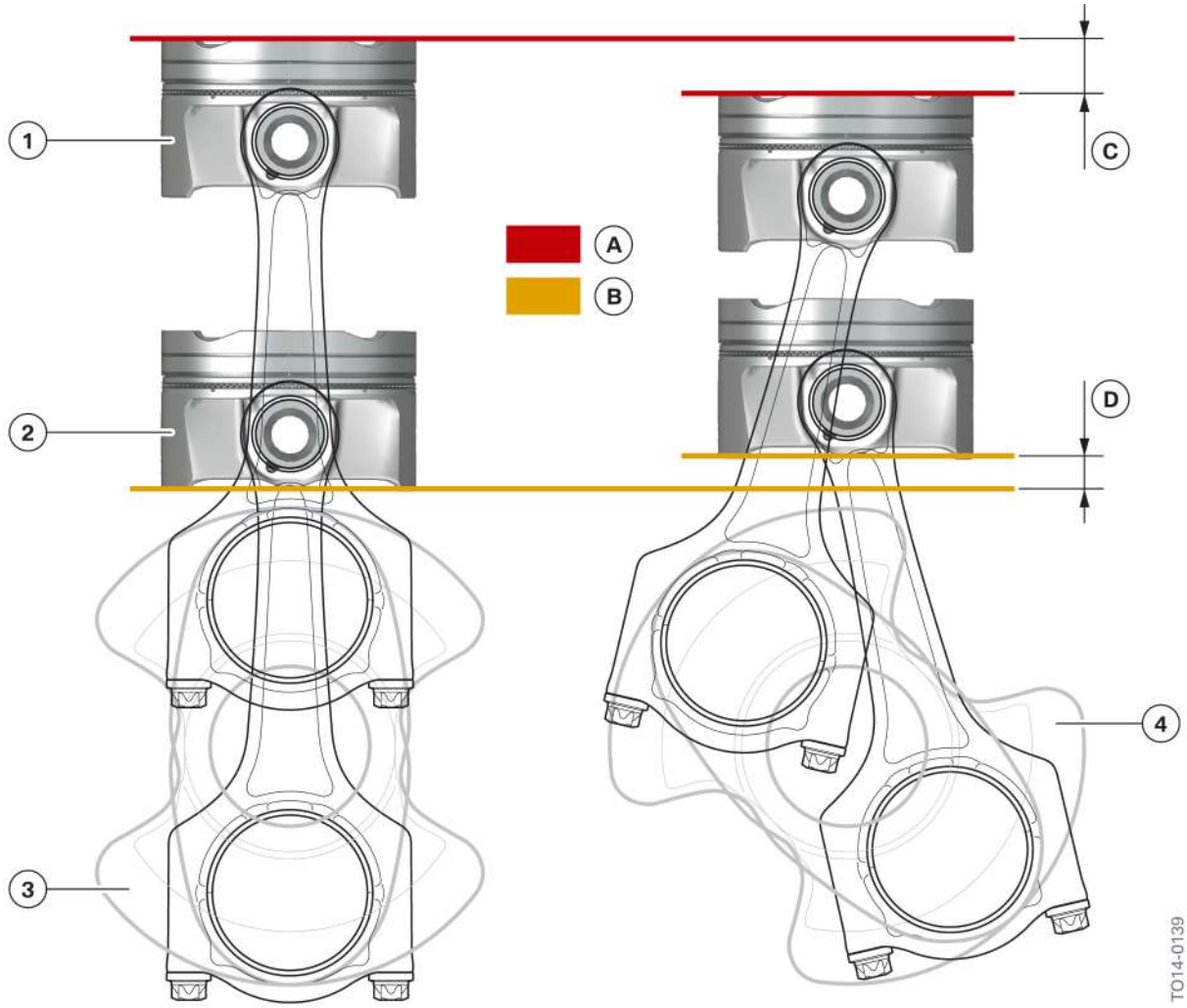



B48 engine, free inertia forces, 1st order

| Index | Explanation |
|---|---|
| $F_1$ | Oscillating and rotating inertia force with operating direction 0° crank angle |
| $F_2$ | Oscillating and rotating inertia force with operating direction 0° crank angle |
| $F_3$ | Oscillating and rotating inertia force with operating direction 180° crank angle |
| $F_4$ | Oscillating and rotating inertia force with operating direction 180° crank angle |
| 1 | Inertia forces in equilibrium (static) |

If you take a look at the crankshaft from the front, then all forces of the 1st order cancel each other. However, inertia forces of the 2nd order occur, which can be explained as follows:

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

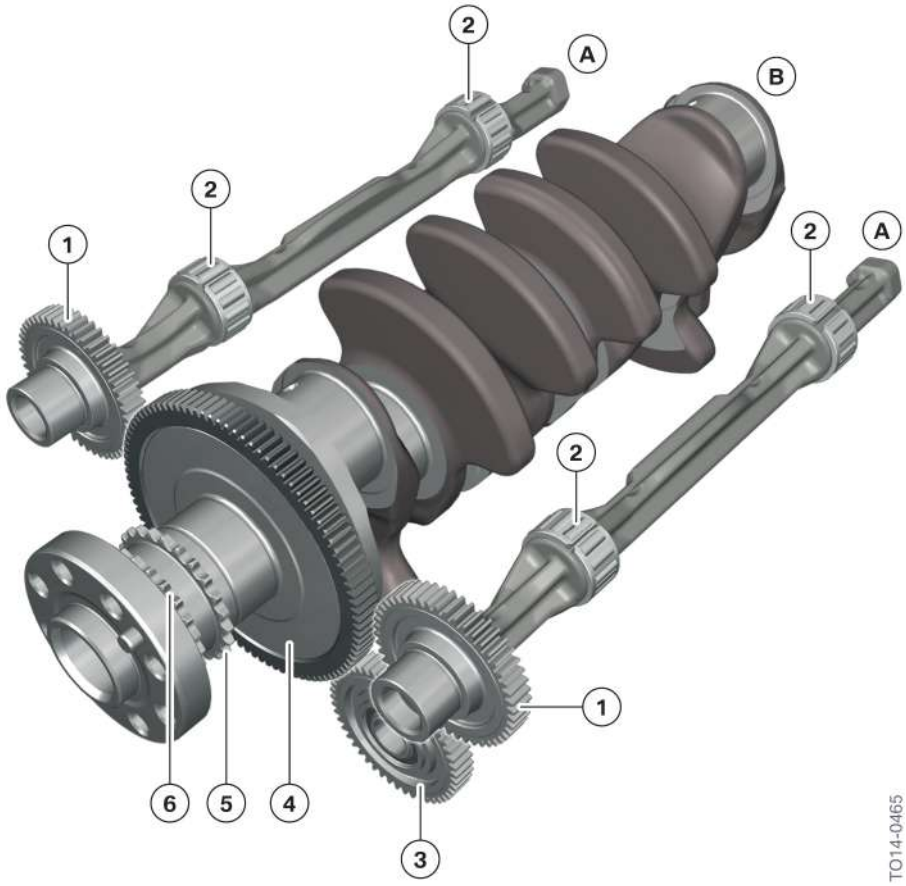



B48 engine, free inertia forces 2nd order

| Index | Explanation |
|---|---|
| A | Position line, piston 1 |
| B | Position line, piston 2 |
| C | Stroke, piston 1 |
| D | Stroke, piston 2 |
| 1 | Piston 1 |
| 2 | Piston 2 |
| 3 | Crankshaft angle 0° |
| 4 | Crankshaft angle 45° after top dead centre |

As can be seen in the previous graphic, both pistons are at top or bottom dead centre at the same time. If the crankshaft is rotated 45° counter-clockwise and the two piston positions are compared, it becomes clear that the paths covered by both pistons differ. Piston 1, which started from the top dead centre position, covers a greater distance than piston 2 from the bottom dead centre position. As both

pistons cover a different distance in the same period, they must have travelled at different speeds. This speed difference affects the acceleration and deceleration of the oscillating masses, thus leading to undesired oscillations. As this process occurs twice per crankshaft revolution, these are referred to as free inertia forces of the 2nd order.




B48 engine, overview counterbalance shafts

| Index | Explanation |
|---|---|
| A | Counterbalance shaft |
| B | Crankshaft |
| 1 | Straight tooth gearing |
| 2 | Needle bearing |
| 3 | Intermediate gear |
| 4 | Output drive gear, crankshaft |
| 5 | Output pinion, timing chain |
| 6 | Output pinion, chain, oil pump |

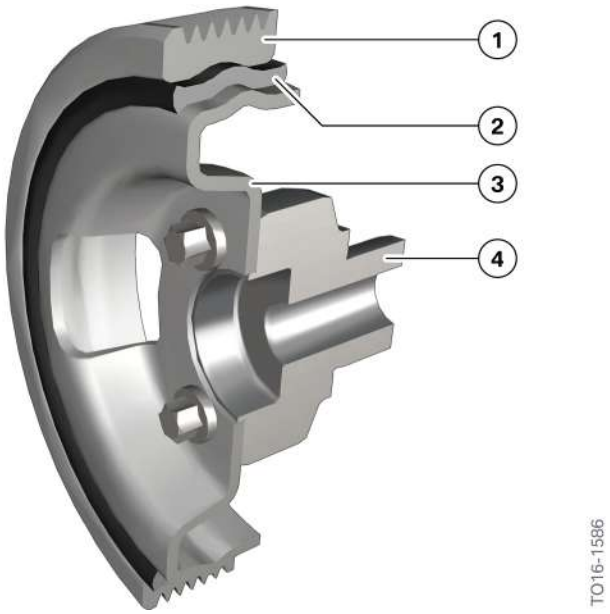
The B48 engine has 2 counterbalance shafts which rotate at twice the speed of the crankshaft. A gear between the crankshaft and the counterbalance shaft effects the change of rotational direction of the right counterbalance shaft.

A correct setting of the two counterbalance shafts is a prerequisite for reduction of the oscillations.

## 4.9. Torsional vibration damper

Due to the irregular application of force during combustion, an irregular rotation arises, which impacts comfort and results in strain on the components. This torsional vibration damper counters this irregular rotation.

The torsional vibration damper always consists of a primary mass and a secondary mass. The primary mass is rigidly coupled with the crankshaft and the secondary mass is elastically coupled with the crankshaft.




N12 engine, torsional vibration damper

| Index | Explanation |
|---|---|
| 1 | Belt pulley |
| 2 | Damping element |
| 3 | Belt pulley carrier |
| 4 | Hub |

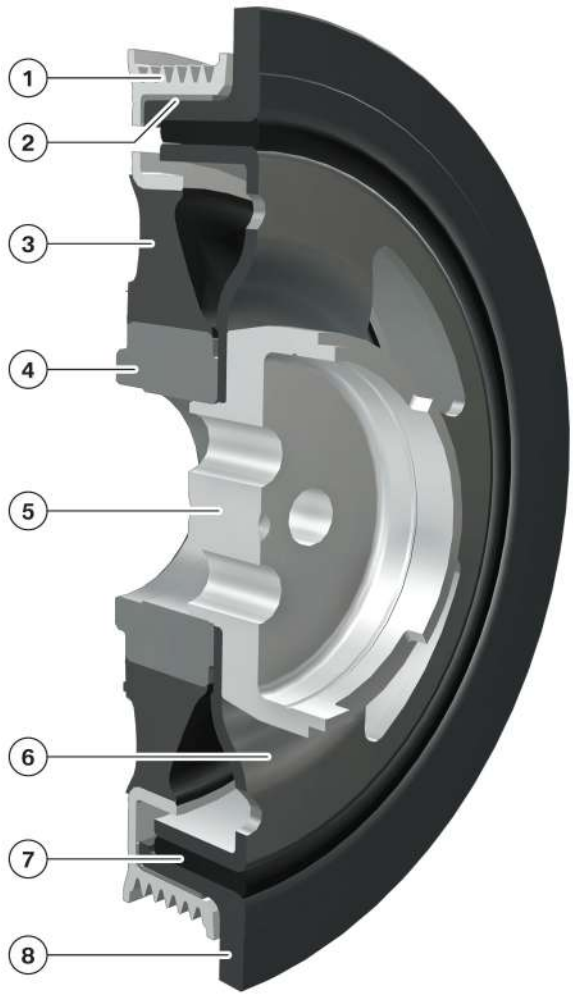
The torsional vibration damper frequently consists of a hub with belt pulley carrier and a belt pulley. Both are connected by a damping element and can therefore rotate freely by a few angular degrees. The hub is bolted to the front end face of the crankshaft.

The torsional vibration damper thus reduces torsional vibrations of the crankshaft. During sudden acceleration of the crankshaft, the belt pulley lags behind the crankshaft by several angular degrees, while it advances correspondingly for deceleration. The loads due to torsional vibration on the belt drive are thereby reduced.

# ENGINE MECHANICS – PETROL ENGINE.
# 4. CRANKSHAFT DRIVE.

To further reduce the load of the crankshaft and noise emission of the engine, additional vibration masses are used on several engines. These are also connected with the crankshaft via damping discs. Examples of this are the N74 engine or the new B engine generation.

On the N74 engine, the belt pulley is rigidly coupled and the additional vibration masses are elastically coupled with the crankshaft. On the new B engine generation, both the belt pulley as well as the additional vibration masses are elastically coupled with the crankshaft.



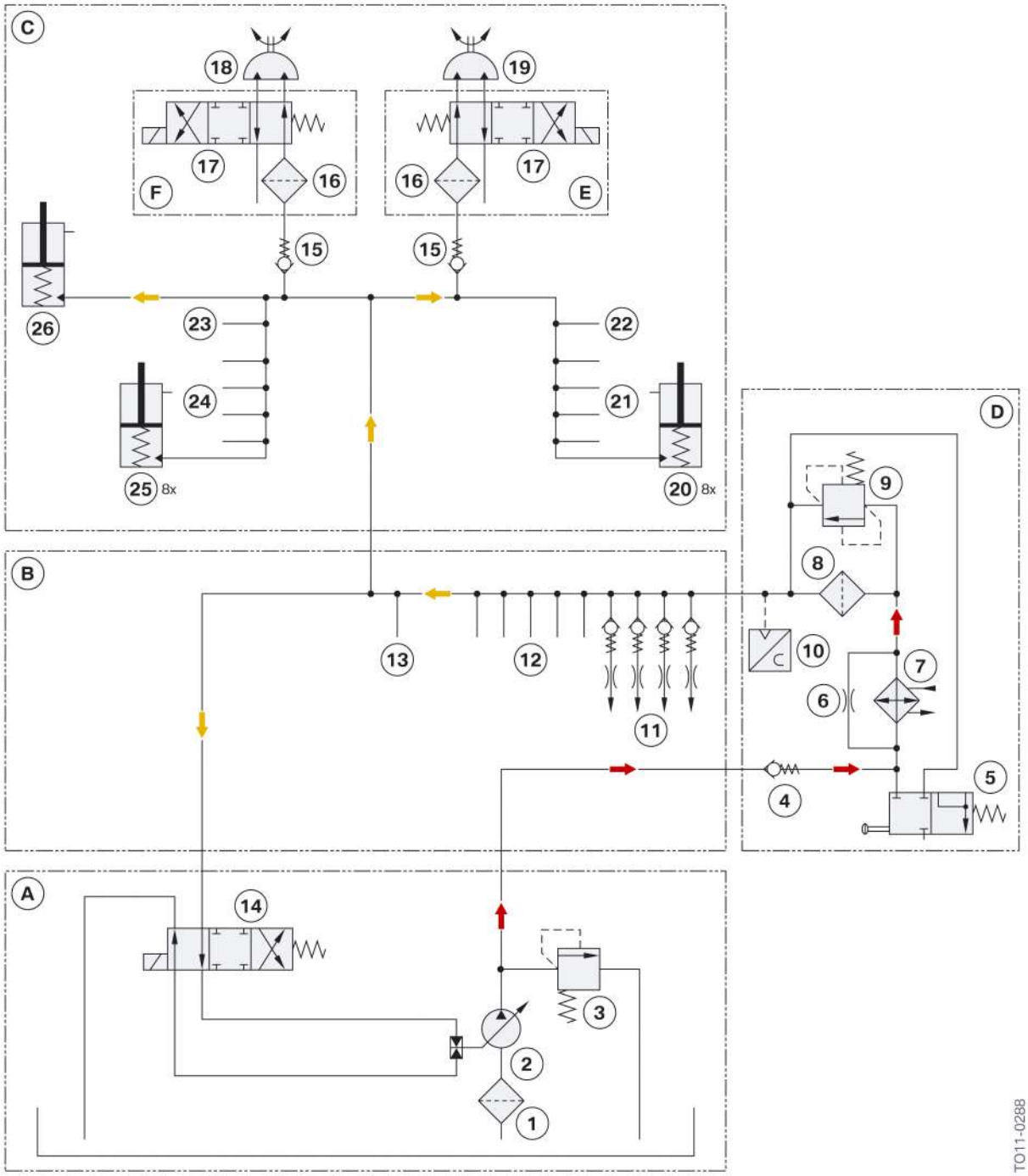

Bx8 engine, torsional vibration damper version

| Index | Explanation |
|---|---|
| 1 | Belt pulley |
| 2 | Plain bearing |
| 3 | Damping element |
| 4 | Belt pulley hub |

| Index | Explanation |
|---|---|
| 5 | Hub |
| 6 | Hub vibration masses |
| 7 | Damping element |
| 8 | Vibration mass |

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

## 5.1. General




Hydraulic circuit diagram oil supply using the N13 engine as an example

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

| Index | Explanation |
|---|---|
| A | Oil pump with oil sump |
| B | Crankcase |
| C | Cylinder head |
| D | Oil filter module |
| E | VANOS solenoid valve, exhaust camshaft |
| F | VANOS solenoid valve, intake camshaft |
| 1 | Strainer |
| 2 | Oil pump |
| 3 | Pressure-limiting valve (cold start valve) |
| 4 | Non-return valve |
| 5 | Discharge valve |
| 6 | Permanent bypass |
| 7 | Oil-to-water heat exchanger |
| 8 | Oil filter |
| 9 | Filter bypass valve |
| 10 | Oil pressure sensor |
| 11 | Oil spray nozzles for piston crown cooling |
| 12 | Lubrication points, crankshaft and connecting rods |
| 13 | Lubrication point, exhaust turbocharger |
| 14 | Oil pressure control valve |
| 15 | Non-return valve |
| 16 | Filter |
| 17 | VANOS solenoid valve |
| 18 | VANOS unit, intake camshaft |
| 19 | VANOS unit, exhaust camshaft |
| 20 | Hydraulic valve clearance compensation (HVCC), exhaust side |
| 21 | Lubrication points, bearings, exhaust camshaft |
| 22 | Lubrication point, vacuum pump |
| 23 | Lubrication point, high-pressure pump |
| 24 | Lubrication point, bearing, intake camshaft |
| 25 | Hydraulic valve clearance compensation (HVCC), intake side |
| 26 | Chain tensioner, timing chain |

With the help of the oil supply, all locations in the engine that need to be cooled are supplied with oil. Most BMW and MINI engines use force-fed circulation lubrication for this. In the process, the oil is drawn out of the oil sump by the oil pump through an intake pipe and conveyed into the circuit. The oil passes through the engine oil cooler, the oil filter and from there into the main oil duct, which runs in

the engine block parallel to the crankshaft. Branch ducts lead to the crankshaft main bearings. There are bore holes between the main bearings of the crankshaft and the connecting rod bearing journal which admit the oil to the lubricating points of the connecting rod bearing.

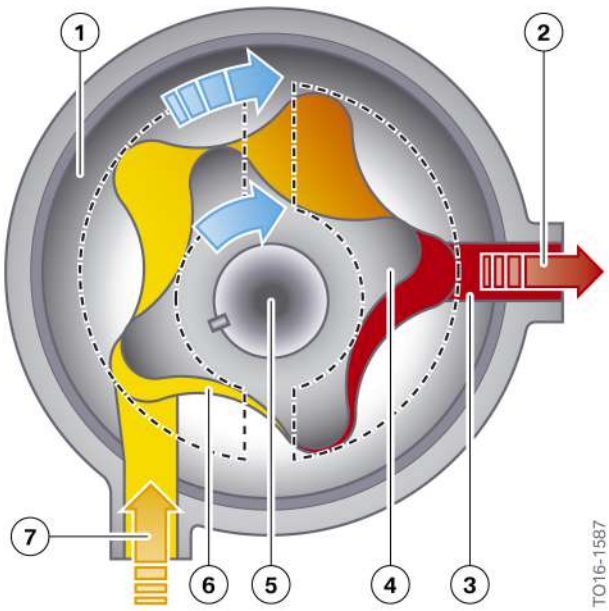
Some of the oil is diverted from the main oil duct and directed to the cylinder head to the relevant lubricating points and adjustment units. When the engine oil flows through the consumers, it either returns to the oil sump via the return ducts or it drips back freely.

## 5.2. Oil pump

Various oil pumps are used in current BMW and MINI engines. The following table provides a brief overview of the oil pumps used:

| Oil pump | Engine |
|---|---|
| Gear pump (single-stage) | W10, W11 |
| Gear pump (two-stage) | N42, N43, N45, N46, N62, N62TU, N73 |
| Spur gear pump (volume flow-controlled) | N12, N14 |
| Spur gear pump (characteristic map-controlled) | N13, N16, N18 |
| Pendulum slide cell pump (volume flow-controlled) | N52, N54, N63, N74 |
| Pendulum slide cell pump (characteristic map-controlled) | B38, B48, B58, N20, N53, N55, N63TU, N63TU2 |

### 5.2.1. Gear pump




Gear pump (rotor oil pump)

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

| Index | Explanation |
|---|---|
| 1 | External gear |
| 2 | Pressure oil |
| 3 | Pressure chamber |
| 4 | Internal gear |
| 5 | Drive shaft |
| 6 | Suction chamber |
| 7 | Suction oil |

The gear pump or rotor pump consists of an outer rotor with internal toothing and an inner rotor with external toothing. The inner rotor is driven. The outer rotor rolls over the teeth of the inner rotor and in this way rotates in the pump housing.

The inner rotor has one tooth fewer than the outer rotor so that in the course of a rotation liquid is transported from one tooth gap to the next on the outer rotor.

During the rotational movement the spaces on the suction side increase in size, while those on the pressure side decrease in size accordingly. With this design, high pressures can be generated at higher rates of delivery.

### Single-stage gear pump



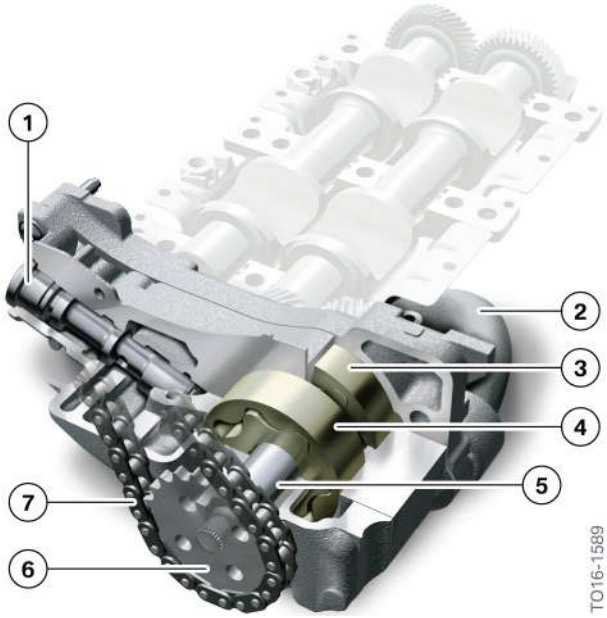

W10 engine, single-stage gear pump

| Index | Explanation |
|---|---|
| 1 | External gear |
| 2 | Internal gear |
| 3 | Guide pin |
| 4 | Flat area for oil pump drive |

Single-stage gear pumps are used, for example in the W10 and W11 engine. Oil pump and pressure limiting valve are located inside on the front cover. The oil pump consists of a gear pair. The internal gear is driven directly by the crankshaft via the 2 flat areas.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

### Two-stage gear pump




N42 engine, two-stage gear pump

| Index | Explanation |
|---|---|
| 1 | Oil pressure control valve |
| 2 | Suction pipe |
| 3 | Oil pump, stage 2 |
| 4 | Oil pump, stage 1 |
| 5 | Drive shaft |
| 6 | Sprocket |
| 7 | Chain |

With the use of the N42 engine, a two-stage oil pump was used. This was necessary due to the increased oil demand of the VANOS with vane motor.

With the two-stage gear pumps, 2 gear pairs are switched consecutively (e.g. N42 engine) or in parallel (e.g. N62 engine). They are located below the crankshaft and are driven by this using a chain.

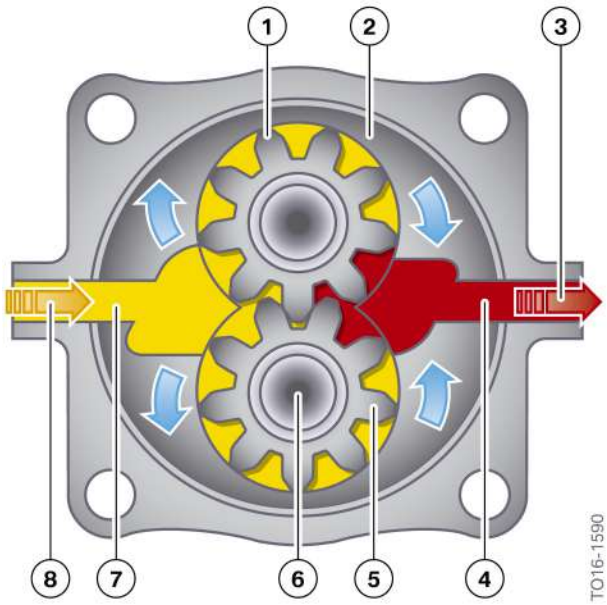
The oil pressure control valve regulates the oil pressure for both gear pairs. The second stage is only active in the lower engine speed range in order to always have sufficient oil pressure for the variable camshaft timing control (VANOS).



On several engines (e.g. the N42 engine), the oil pump is screwed onto the counterbalance shaft housing.

Never open the screw connections between the oil pump and counterbalance shaft housing since the tooth backlash is adjusted via this screw connection and can only be performed at the factory. The counterbalance shafts or the oil pump may only be replaced as common unit.

### 5.2.2. Spur gear pump

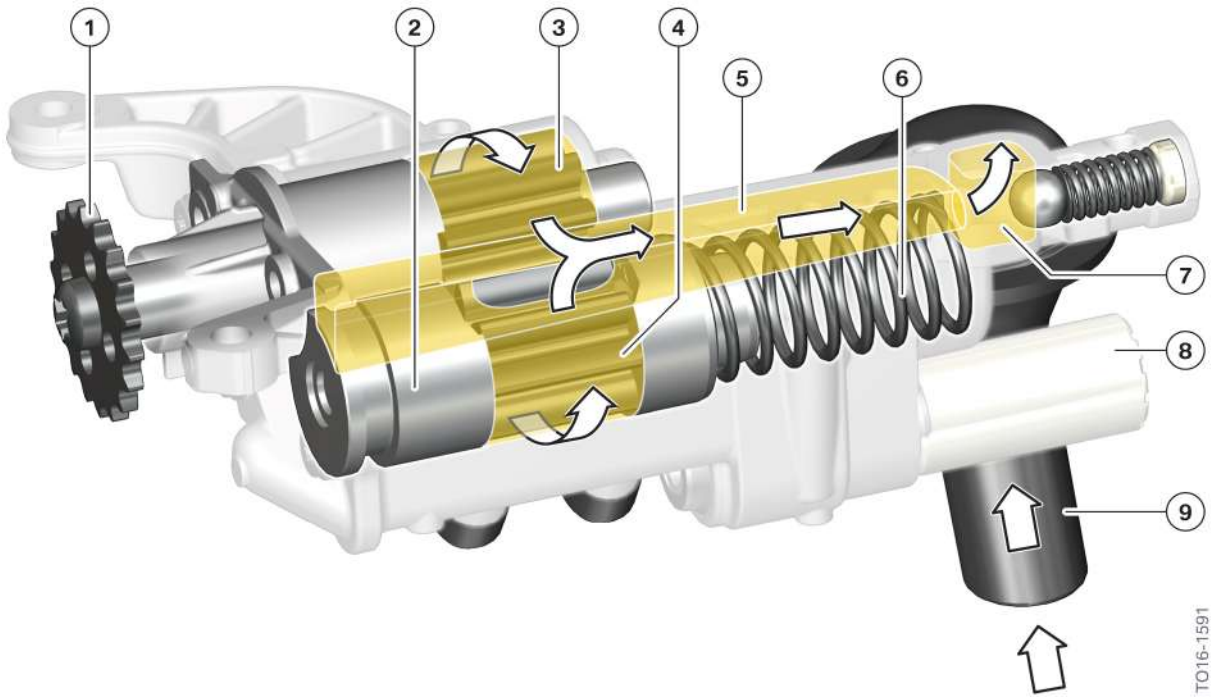



Spur gear pump

| Index | Explanation |
|---|---|
| 1 | Gear |
| 2 | Oil pump housing |
| 3 | Pressure oil |
| 4 | Pressure chamber |
| 5 | Gear |
| 6 | Drive shaft |
| 7 | Suction chamber |
| 8 | Suction oil |

In this oil pump, 2 gears with external teeth mesh with each other, one of these gears being driven. The crowns of the teeth not in use move along in the pump housing, delivering oil from the suction chamber to the pressure chamber.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.



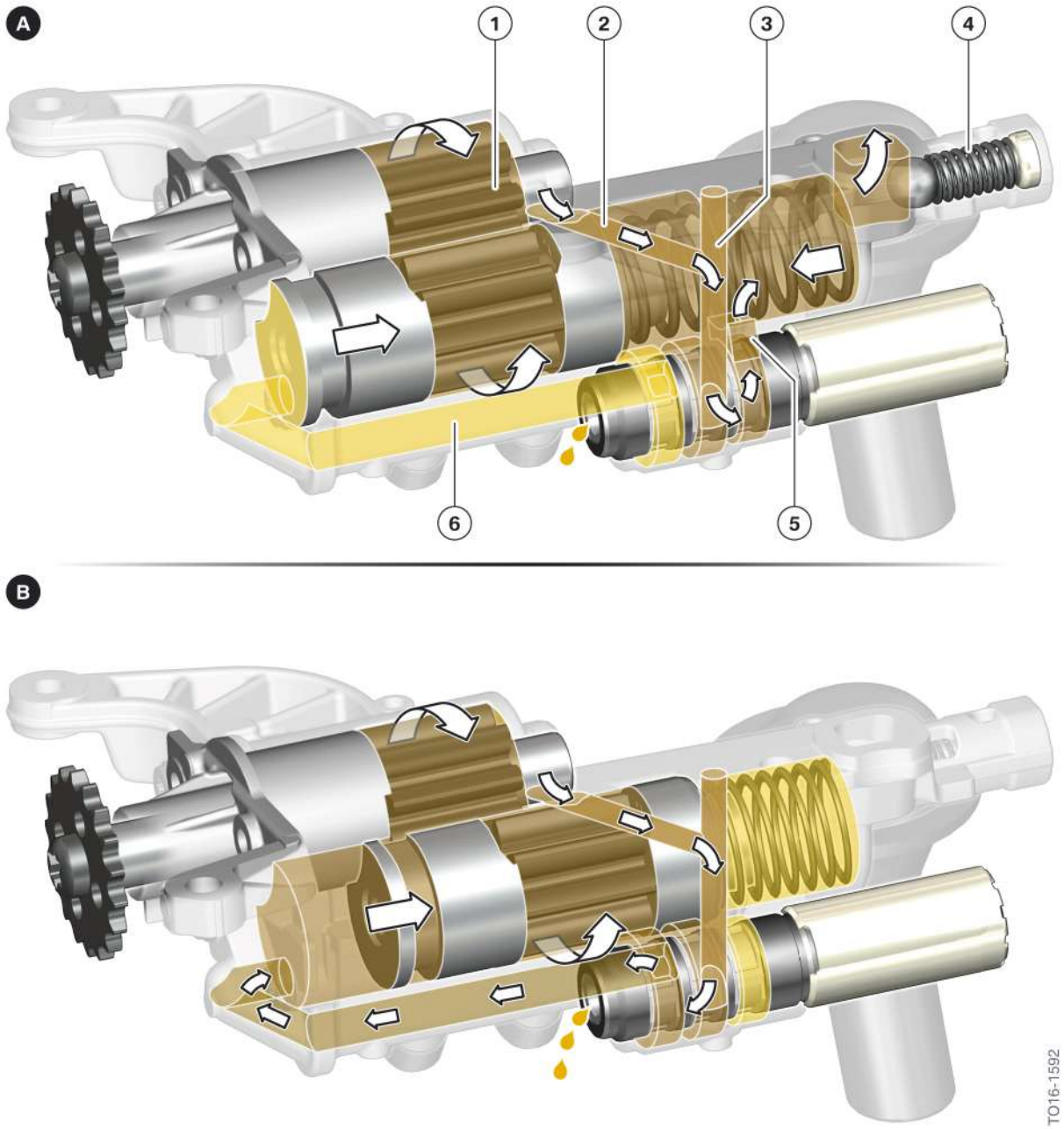

N13 engine, characteristic map-controlled spur gear pump

| Index | Explanation |
|---|---|
| 1 | Sprocket |
| 2 | Control plunger |
| 3 | Gear, oil pump |
| 4 | Gear, oil pump |
| 5 | Raw oil duct |
| 6 | Compression spring |
| 7 | Raw oil duct to bedplate |
| 8 | Oil pressure control valve |
| 9 | Intake neck |

The oil pump is driven via a chain by the crankshaft. The gear ratio of the oil pump to the crankshaft is dependent on the number of teeth on the respective sprockets.

The non-driven oil pump gear can be axially shifted in this pump, thereby varying the delivery rate. The axial shift occurs by means of oil pressure, which is able to be varied by oil pressure control valve. The oil pressure control valve is supplied with filtered oil from the main oil duct. The operating principle of the oil pump ensures that the required oil quantity in each case and the oil pressure are supplied.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.




N13 engine, maximum delivery of the oil pump

| Index | Explanation |
|---|---|
| A | Maximum delivery |
| B | Minimum delivery |
| 1 | Gear, oil pump |
| 2 | Clean oil duct |

| Index | Explanation |
|---|---|
| 3 | Clean oil duct |
| 4 | Pressure-limiting valve |
| 5 | Oil duct to rear end of control piston |
| 6 | Oil duct to front end of control piston |

In the basic position, the compression spring presses the control piston into the position for maximum delivery. In addition, this position can be actively moved to via the oil pressure control valve. For this, the oil pressure control valve directs the oil pressure from the filtered oil duct to the rear side of the control piston and presses it into the position for maximum delivery. At the same time, the oil from the front of the control piston can drain into the oil sump.

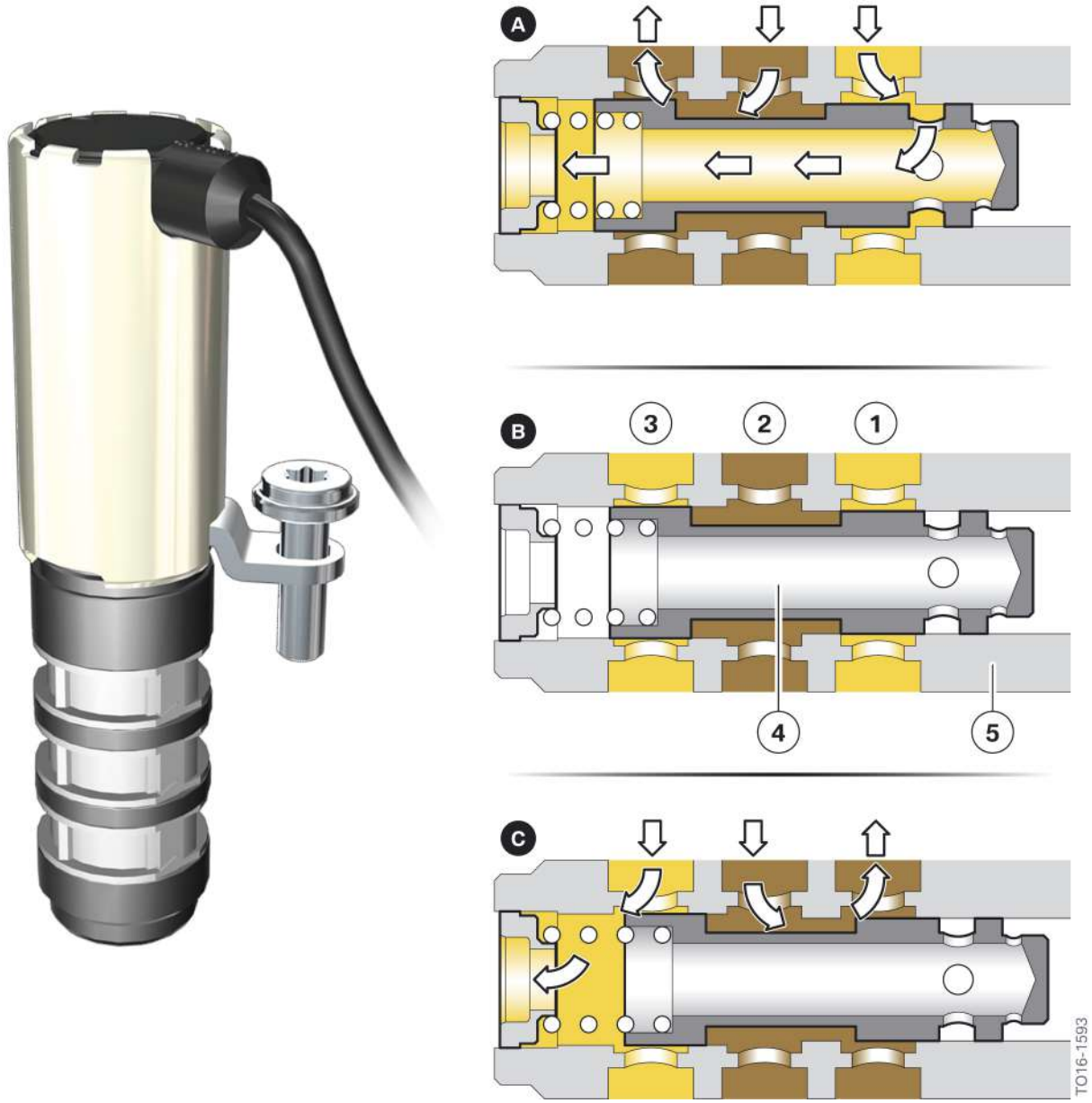
For the minimum delivery, the oil pressure control valve directs the oil pressure from the filtered oil duct to the front side of the control piston and presses it into the position for minimum delivery. At the same time, the oil from the rear of the control piston can drain into the oil sump.

**Map control**

An oil pressure sensor detects the oil pressure and sends the data to the Digital Motor Electronics (DME). The DME uses a stored characteristic map to determine the required oil delivery quantity and controls the oil pressure control valve accordingly.

The oil pressure control valve is located on the oil pump and engages the oil ducts in the oil pump to increase or reduce the oil delivery quantity. This is dependent on the engine speed and the position of the oil pressure control valve.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.




N13 engine, oil pressure control valve

| Index | Explanation |
|---|---|
| A | Reduce delivery quantity |
| B | Hold delivery quantity |
| C | Increase delivery quantity |
| 1 | Oil duct to rear end of control plunger in oil pump |

| Index | Explanation |
|---|---|
| 2 | Clean oil duct |
| 3 | Oil duct to front end of control plunger in oil pump |
| 4 | Control plunger, oil pressure control valve |
| 5 | Housing, oil pressure control valve |

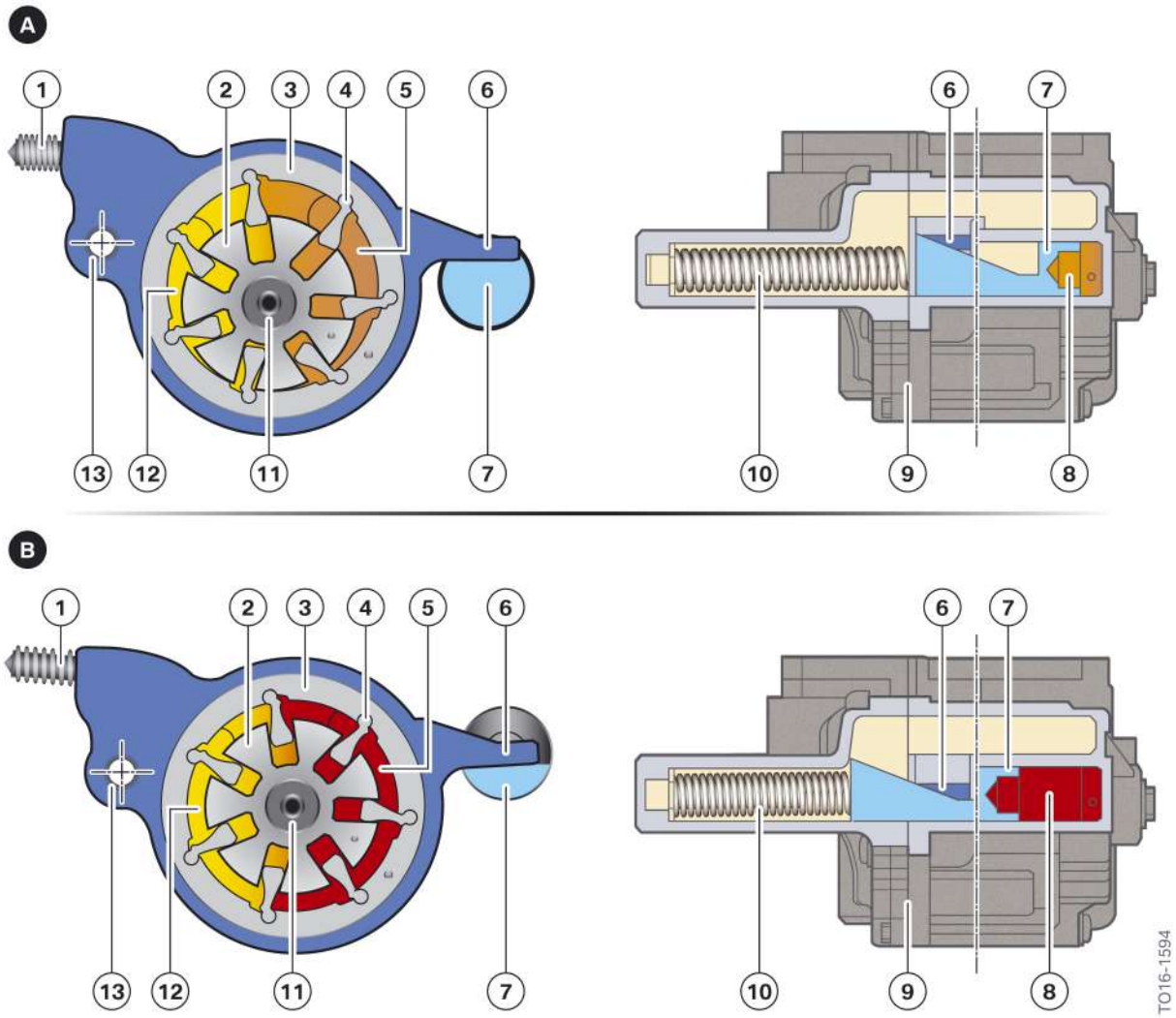
**Volumetric flow control**

The control piston in the oil pressure control valve is shaped in such a way as to integrate a volumetric flow control. This is also used as emergency operation for the characteristic map regulation.

Due to different diameters of the control piston, the groove on the side of the spring has a larger side wall area than on the side of the solenoid valve. If the oil pressure increases, the force that acts against the spring also increases. Thus, the control piston is pressed against the spring force in the direction of "reduce delivery rate".

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

## 5.2.3. Pendulum slide cell pump




Pendulum slide cell pump

| Index | Explanation |
|---|---|
| A | Maximum delivery rate |
| B | Minimum delivery rate |
| 1 | Compression spring |
| 2 | Inner rotor |
| 3 | Outer rotor |
| 4 | Pendulum |
| 5 | Pressure chamber |
| 6 | Adjusting ring (pendulum slide) |
| 7 | Control plunger |

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

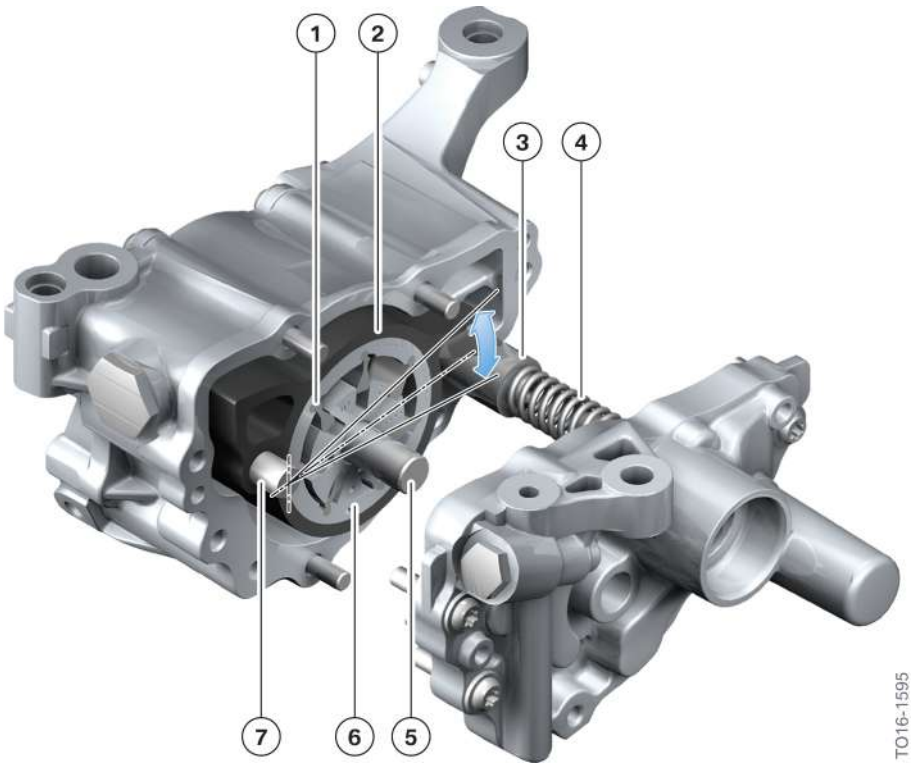
| Index | Explanation |
|---|---|
| 8 | Oil pressure from the pressure chamber |
| 9 | Housing |
| 10 | Compression spring |
| 11 | Pump shaft |
| 12 | Suction chamber |
| 13 | Bearing journal (rotational axis) |

The pendulum slide cell pump only conveys as much oil as the respective operating range of the engine necessitates. In low load operating ranges, the pump does not deliver excess oil quantities. This lowers the fuel consumption of the engine and slows down the deterioration of the oil.

An inner rotor with pendulum rotates as shown in the graphic on the pump shaft. The eccentric position of the inner rotor creates a crescent-shaped cavity. This is divided into chambers with different volumes by the pendulum. During this process, the oil is drawn into the expanding chamber (intake side) and is delivered via the contracting chamber (pressure side). The change of position of the adjusting ring allows the delivery rate of the pendulum slide cell pump to be varied.

You can find out more about the adaptation of the various development stages of the pendulum slide cell pump in the following.

### Volumetric flow control




N52 engine, volume-flow controlled pendulum slide cell pump

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

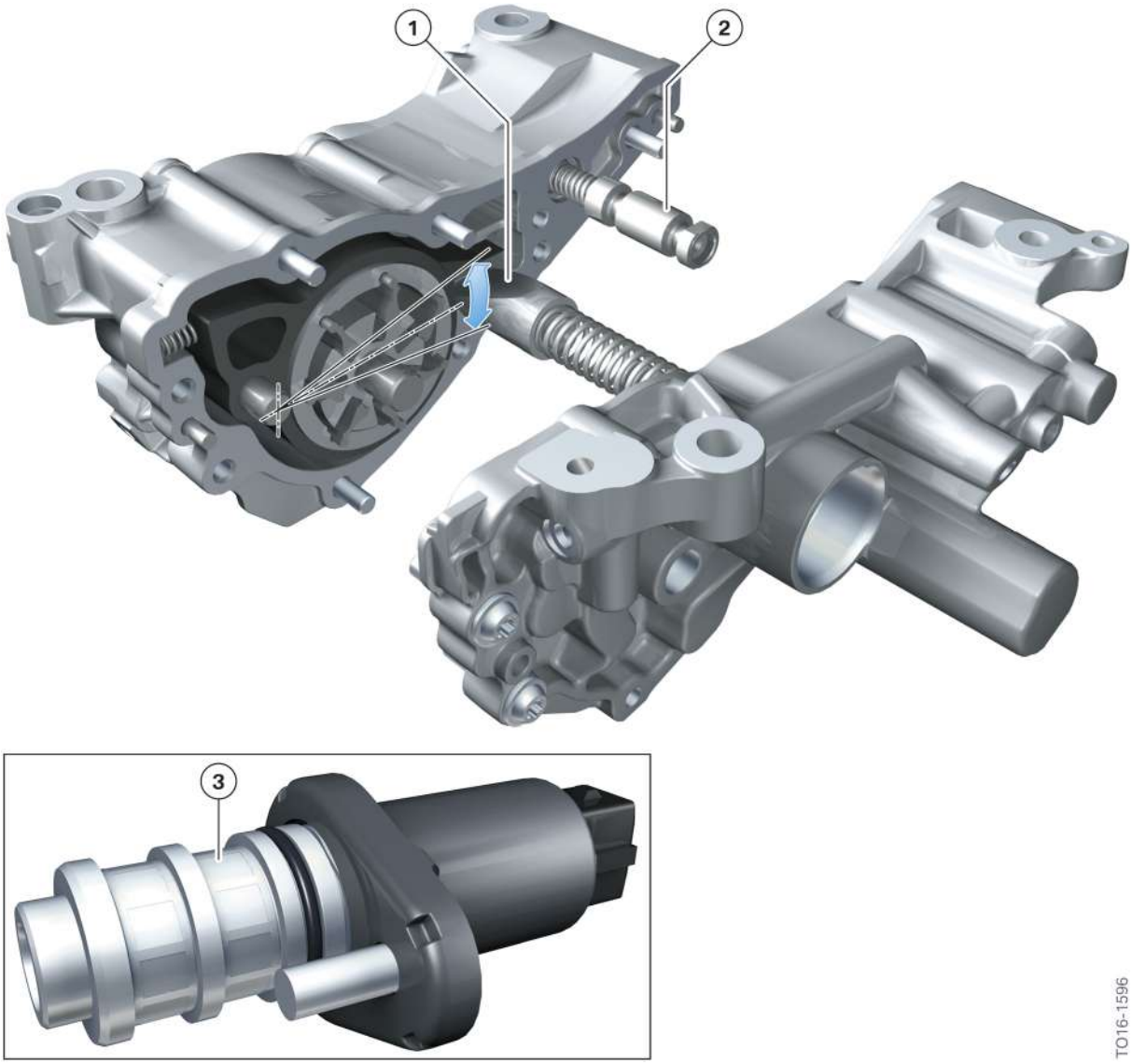
| Index | Explanation |
|---|---|
| 1 | Pendulum |
| 2 | Adjusting ring (pendulum slide) |
| 3 | Control plunger |
| 4 | Compression spring |
| 5 | Pump shaft |
| 6 | Outer rotor |
| 7 | Bearing journal (rotational axis) |

The volume-flow controlled pendulum slide cell pump introduced in the N52 engine is driven by a chain from the crankshaft. The generated oil pressure also acts on the control piston against the force of a compression spring. The position of the adjusting ring is varied via the slanted contact surface on the control piston. If the pump shaft is sitting more centred in the outer rotors, the changes in volume are small and the delivery rate is low. If the pump shaft is off-centre, the changes in volume and the delivery rate are larger.

If the oil consumption of the engine increases, for example, due to a regulating intervention of the VANOS, the pressure in the lubricating system falls and thus also the pressure at the control piston. The oil pump increases the delivery volume and restores the previous pressure ratios again. If the oil consumption of the engine decreases, the oil pump regulates a lower delivery volume accordingly in the direction 0-delivery.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

### Map control



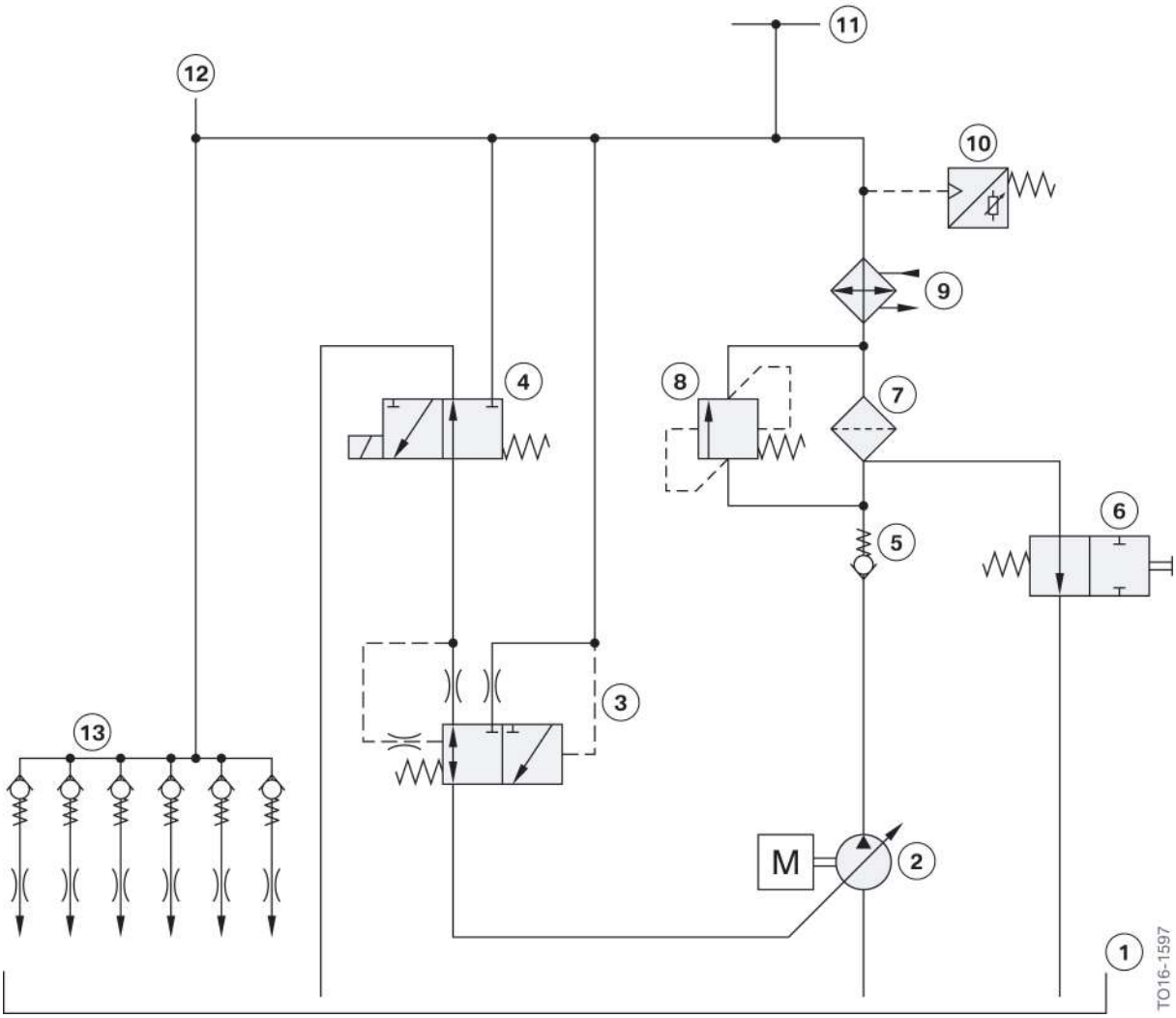

N53 engine, characteristic map-controlled pendulum slide cell pump

| Index | Explanation |
|---|---|
| 1 | Control plunger |
| 2 | Control piston (additional control unit) |
| 3 | Map control valve |

In the N53 engine, the volumetric flow is supplemented by a characteristic map control. For this, the Digital Motor Electronics (DME) uses a sensor to monitor the actual oil pressure in the main oil duct. In addition, it uses the engine speed and the fuel injection rate to calculate the required setpoint oil pressure. According to the deviation, it sends a signal to the characteristic map control valve.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

The valve slider in the characteristic map control valve is opened variably according to the actuation. Thus, a control oil pressure is generated, which activates an additional control unit in the oil pump. This, in turn, passes the pressure to the control piston.




N53 engine, detail of simplified oil circuit

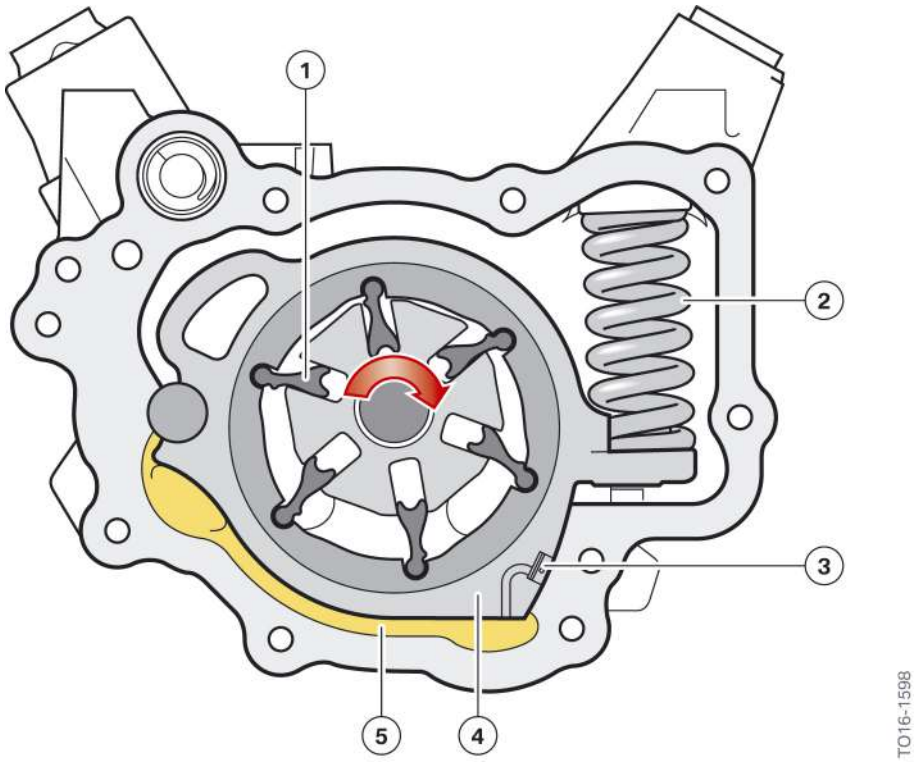
| Index | Explanation |
|---|---|
| 1 | Oil sump |
| 2 | Oil pump |
| 3 | Emergency valve (second-level control valve) |
| 4 | Map control valve |
| 5 | Non-return valve |
| 6 | Discharge valve (on the oil filter) |
| 7 | Oil filter |
| 8 | Filter bypass valve |

| Index | Explanation |
|---|---|
| 9 | Engine oil cooler |
| 10 | Oil pressure sensor |
| 11 | Cylinder head lubrication points |
| 12 | Crankcase lubrication points |
| 13 | Piston crown oil spray nozzles |

If the electrical activation of the oil pump fails, the system falls back on the familiar volume-flow regulation.

An hydraulic valve is used for this, which is also referred to as an emergency valve. This is a 3/2-way valve which is able to channel the main oil pressure into the control piston. To this end, oil from the main oil duct forces a piston back against a spring until finally the duct to the oil pump is opened. The oil pressure from the map control valves acts on the other end of the plunger.

**Control via control oil chamber**




N63 engine, volume-flow controlled pendulum slide cell pump

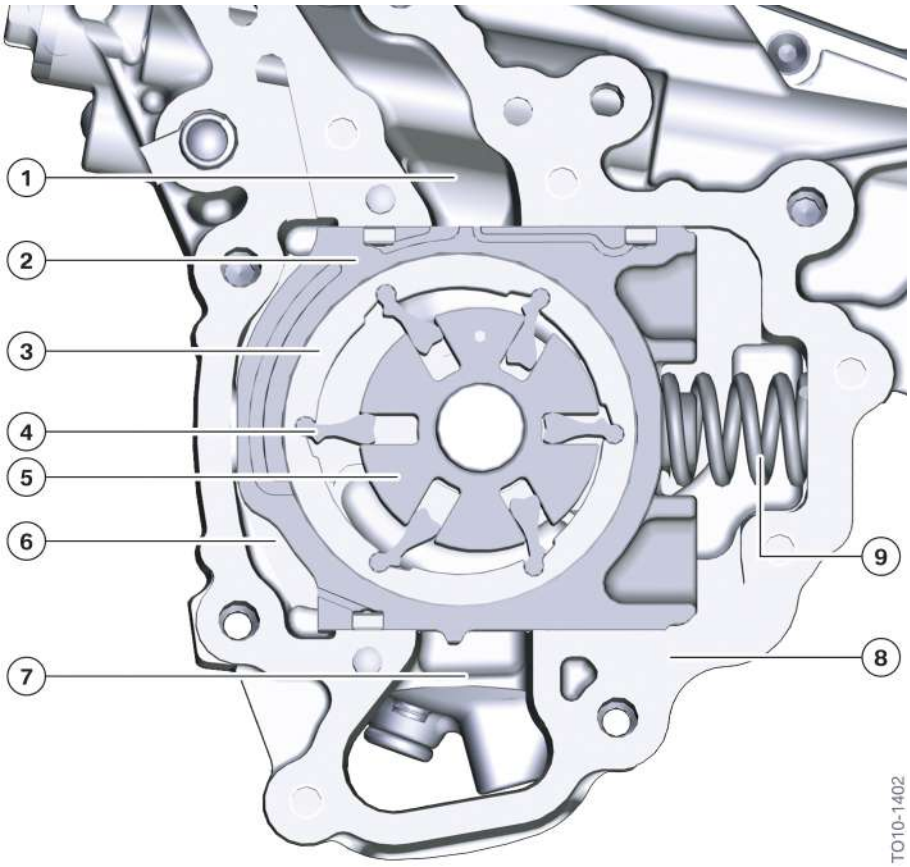
# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

| Index | Explanation |
|---|---|
| 1 | Pendulum |
| 2 | Compression spring |
| 3 | Sealing strip |
| 4 | Adjusting ring (pendulum slide) |
| 5 | Control oil chamber |

An additional change was used in the volume-flow controlled pendulum slide cell pump in the N63 engine. The position of the adjusting ring is now no longer modified by a control piston, but rather the oil pressure acts directly on the adjusting ring. To this end, the oil is channelled into the control oil chamber and acts there against the force of a compression spring on the adjusting ring.

In the N55 engine, a characteristic map-controlled pendulum slide cell pump without control piston is used for the first time. A Duroplast adjusting ring is also used for the first time.

**Special feature N20 engine**




N20 engine, characteristic map-controlled pendulum slide cell pump

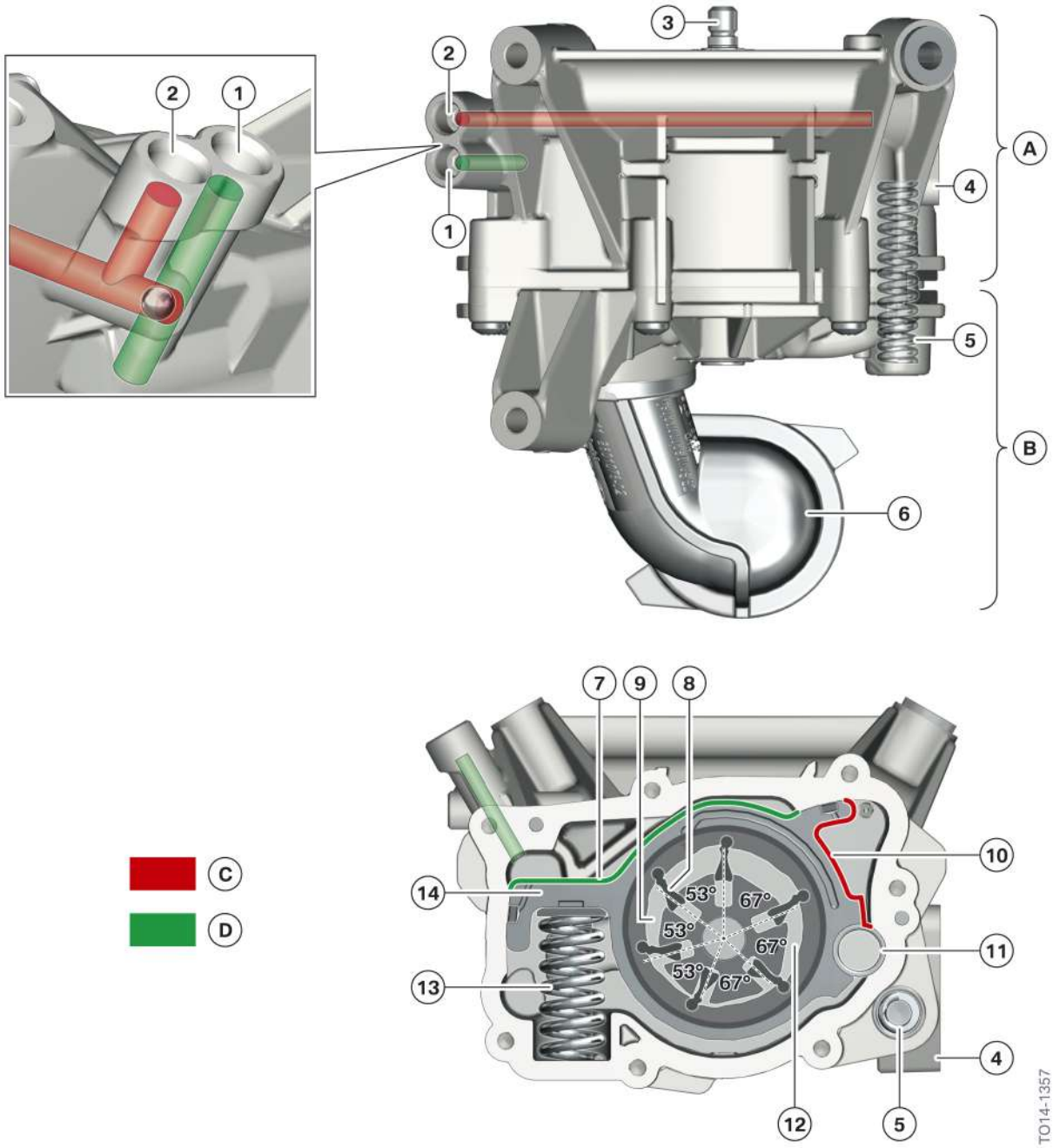
# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

| Index | Explanation |
|---|---|
| 1 | Major thrust face |
| 2 | Slide |
| 3 | Outer rotor |
| 4 | Adjusting ring (pendulum slide) |
| 5 | Inner rotor |
| 6 | Control oil chamber |
| 7 | Suction side |
| 8 | Housing |
| 9 | Main spring |

In the N20 engine, an adjusting ring is used, which no longer rotates around an axis during adjustment, but is instead moved parallel.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

**Oil pump with 2 separate control loops**




N63TU2 engine, oil pump design

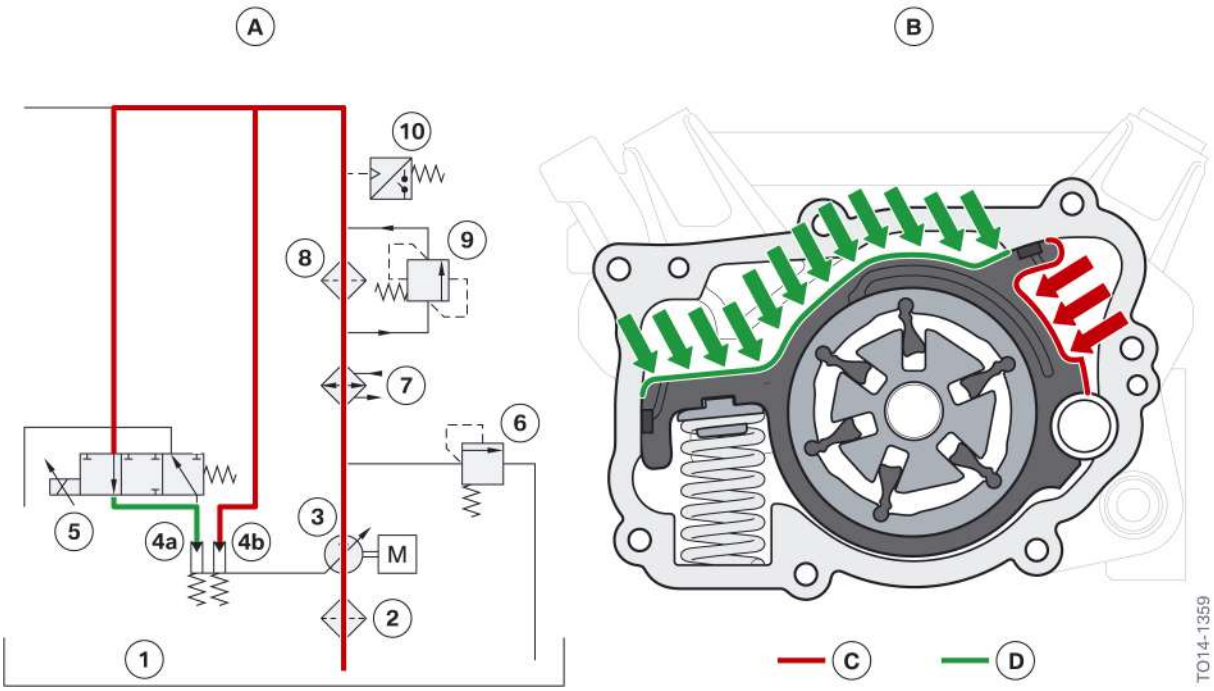
# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

| Index | Explanation |
|---|---|
| A | Oil pump |
| B | Oil pump cover |
| C | Second-level control area (emergency operation) |
| D | Map-controlled control area (normal operation) |
| 1 | Oil duct to map-controlled control chamber |
| 2 | Oil duct to second-level control chamber |
| 3 | Pump shaft |
| 4 | Oil pressure channel, pump output |
| 5 | Pressure-limiting valve |
| 6 | Intake pipe with filter |
| 7 | Map-controlled sensing area |
| 8 | Inner rotor with pendulum |
| 9 | Suction side |
| 10 | Second level control surface |
| 11 | Bearing tube (centre of rotation) |
| 12 | Major thrust face |
| 13 | Adjusting ring spring |
| 14 | Adjusting ring |

On the new B engine generation and in the N63TU2 engine, an adjusting ring is again used, which rotates around an axis. In addition, the emergency valve (second-level control valve) is omitted with these engines. Instead, an additional oil chamber is located in the oil pump so that pressure can be applied to the adjusting ring by means of a second area.

The oil pump has 2 separate control loops in order to guarantee normal operation (characteristic map control operation) and emergency operation (second level control operation).

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

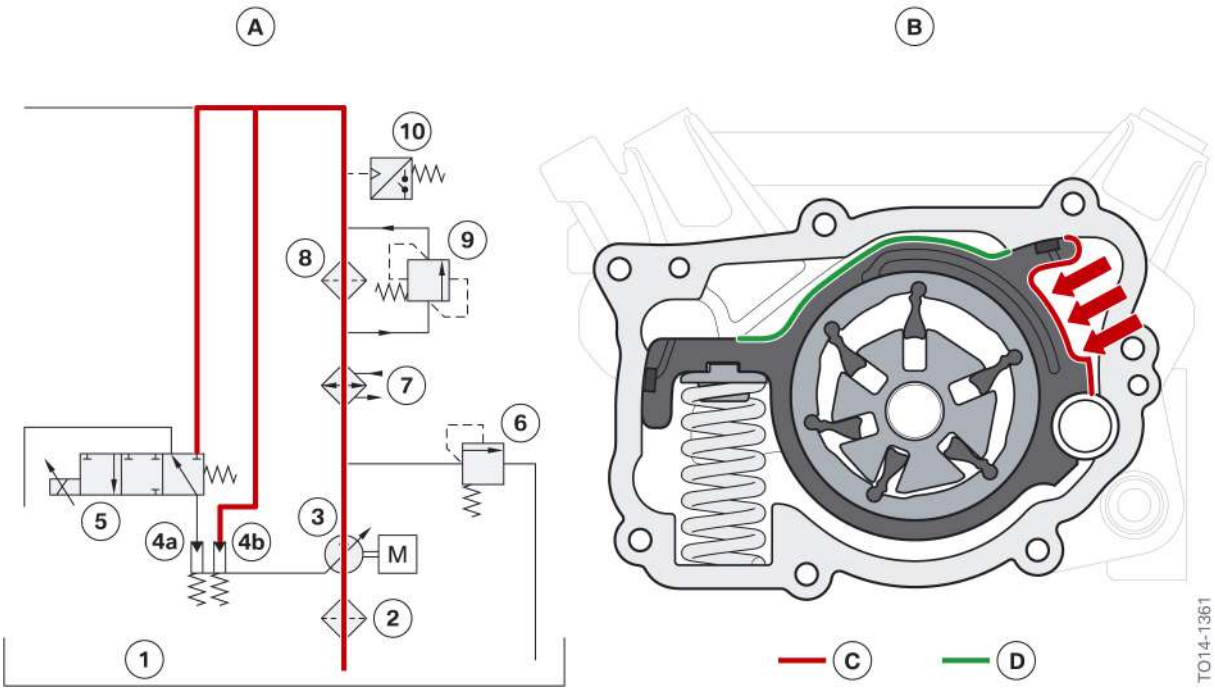



N63TU2 engine, oil circuit in normal operation

| Index | Explanation |
|---|---|
| A | Hydraulic circuit diagram for normal operation |
| B | Oil pump in characteristic map control operation (normal operation) |
| C | Volume flow-controlled oil pressure |
| D | Characteristic map-controlled oil pressure |
| 1 | Oil sump |
| 2 | Strainer |
| 3 | Characteristic map-controlled pendulum slide cell pump |
| 4a | Map-controlled control chamber (normal operation) |
| 4b | Second-level control chamber (emergency operation) |
| 5 | Map control valve |
| 6 | Pressure-limiting valve |
| 7 | Oil-to-water heat exchanger |
| 8 | Oil filter |
| 9 | Filter bypass valve |
| 10 | Oil pressure sensor |

The control loop for the normal operation works with an external characteristic map control valve. The characteristic map control valve controls the oil pressure in the characteristic map control chamber using software in the DME. If the oil pressure in the characteristic map control chamber increases, the surface difference in the control chambers presses the adjusting ring further against the adjusting ring spring, and the pump eccentricity is reduced. This results in a lower volumetric flow.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.



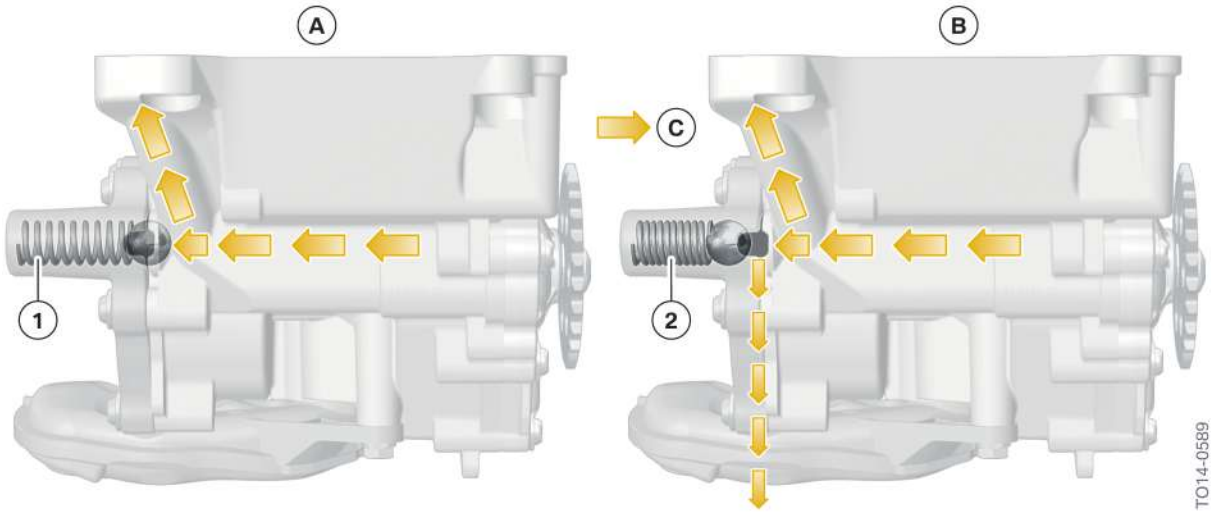

N63TU2 engine, oil circuit in emergency operation

| Index | Explanation |
|---|---|
| A | Hydraulic circuit diagram for emergency operation |
| B | Oil pump in emergency operation (second level operation) |
| C | Volume-flow-controlled oil pressure, oil pump |
| D | Characteristic map-controlled oil pressure |
| 1 | Oil sump |
| 2 | Strainer |
| 3 | Characteristic map-controlled pendulum slide cell pump |
| 4a | Map-controlled control chamber (normal operation) |
| 4b | Second-level control chamber (emergency operation) |
| 5 | Map control valve |
| 6 | Pressure-limiting valve |
| 7 | Oil-to-water heat exchanger |
| 8 | Oil filter |
| 9 | Filter bypass valve |
| 10 | Oil pressure sensor |

During emergency operation, the system operates without the map control by the Digital Engine Electronics (DME). In this operating condition the characteristic map control valve is de-energized and releases the oil duct from the characteristic map control chamber to the oil sump. The purpose of emergency operation is to keep the oil pump at a constant, volume-flow-controlled oil pressure

level. For this purpose, the oil pressure is guided directly from the main oil duct into the second level control chamber. This then conforms to the principle of a volume-flow-controlled oil pump without characteristic map regulation.

## 5.3. Pressure-limiting valve




B38 engine, pressure limiting valve

| Index | Explanation |
|---|---|
| A | Oil pressure < Opening pressure |
| B | Oil pressure > Opening pressure |
| C | Engine oil volumetric flow |
| 1 | Pressure limiting valve closed |
| 2 | Pressure limiting valve open |

The pressure limiting valve protects the oil supply components from excessive oil pressure. In particular, the oil filter, oil pump and engine oil cooler are thus protected. This is necessary above all at low temperatures and when the oil is viscous. The pressure limiting valve is thus also often referred to as cold-start valve.

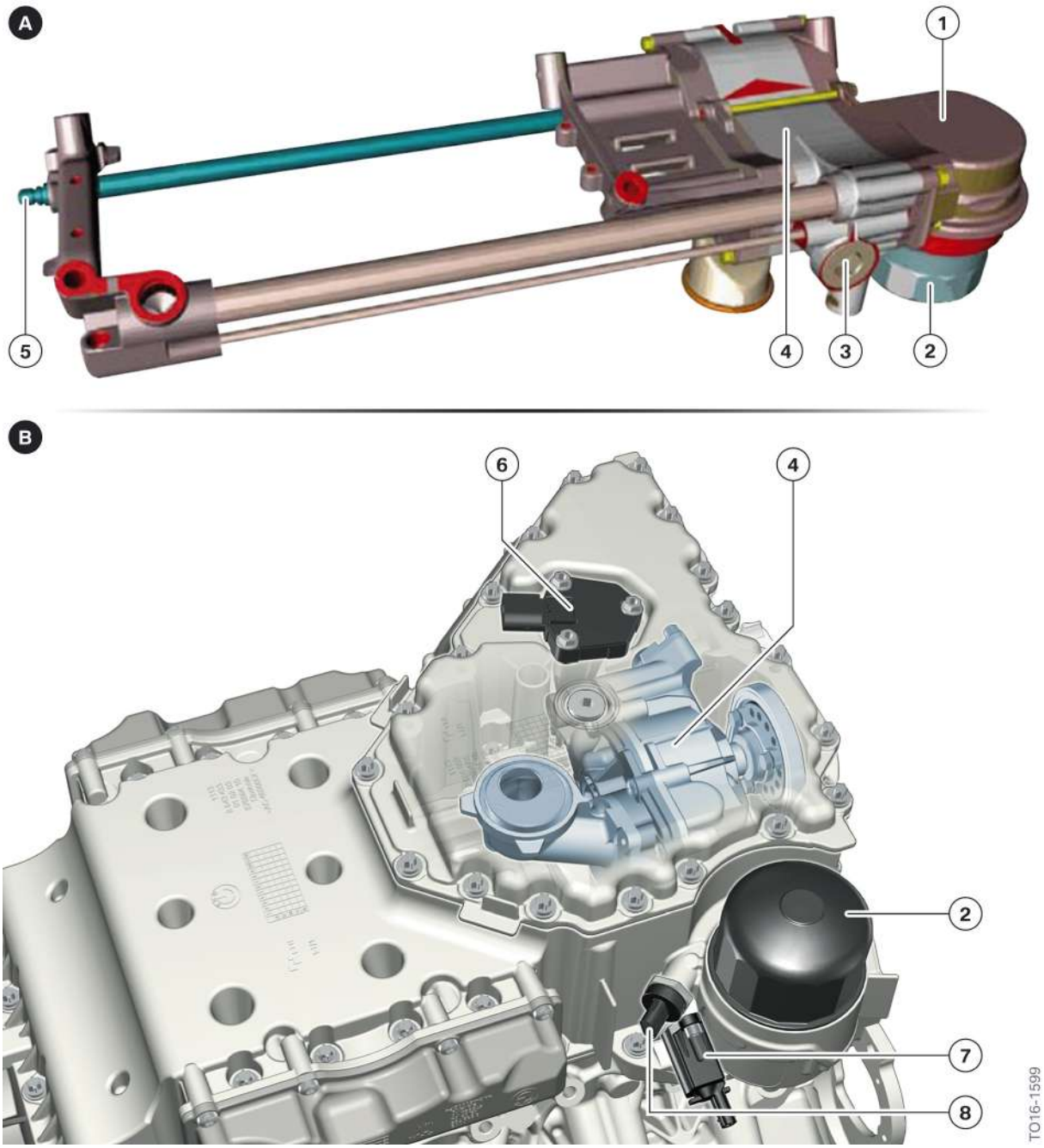
The pressure limiting valve is situated as closely as possible behind the oil pump, in the N47TU engine often directly in the oil pump housing. The opening pressure is dependent on the respective engine.

## 5.4. Oil filter and engine oil cooler

BMW and MINI use full-flow oil filters. This means that the entire oil quantity pumped by the oil pump passes through the filter before it is fed to the lubrication points. Thus, the lubrication points are always provided with filtered oil.

On the current V-engines, the oil filter housing is integrated into the oil pump cover, or as of the N63 engine, in the oil sump:

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.



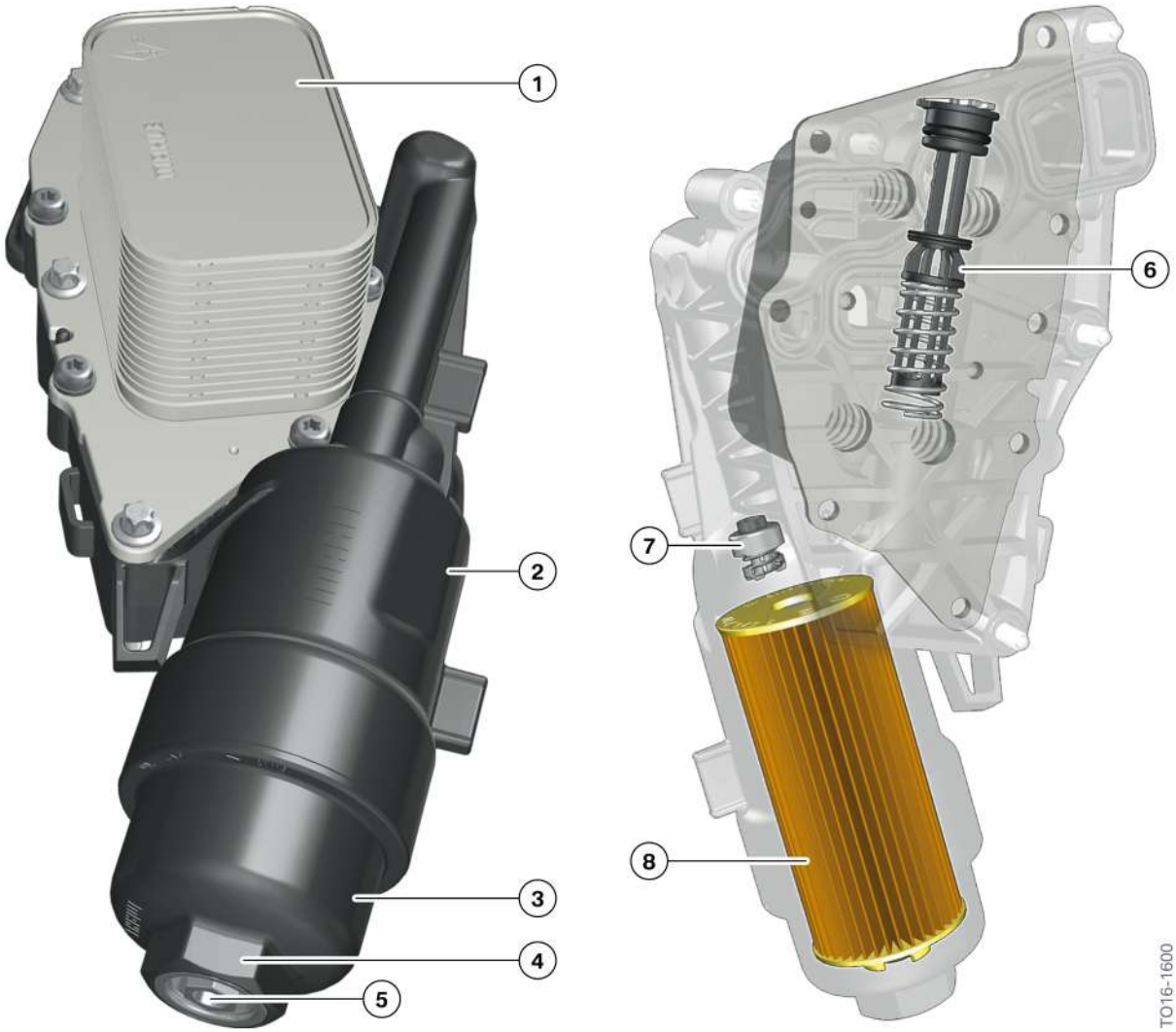

Oil pumps V-engines

| Index | Explanation |
|---|---|
| A | Oil filter housing in oil pump cover, N73 engine |
| B | Oil filter housing in oil sump, N63TU2 |
| 1 | Oil pump cover |
| 2 | Oil filter cover |

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

| Index | Explanation |
|---|---|
| 3 | Control valve |
| 4 | Oil pump |
| 5 | Oil pump drive shaft |
| 6 | Oil-level sensor |
| 7 | Map control valve |
| 8 | Oil pressure sensor |

On all current in-line engines, the oil filter housing is attached to the cylinder head or crankcase:




B48 engine, oil filter module

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.

| Index | Explanation |
|---|---|
| 1 | Oil-to-water heat exchanger |
| 2 | Oil filter housing |
| 3 | Oil filter cover |
| 4 | Hexagon head for opening the oil filter cover |
| 5 | Hexagon head for opening the oil drain plug |
| 6 | Heat exchanger bypass valve |
| 7 | Oil filter bypass valve |
| 8 | Oil filter element |

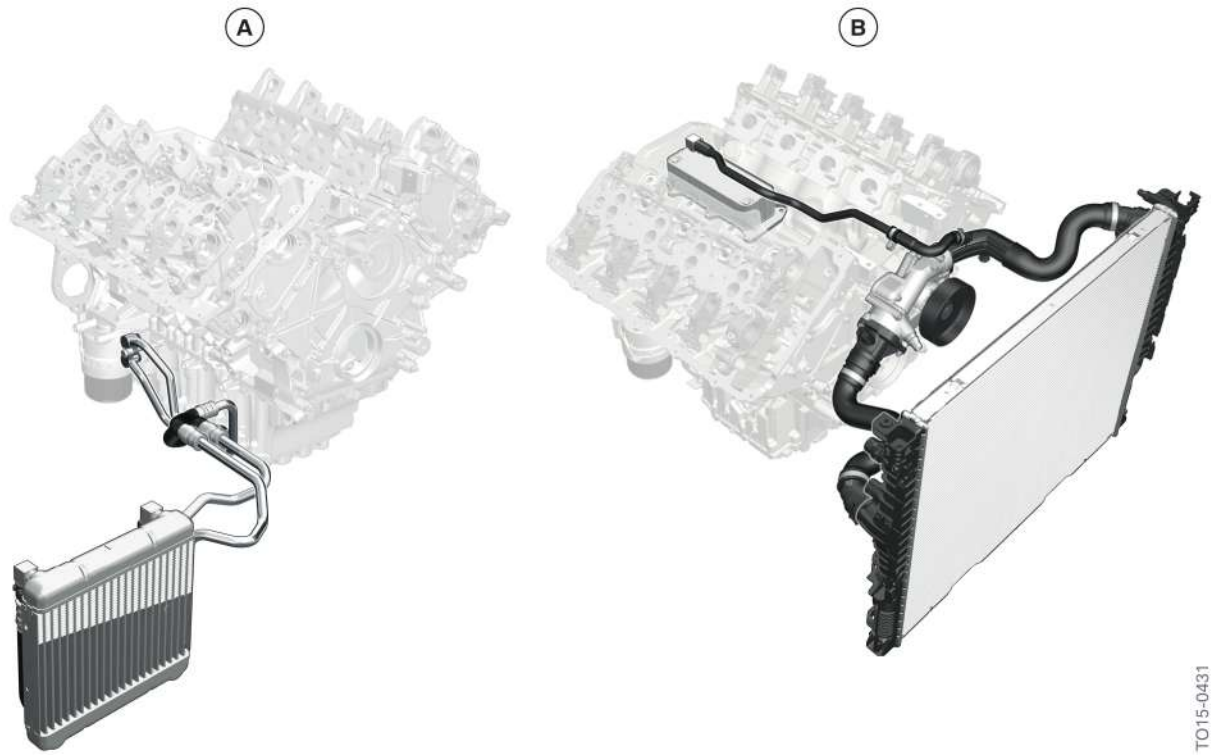
An oil filter bypass valve is arranged parallel to the filter to guarantee that the lubrication points are provided with oil even when the full-flow oil filter is contaminated or clogged. If the oil pressure increases due to a clogged oil filter, the oil filter bypass valve opens and the unfiltered lubricating oil can reach the lubrication points.

On some engines (e.g. Bx8), the installed heat exchanger bypass valve has the same function as the oil filter bypass valve. If the oil pressure increases due to a clogged oil-to-coolant heat exchanger, the heat exchanger bypass valve opens and the uncooled lubricating oil can reach the lubrication points.

On other engines (e.g. N18, N20 and N55) a permanent bypass is used instead of the heat exchanger bypass valve. This is a permanently open bypass around the oil-to-coolant heat exchanger. The bypass incorporates a flow restrictor to ensure that the majority of the oil nevertheless flows through the oil-to-coolant heat exchanger.

Additional engine oil coolers are frequently required for the cooling of the engine oil. Both oil-to-air heat exchangers as well as oil-to-coolant heat exchangers are used here.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.




N63TU/N63TU2 engine, arrangement of engine oil cooling components

| Index | Explanation |
|---|---|
| A | Oil-to-air heat exchanger in the N63TU engine |
| B | Oil-to-coolant heat exchanger in the N63TU2 engine |

The oil-to-air heat exchanger is usually located in the area of the cooling module. Depending on the engine oil temperature, a thermostat enables the oil flow to the oil cooler and thus ensures that the engine oil heats up quickly.

The oil-to-coolant heat exchangers are usually attached to the oil filter housing. The entire unit is known as the oil filter module. One exception, for example, is the N63TU2 engine where the oil-to-coolant heat exchanger is attached directly in the V-space of the crankcase.

In Service, the specified torques for the oil drain plug and the oil filter cover must be complied with. The O-rings must be replaced each time the oil filter cover and the oil drain plug are opened. O-rings are included in the oil filter service kit.

To ensure the engine runs perfectly, only engine oils approved by BMW can be used.

# ENGINE MECHANICS – PETROL ENGINE.
# 5. OIL SUPPLY.



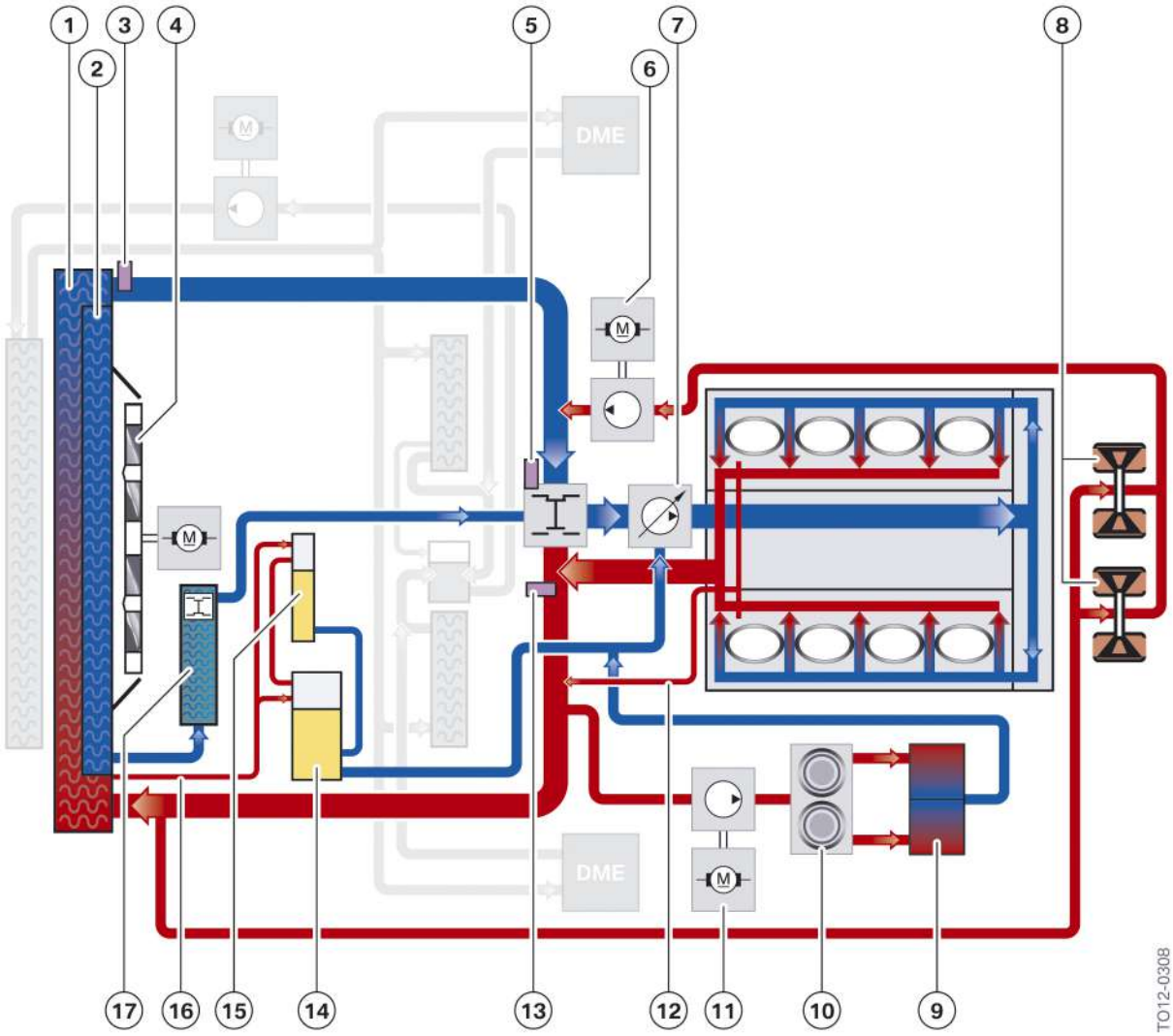
When removing and installing the oil-to-coolant heat exchanger, no contamination is allowed to occur. Coolant should not enter the bore holes for the engine oil, and engine oil should not enter the coolant bore holes. Seal the corresponding bore holes on the crankcase and on the oil-to-coolant heat exchanger immediately with suitable plugs.

# ENGINE MECHANICS – PETROL ENGINE.
# 6. COOLING.

## 6.1. Cooling system

In addition to the engine cooling, the job of modern coolant circuits is also the efficient regulation of the operating temperatures of a wide variety of components depending on the operating condition. Depending on the vehicle, the coolant circuits may be divided into multiple separate coolant circuits.

On various engines, the coolant circuit is divided into 2 areas – the high-temperature and the low-temperature coolant circuit. The high-temperature coolant circuit is mainly responsible for the engine cooling. The low-temperature coolant circuit ensures, e.g. the cooling of the charge air, the transmission oil or the engine control units.

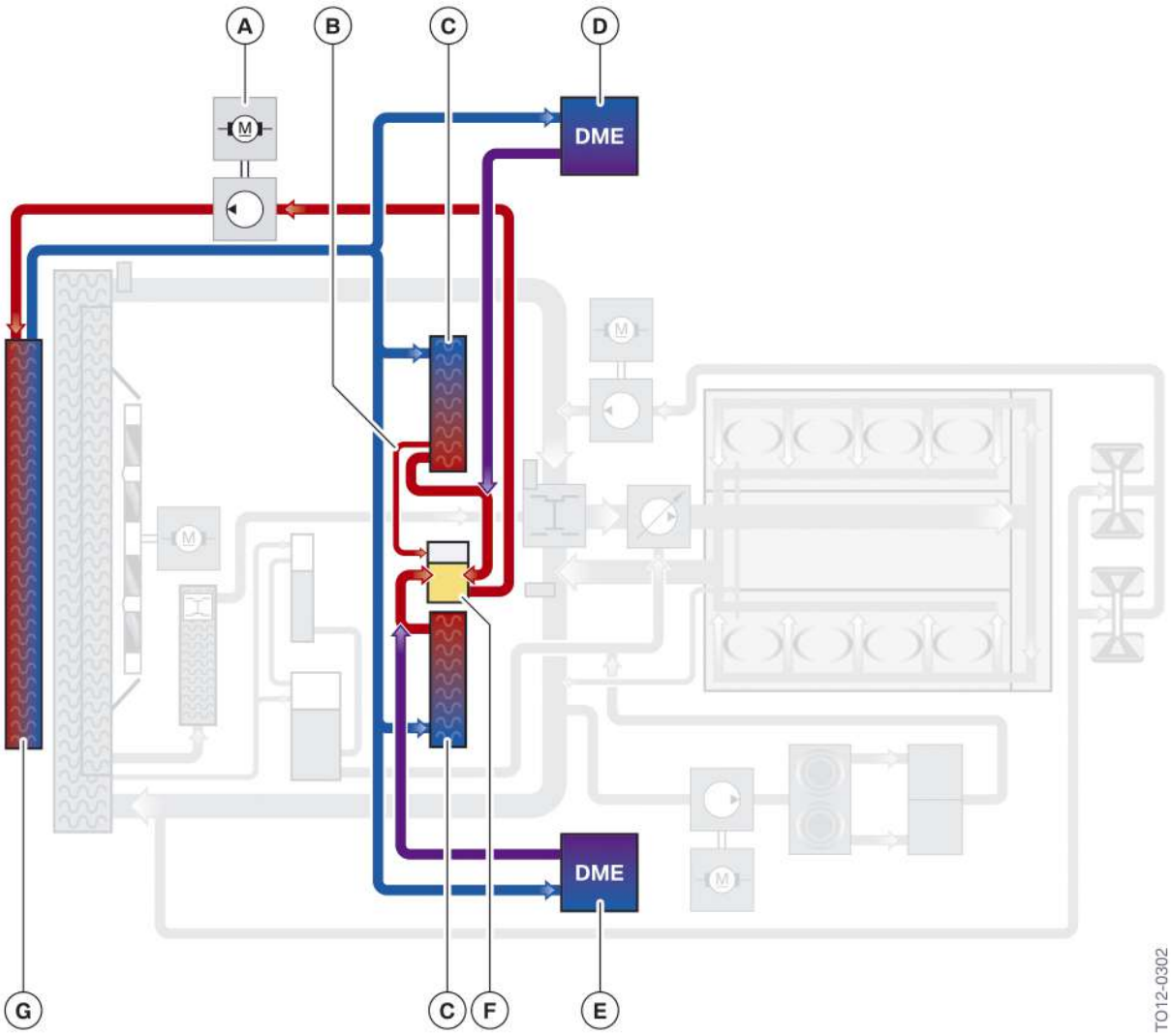



N63TU engine, high-temperature coolant circuit (European version)

# ENGINE MECHANICS – PETROL ENGINE.
# 6. COOLING.

| Index | Explanation |
|---|---|
| 1 | Radiator |
| 2 | Transmission oil cooler |
| 3 | Coolant temperature sensor (radiator outlet) |
| 4 | Electric fan |
| 5 | Data-map thermostat |
| 6 | Electrical auxiliary coolant pump (cooling of exhaust turbocharger) |
| 7 | Coolant pump |
| 8 | Exhaust turbocharger |
| 9 | Heat exchanger |
| 10 | Heater valve |
| 11 | Electrical auxiliary coolant pump (heating) |
| 12 | Ventilation line (cylinder banks) |
| 13 | Coolant temperature sensor (engine outlet) |
| 14 | Expansion tank |
| 15 | Coolant filler pipe |
| 16 | Tank ventilation line |
| 17 | Oil-to-water heat exchanger (transmission oil) |

# ENGINE MECHANICS – PETROL ENGINE.
# 6. COOLING.

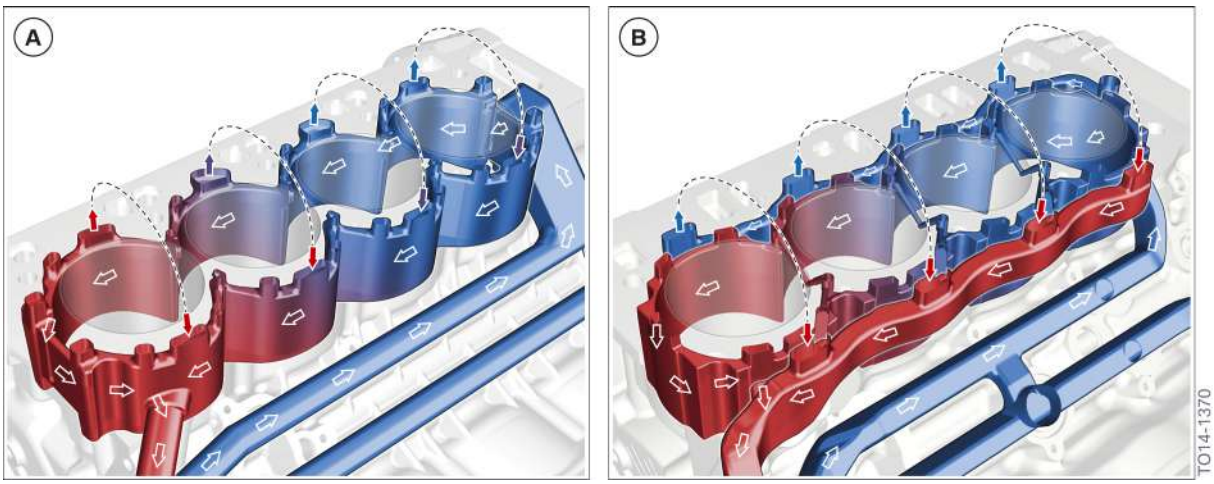



N63TU engine, low-temperature coolant circuit (European version)

| Index | Explanation |
|---|---|
| A | Electric coolant pump |
| B | Tank ventilation line |
| C | Charge air cooler |
| D | Digital Engine Electronics DME 1 |
| E | Digital Engine Electronics DME 2 |
| F | Expansion tank |
| G | Air-coolant heat exchanger (low-temperature coolant circuit) |

For bleeding the low-temperature coolant circuit, some engines (e.g. N63 and N74) have their own cooling system bleeding routine.

## 6.2. Split Cooling Combined




N63TU2 engine, coolant flows

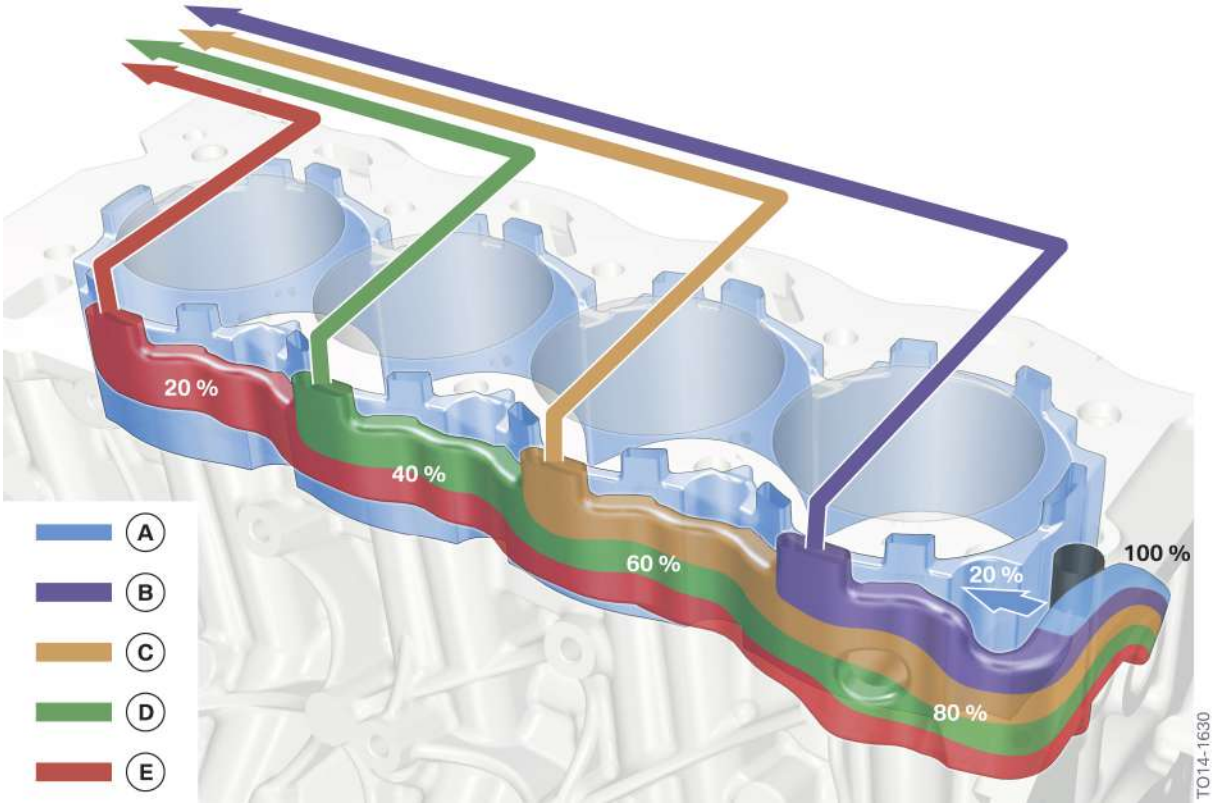
| Index | Explanation |
|---|---|
| A | Longitudinal distribution on N63TU engine |
| B | Parallel configuration on N63TU2 engine |

In the N63TU2 engine, the so-called SCC concept (Split-Cooling-Combined) is employed. That means the coolant circuits for cylinder head and crankcase are grouped in a parallel circuit. In the process, during the cooling-down period with opened thermostat 80% of the available coolant quantity flows through the cylinder head and 20% flows through the crankcase. In the warm-up phase, the coolant pump only operates with 10% volumetric flow, whereby the cylinder head is supplied with 8% and the crankcase with 2 % of the available coolant quantity.

As such, the SCC concept allows the warm-up phase to be significantly shortened, thereby optimising fuel consumption and emission behaviour and, in particular, the $CO^2$ emissions.

The coolant for cooling the cylinder head is conveyed along the crankcase via an outer gallery. The coolant for cooling the cylinder head is conveyed through defined openings in the cylinder head gasket. The coolant then flows through the cylinder head from the intake side to the outlet side and is then collected on the outlet side in a collecting duct together with the coolant for cooling the crankcase, and conveyed to the coolant pump again.

The V-shaped, drilled bridge cooling in the crankcase is supplied from the cylinder head through separate bore holes. The outflow runs through short coolant ducts into the coolant gallery in the cylinder jacket on the outlet side.




N63TU2, coolant distribution, cylinder bank 2

| Index | Explanation |
|---|---|
| A | Coolant section, crankcase |
| B | Coolant section, cylinder head-cylinder 8 |
| C | Coolant section, cylinder head-cylinder 7 |
| D | Coolant section, cylinder head-cylinder 6 |
| E | Coolant section, cylinder head-cylinder 5 |

To ensure that all cylinders in the cylinder head receive a sufficient quantity of coolant, the flow rates and the coolant routes are adapted to the specific cylinders by the geometric design of the coolant ducts in the crankcase. The geometry of the coolant ducts has been selected such that all paths for the coolant to each individual cylinder have the same length, but the cross-sections of the coolant ducts are adapted in line with the coolant quantity required by each cylinder.

## 6.3. Heat management

The coolant temperature influences the fuel consumption, power, quality of the mixture preparation, pollutant emission as well as the mechanical loading of the components. The optimisation of these parameters does not allow for any fixed temperature value for the various speed and load conditions, but rather requires a corresponding temperature range to the respective operating condition.

For this reason, there are various thermal management functions integrated into the Digital Motor Electronics. These comprise an independent control of various electrical cooling components such as:

- Electric fan
- Data-map thermostat
- Electric coolant pump
- Switchable coolant pump.

## 6.4. Coolant pump

Different coolant pumps are used on BMW and MINI engines. The following table provides a brief overview:

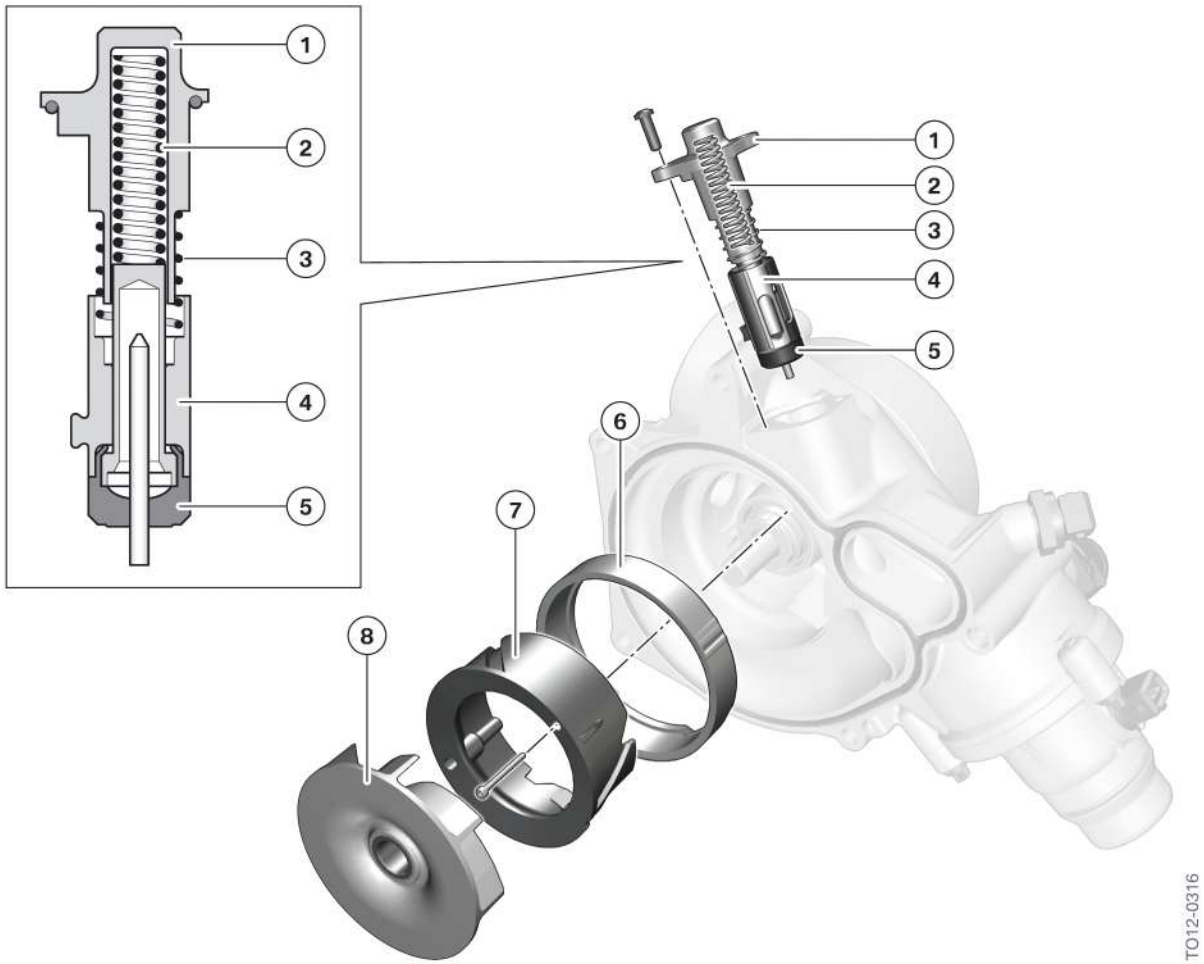
| Coolant pump | Engines |
|---|---|
| Mechanical coolant pump – not switchable | B38, B48A (transverse mounting), B58, N12, N13, N14, N16, N18, N42, N45, N46, N62, N62TU, N63, N73, N74, W10, W11 |
| Mechanical coolant pump – temperature-dependent adjustment via ring valve | N63TU |
| Mechanical coolant pump – characteristic map dependent adjustment via ring valve | N63TU2 |
| Mechanical coolant pump – adjustment via switching element | B48B (longitudinal mounting) |
| Electric coolant pump | N20, N43, N52, N53, N54, N55 |

Depending on the engine design, electrically driven auxiliary coolant pumps are also used.

### 6.4.1. Mechanical coolant pump

# ENGINE MECHANICS – PETROL ENGINE.
# 6. COOLING.

### Temperature-dependent adjustment via ring valve



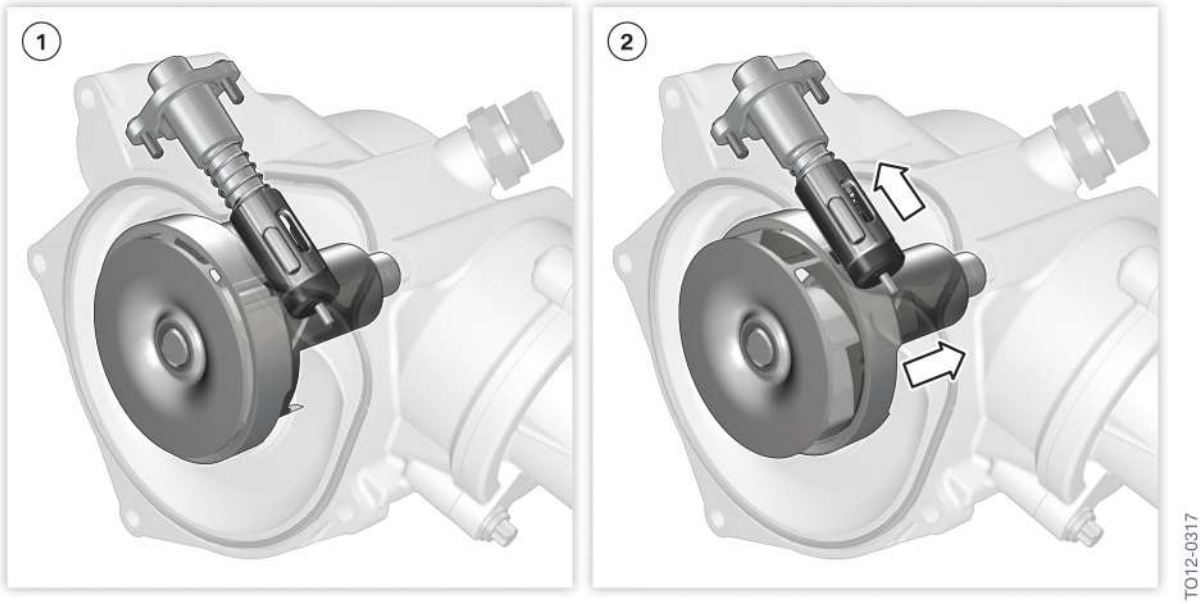

N63TU engine, coolant pump with temperature-dependent adjustment via ring valve

| Index | Explanation |
|---|---|
| 1 | Cover |
| 2 | Spring for thermal actuator |
| 3 | Spring for adjusting piston |
| 4 | adjusting piston |
| 5 | Thermal actuator |
| 6 | Ring valve |
| 7 | Ring insert |
| 8 | Impeller |

# ENGINE MECHANICS – PETROL ENGINE.
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A thermal actuator with an adjusting piston made of plastic has been integrated in the cooling pump of the N63TU engine. The adjusting piston is positioned via a preloaded spring and it is supported at the lower end by the thermal actuator. The adjusting piston can thus be moved upwards by the thermal actuator, but remains free to move axially. The thermal actuator is fixed by a long preloaded spring through the cover in the housing.

The adjusting piston engages in the annular valve groove by means of a lug, The annular valve is able to move via guiding grooves towards the outside on the fixed ring insert. It is thus possible to axially cover the impeller with the annular valve.




N63TU, function of the temperature-dependent adjustment

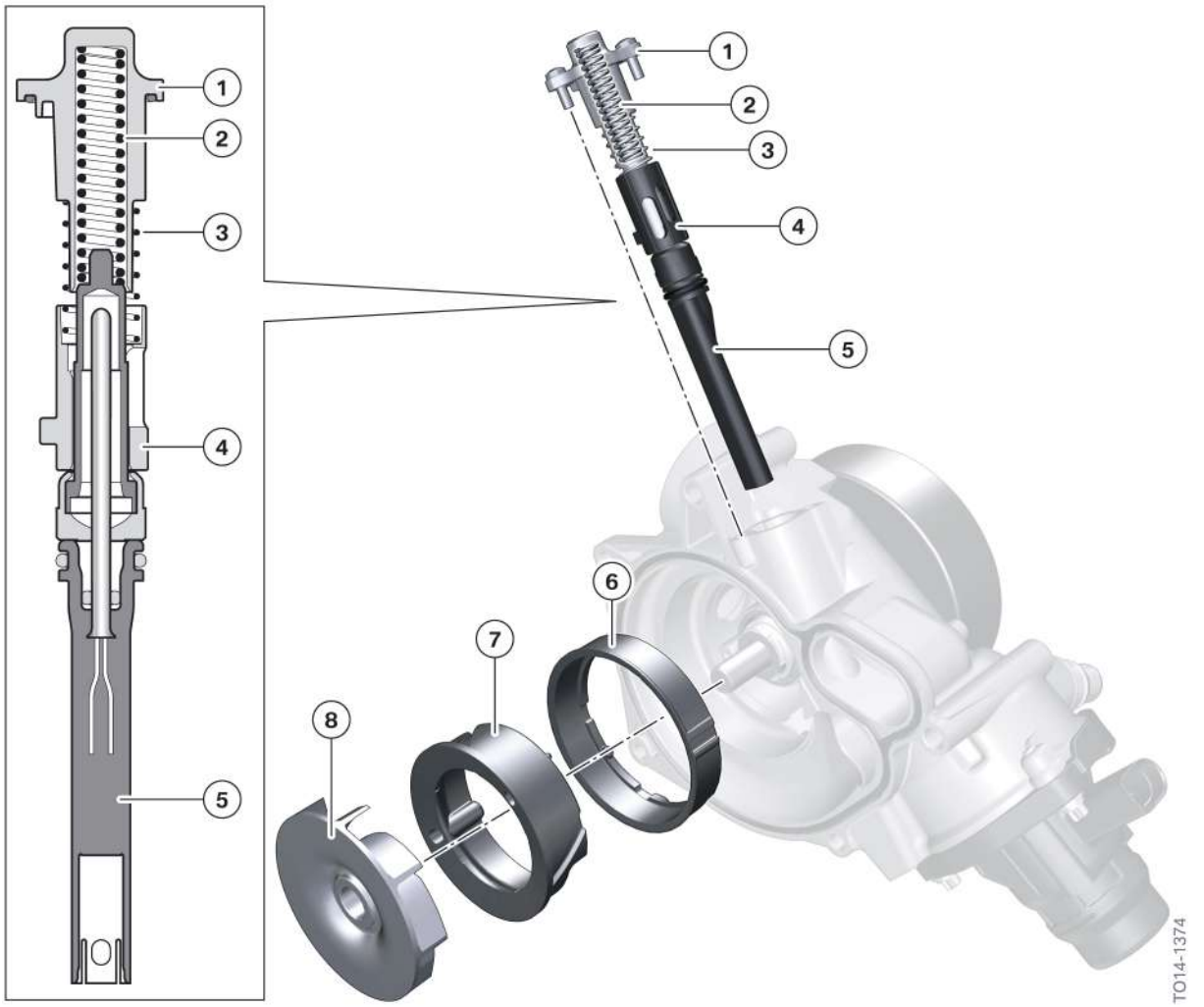
| Index | Explanation |
|---|---|
| 1 | 10% volumetric flow of the coolant, ring valve encases the impeller |
| 2 | 100% volumetric flow, ring valve releases the impeller |

Upon a cold start of the engine the ring valve is positioned over the impeller so that only a small volumetric flow of 10% arises. Thus, the operating temperature of the engine is reached more quickly, which has a positive effect on the fuel consumption and the emission behaviour. If a specified operating temperature of the coolant is exceeded, the thermal actuator moves the adjusting piston. This twists the ring valve, whereby the impeller is enabled and the volumetric flow increases to 100 %.

In addition to the temperature-dependent switching point of the thermal actuator, a speed-dependent security function is also integrated. This serves to protect the engine from overheating if a very high engine performance is demanded during a cold-start phase. From an engine speed of 3500 rpm, the impeller is released completely via another adjusting mechanism so that a delivery rate of 100% is immediately available. This is realised via return of the coolant pressure from the pressure auger directly to the ring valve. The adjusting piston is axially disconnected from the thermal actuator as described above and acts against a second preloaded spring. In the vane chamber a speed-dependent coolant pressure builds up which presses the ring valve against the spring force of the adjusting piston in an open direction. The flow rate increase is effected without delay.

# ENGINE MECHANICS – PETROL ENGINE.
# 6. COOLING.

### Characteristic map dependent adjustment via ring valve




N63TU2 engine, coolant pump with characteristic map dependent adjustment via ring valve

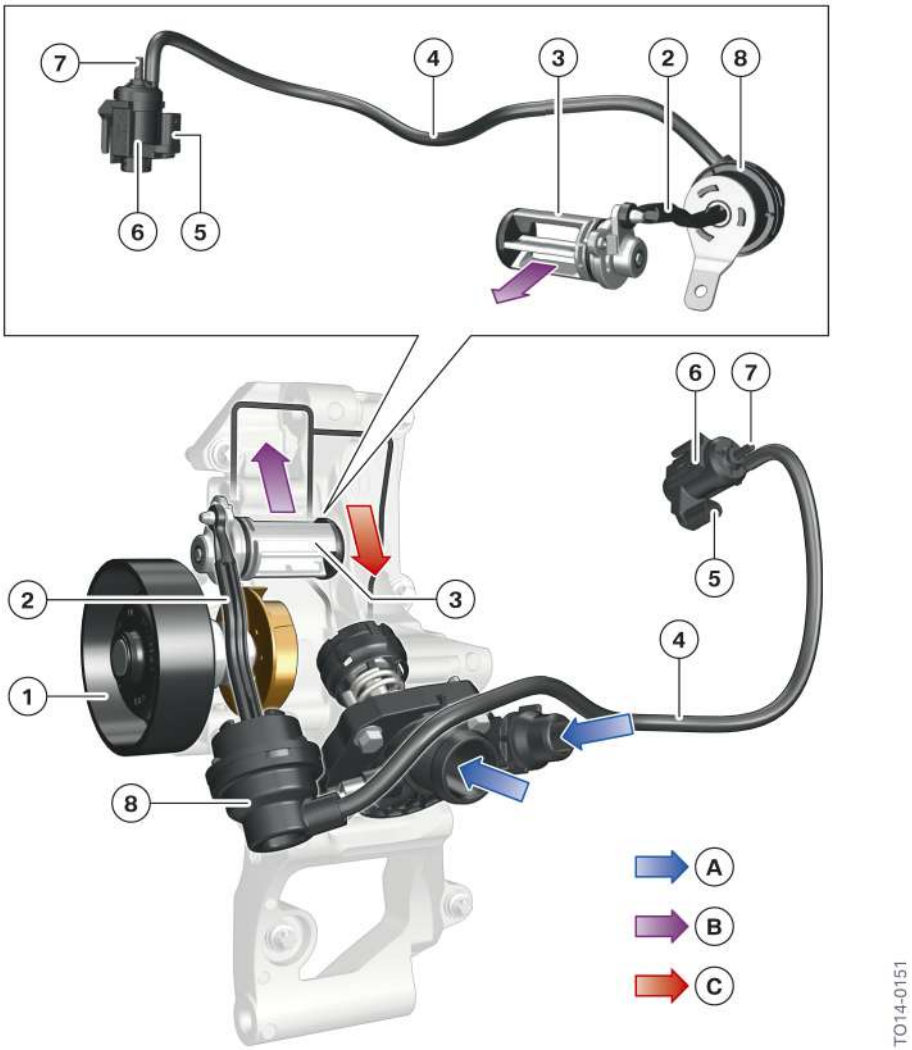
| Index | Explanation |
|---|---|
| 1 | Cover |
| 2 | Spring for thermal actuator |
| 3 | Spring for adjusting piston |
| 4 | Adjusting piston made out of plastic |
| 5 | Thermal actuator with electrical heating element |
| 6 | Ring valve |
| 7 | Ring insert |
| 8 | Impeller |

# ENGINE MECHANICS – PETROL ENGINE.
# 6. COOLING.

A characteristic map dependent switching coolant pump is used in the N63TU2 engine. In its main function it corresponds to the temperature-dependent switching coolant pump from the N63TU engine.

The integration of a heating element is new, whereby this is able to be influenced by the Digital Motor Electronics (DME) in dependence on the characteristic map.

**Adjustment via switching element**




B48B engine, coolant pump with adjustment via switching element

| Index | Explanation |
|---|---|
| A | Cooled coolant |
| B | Warmed coolant |
| C | Heated coolant |
| 1 | Belt pulley |

| Index | Explanation |
|---|---|
| 2 | Linkage |
| 3 | Shift element |
| 4 | Vacuum line |
| 5 | Electrical connection |
| 6 | Electric changeover valve |
| 7 | Connection for vacuum supply |
| 8 | Vacuum unit |

A switchable coolant pump is used on BMW vehicles with B48B engine (longitudinal mounting position). This reduces the warm-up phase of the coolant which means the engine reaches its operating temperature more quickly.

As long as there is no vacuum in the vacuum unit, the shift element of the coolant pump remains open. In this state the cooling circuit is identical to the vehicles without a switchable coolant pump.

Using the electric changeover valve, the Digital Motor Electronics (DME) can connect the vacuum from the vacuum supply connection to the vacuum unit. The vacuum pulls the linkage in the direction of the vacuum unit. This closes the cooling circuit on the pressure side of the coolant pump via the shift element.

In the warm-up phase, the radiator is disconnected from the engine cooling circuit via the characteristic map thermostat. In this situation, the coolant is only circulated via the crankcase (small cooling circuit).

When the shift element of the switchable coolant pump is closed, the crankcase cooling circuit (small cooling circuit) is also interrupted. Coolant is no longer circulated. This shortens the warm-up phase which in turn reduces the fuel consumption and exhaust emissions.

The belt pulley of the coolant pump is permanently connected to the impeller and cannot be uncoupled.

In the event of a leak in the vacuum system, or a failure of the electric changeover valve, the cooling circuit can no longer be closed via the shift element. In this case the cooling circuit is only adjusted via the characteristic map thermostat.

### 6.4.2. Electric coolant pump




N20 engine, electric coolant pump

If the electric coolant pump is removed and then to be reused, it is important to ensure that it is set down still filled with coolant. Drying out may cause the bearings positions to stick and the coolant pump will no longer start up. This can cause engine damage.

Before installing, turn the impeller manually to ensure that it moves freely.

## 6.5. Thermostat/heat management module

In engines with a thermostat with expansion element the coolant heats up the wax inside an operating housing. The wax liquefies from a certain, specified temperature. In this process, it expands and acts on a working piston in the housing, which releases the coolant flow towards the radiator using a poppet valve. If the coolant temperature falls below the opening temperature, a spring presses the poppet back into its initial position and closes the coolant flow towards the radiator. The engine maintains a specified temperature range.

With the exception of the following engines, all current BMW and MINI engines use thermostats with characteristic map regulation:

- W10, W11, N14 and N45 engine: Thermostat without characteristic map regulation
- B58 engine: Heat management module.

### 6.5.1. Data-map thermostat

In the characteristic map thermostat, an electrical heating resistor is additionally integrated into the expansion element. In this way, the expansion material not only heats up from the coolant flowing by but can also be actively influenced by the Digital Motor Electronics (DME).

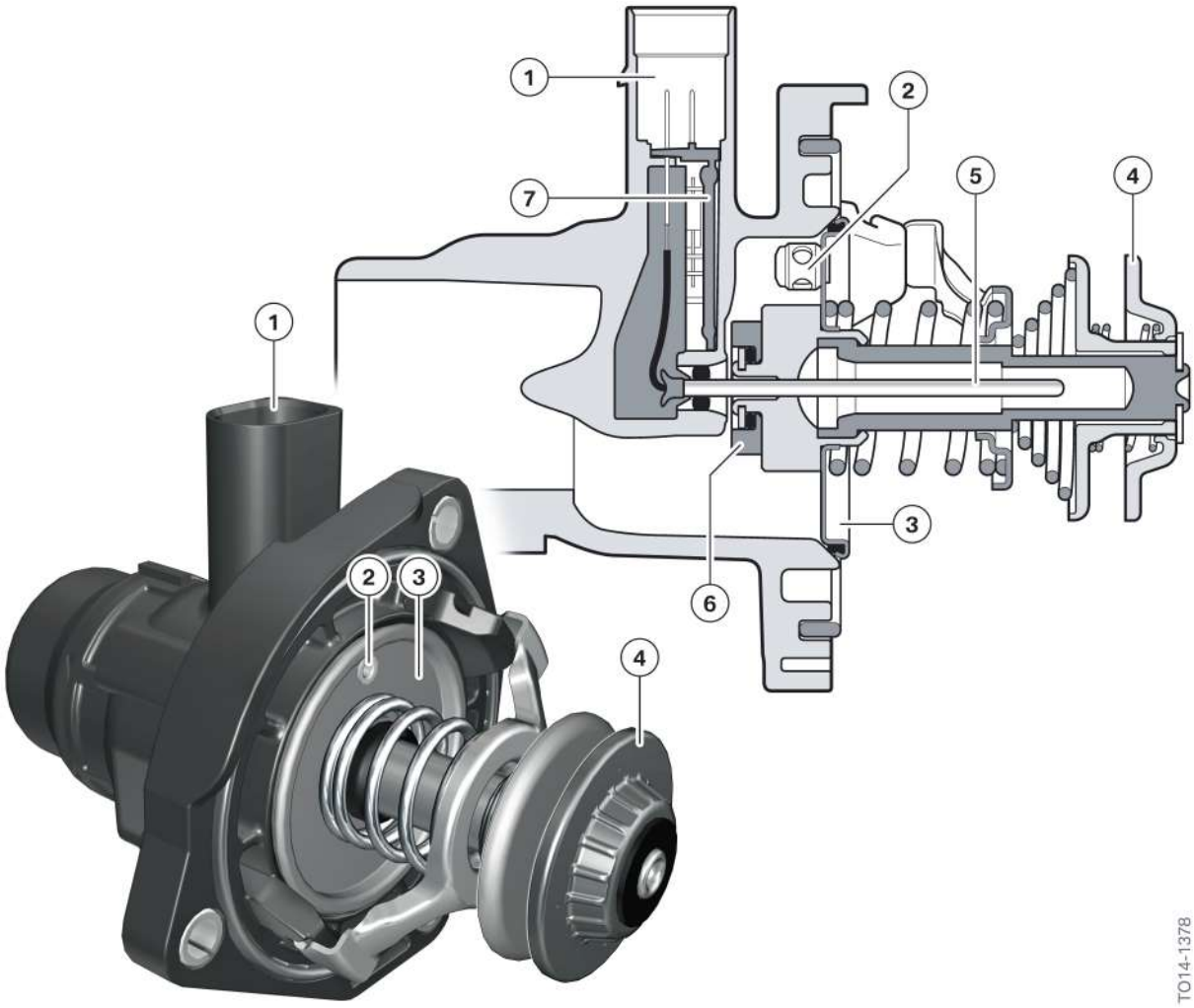
The DME controls the coolant temperature in dependence on a characteristic map, which can be determined by means of the following input variables, for example:

# ENGINE MECHANICS – PETROL ENGINE.
# 6. COOLING.

- Load condition
- Engine speed
- Vehicle speed
- Intake air temperature
- Coolant temperature.

This control allows the coolant temperature in the partial load range to be selectively increased. This leads to clear consumption and emission advantages.

A characteristic map thermostat is used with a stroke sensor and non-return valve in the N63TU2 engine.




N63TU2 engine, characteristic map thermostat

| Index | Explanation |
|---|---|
| 1 | Electrical connection |
| 2 | Vent valve |
| 3 | Thermostat main disc |
| 4 | Non-return valve |
| 5 | Thermocouple with electrical heating |
| 6 | Neodymium magnet for travel sensor |
| 7 | Hall effect sensor for travel sensor |

**Travel sensor**

The characteristic map thermostat in the N63TU2 engine is provided with an opening detection. As a result, it is now possible to monitor the opening function and thus detect a malfunction in the thermostat (OBD emission monitoring), including via diagnosis.

A hall effect sensor monitors the opening status. The neodymium magnet required for the hall effect sensor is attached to the thermocouple rod. When opening the thermostat, the opening status of the main disc relative to the closing point can be accurately measured on the basis of the strength of the magnetic field and the resulting voltage.

The electronic components are fully insulated from the coolant and are cast in an epoxy resin material.

The travel sensor in the characteristic map thermostat allows the following OBD-relevant functional problems to be recorded:

- **Thermostat jams open.** Can even be recorded before the engine starts (no emissions). (Relevant to OBD emission monitoring)
- **Thermostat does not open.** Can be detected much sooner and the risk of overheating can be prevented.



As the travel sensor is a linear hall effect sensor, a resistance measurement for checking the sensor is not permitted.

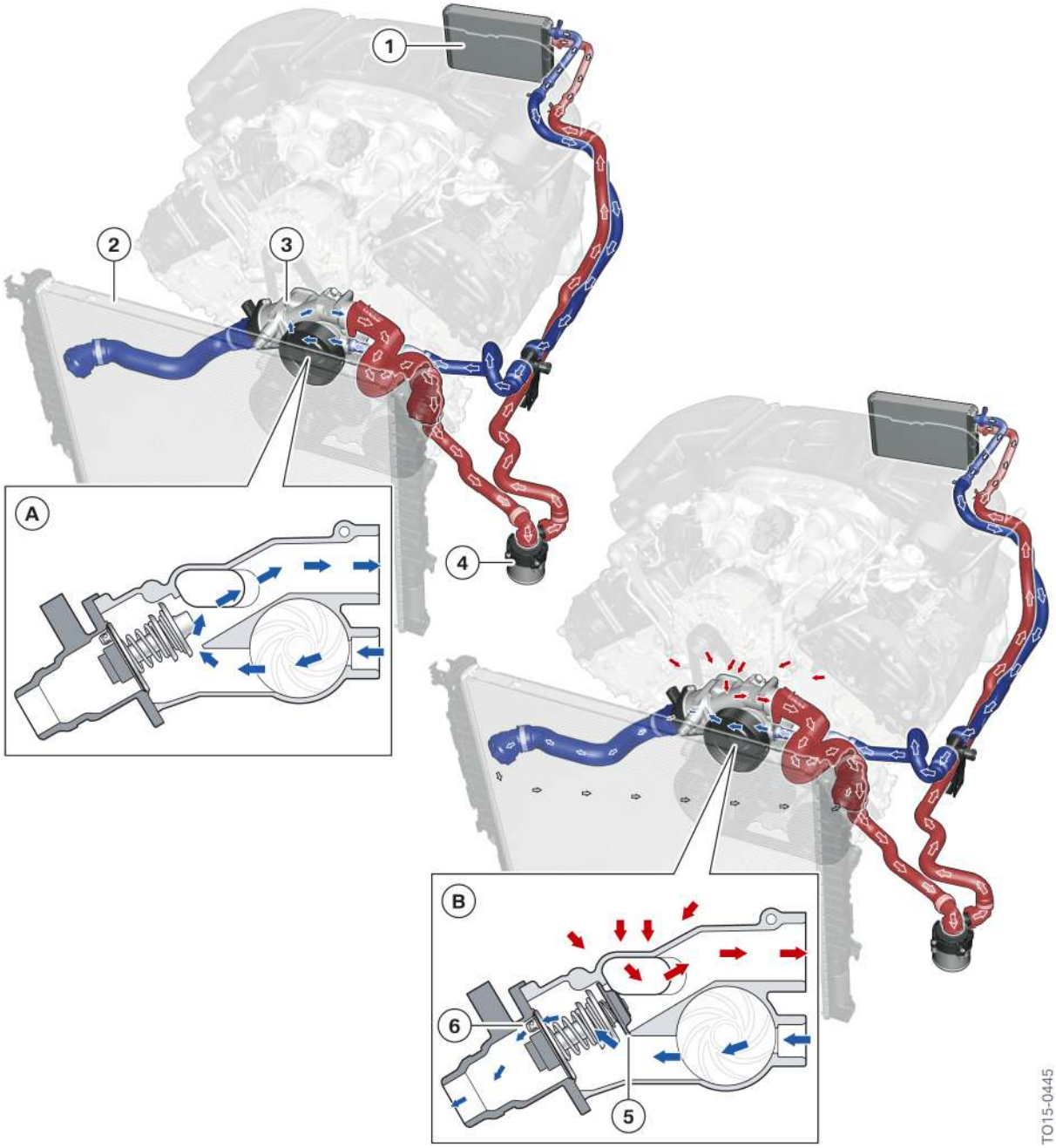
**Non-return valve**

The non-return valve takes the form of an additional valve head on the thermostat. When the engine (and thus also the mechanical coolant pump) are switched off, this valve head closes the short-circuit cross section between the engine outlet and thermostat mixing chamber. However, the thermostat mixing chamber simultaneously acts as the suction chamber for the electrical auxiliary coolant pump for the exhaust turbochargers.

The non-return valve influences the following 2 coolant circuits:

- Heating in vehicle interior
- Exhaust turbocharger cooling.

# ENGINE MECHANICS – PETROL ENGINE.
# 6. COOLING.




N63TU2 engine, non-return valve for heating

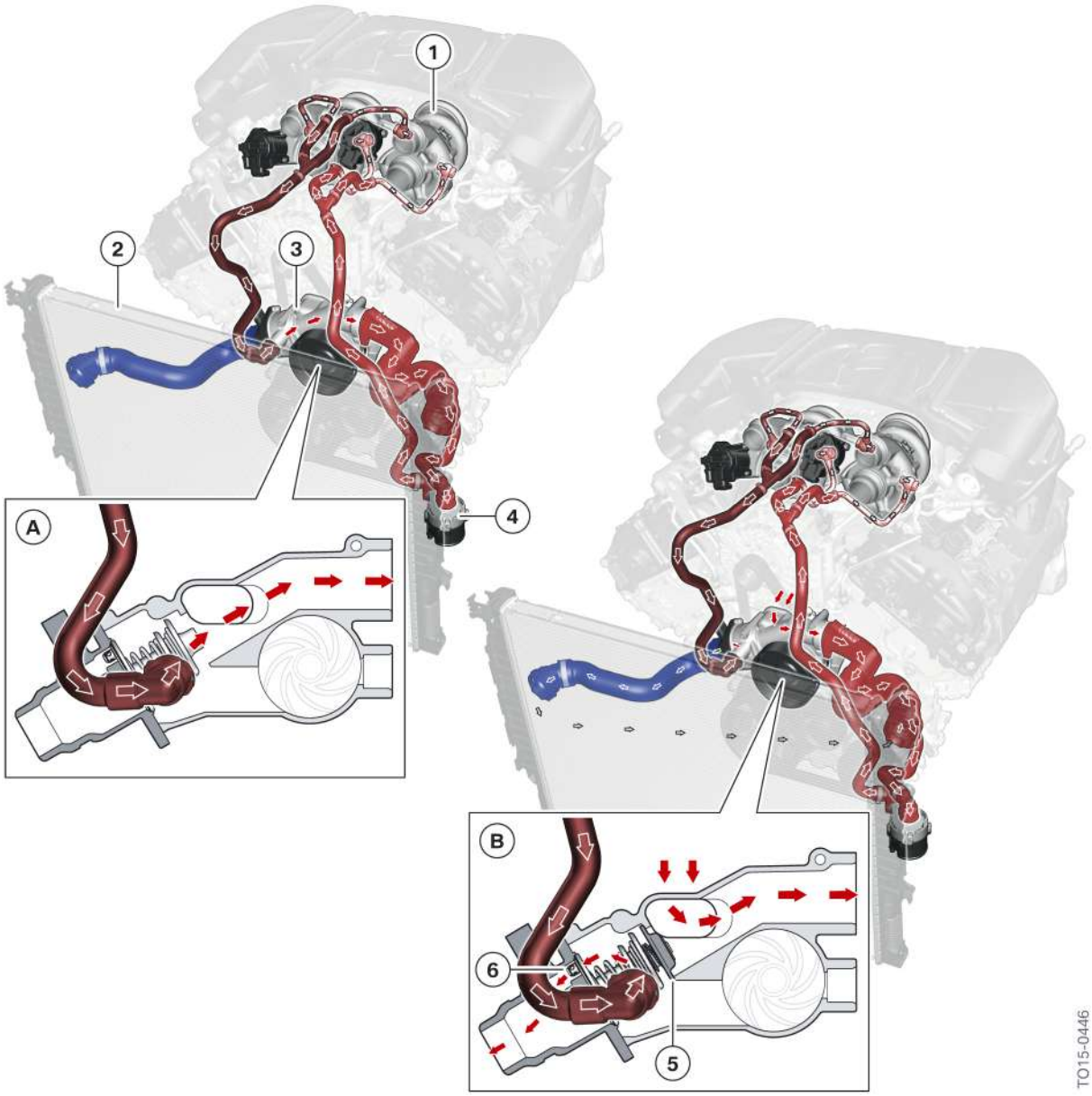
| Index | Explanation |
|---|---|
| A | Heating coolant flow without non-return valve |
| B | Heating coolant flow with non-return valve |
| 1 | Heat exchanger |
| 2 | Radiator |

| Index | Explanation |
|---|---|
| 3 | Coolant pump |
| 4 | Electrical auxiliary coolant pump (vehicle heating) |
| 5 | Non-return valve |
| 6 | Vent valve |

If the non-return valve were not fitted, when the engine is at operating temperature or stopped the electric coolant pump would always pump the same coolant to supply the heat exchanger for the heating system. This is because the coolant would always take the path of least resistance.

This is prevented by the non-return valve as it closes this path of least resistance. As a result, the electric coolant pump always draws the coolant (at operating temperature) required for heating from the crankcase and cylinder head. The coolant – which is at operating temperature and available in a larger quantity in the cylinder head and crankcase ensures that the heating can be maintained for a longer period of time. This measure allows e.g. longer engine stop times of the automatic engine start-stop to be achieved, which in turn saves fuel and reduces carbon dioxide emissions.

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N63TU2 engine, non-return valve exhaust turbocharger cooling

| Index | Explanation |
|---|---|
| A | Coolant flow for exhaust turbocharger cooling without non-return valve |
| B | Coolant flow for exhaust turbocharger cooling with non-return valve |
| 1 | Exhaust turbocharger |
| 2 | Radiator |

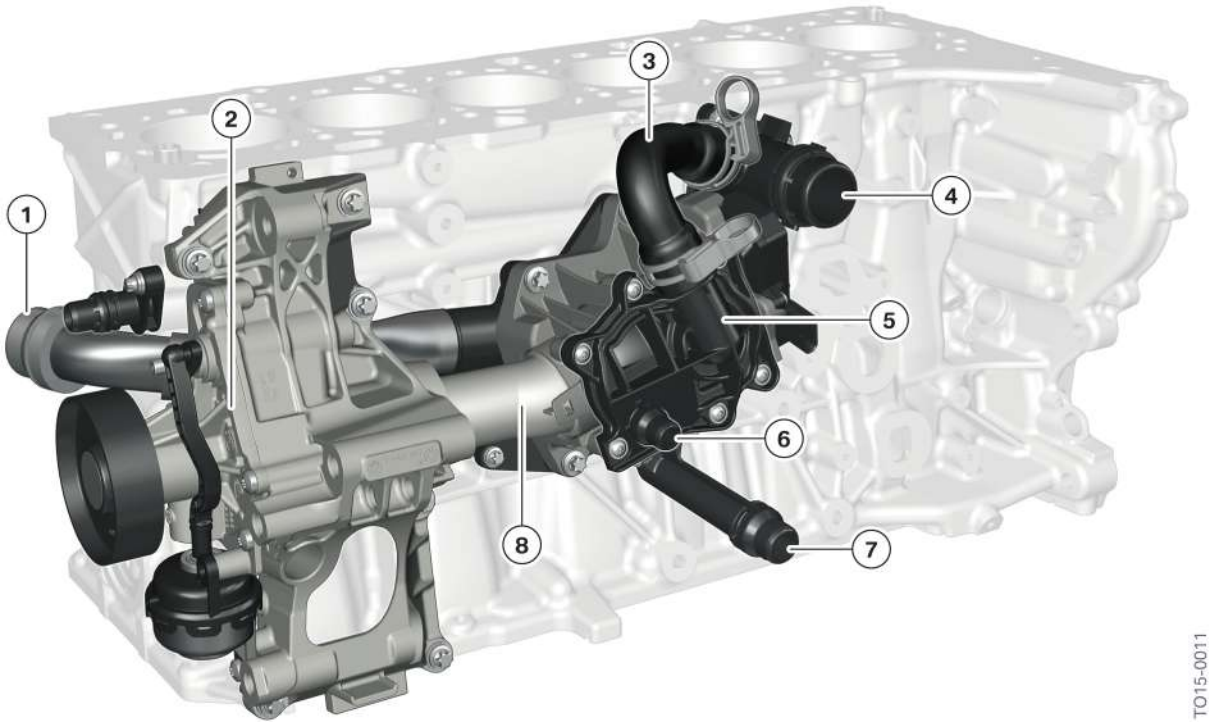
| Index | Explanation |
|---|---|
| 3 | Coolant pump |
| 4 | Electric auxiliary coolant pump (exhaust turbocharger cooling) |
| 5 | Non-return valve |
| 6 | Vent valve |

The non-return valve is also active during the after-run if the electrical auxiliary coolant pump is used when the ignition is switched off.

During after-run, the non-return valve prevents the coolant from being permanently pumped to the exhaust turbochargers in a small coolant circuit as this would produce a temperature difference between the exhaust turbochargers and the remaining engine components. The non-return valve closes the bypass such that the coolant for cooling the exhaust turbochargers is always pumped via the radiator. This ensures that the larger quantity of available coolant (at operating temperature) is used in the radiator circuit to provide more even cooling between the exhaust turbochargers and the rest of the engine components. As a result, the temperature of the exhaust turbochargers can be maintained for longer and is aligned to the other engine components. This rules out negative temperature differences between the individual components.

The non-return valve does not function during the warm-up phase (when the main disc of the thermostat is closed) or in the cooling-down period (when the thermostat is open). During the warm-up phase (when the bypass is active), the spring force of the non-return valve is overridden meaning the non-return valve opens. The coolant can then circulate as before in the small coolant circuit via the opened non-return valve.

## 6.5.2. Heat management module




B58 engine, heat management module

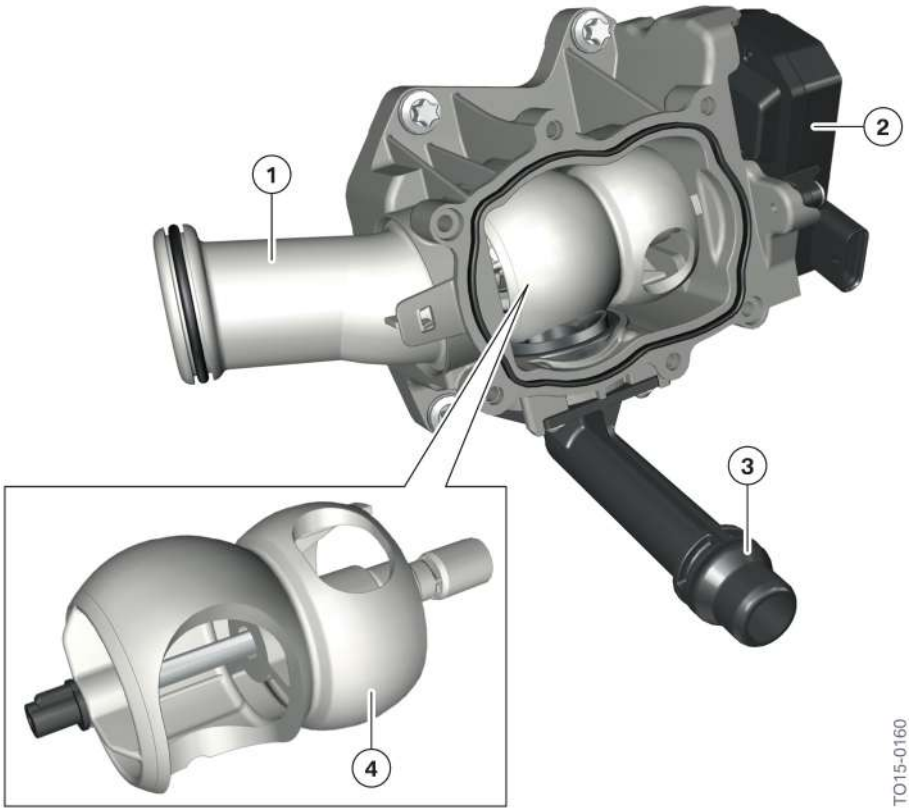
| Index | Explanation |
|---|---|
| 1 | Radiator return |
| 2 | Coolant pump (with alternator fixture and air conditioning compressor) |
| 3 | Short-circuit line |
| 4 | Crankcase coolant output |
| 5 | Heat management module |
| 6 | Expansion tank return |
| 7 | Heating return |
| 8 | Coolant pump connection |

The conventional thermostat in the B58 engine is replaced by a purely electrically operated heat management module. In contrast to a characteristic map thermostat with expansion element, there is no direct, physical connection to the coolant temperature.

The opening cross-sections of the various cooling ducts can be variably opened and closed using a rotary valve. The Digital Motor Electronics (DME) require information including the coolant temperature from the coolant temperature sensor and the material temperature of the cylinder head from the component temperature sensor to correctly position the rotary valve.

A position sensor in the electrical actuator of the heat management module forwards the current position of the rotary valve to the Digital Motor Electronics (DME). As a result, the exact position of the rotary valve can be determined so that it releases or seals an precisely defined cross-section towards

the different coolant ducts. The adjustment of cross-sections ideally adapts the flow rates of the cooling ducts connected to the heat management module to the operating points accordingly. Warm-up and cooling of the engine and the supply of additional ancillary components can be implemented as required and hence consumption can be optimised.




B58 engine heat management module

| Index | Explanation |
|---|---|
| 1 | Coolant output towards the coolant pump |
| 2 | Electrical actuator |
| 3 | Heating return |
| 4 | Rotary valve |

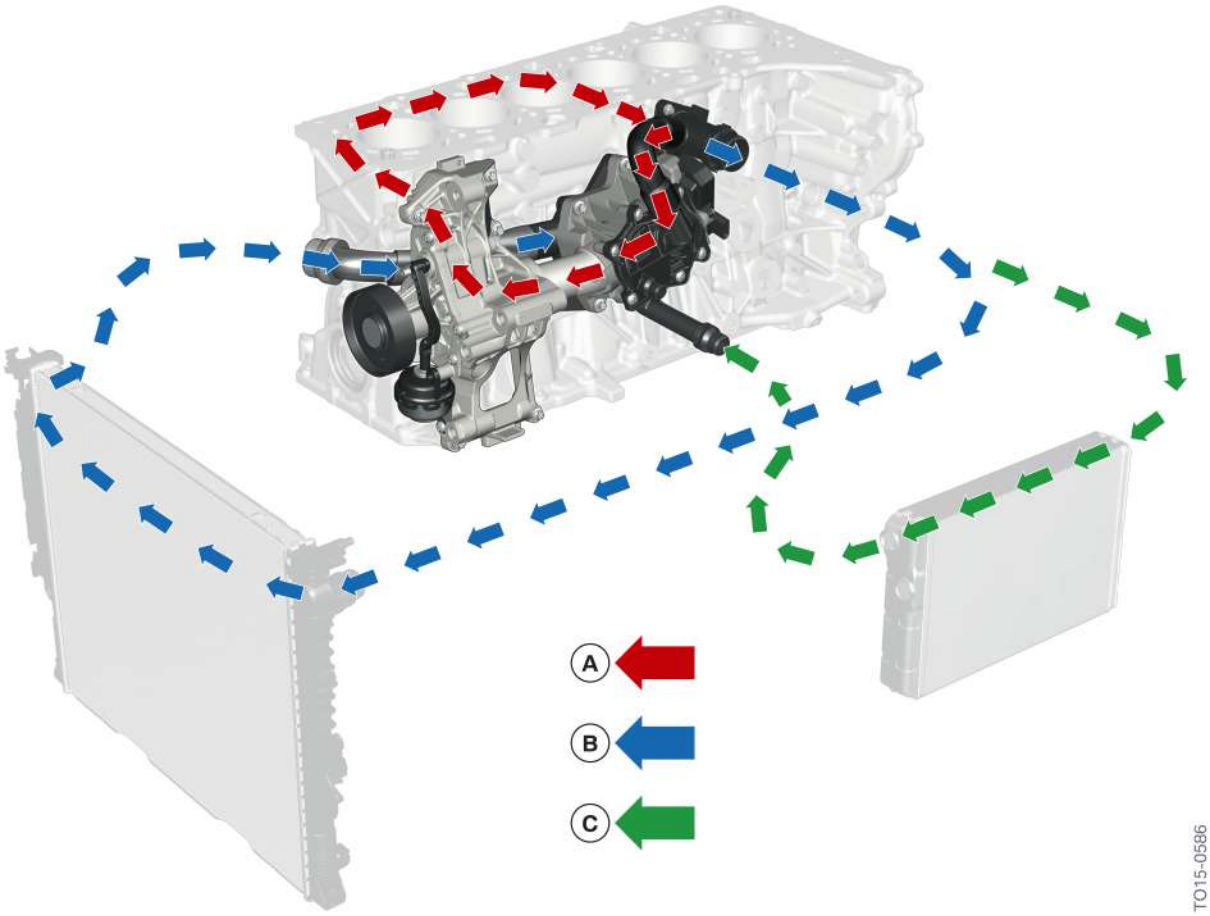
The heat management module consists of the following components which serve to regulate the cooling requirement:

- Rotary valve
  Connects or seals individual coolant connections
- Direct current motor (DC)
  Drive to adjust the rotary valve
- Position sensor
  Position feedback from the rotary valve to the engine control unit (DME)
- Transmission

Transforms the torque of the direct current motor (DC).

A service function in the BMW diagnosis system enables the end stops of the position sensor to be taught-in again.

**Cooling circuits**




B58 engine, coolant flow of the small and large coolant circuit

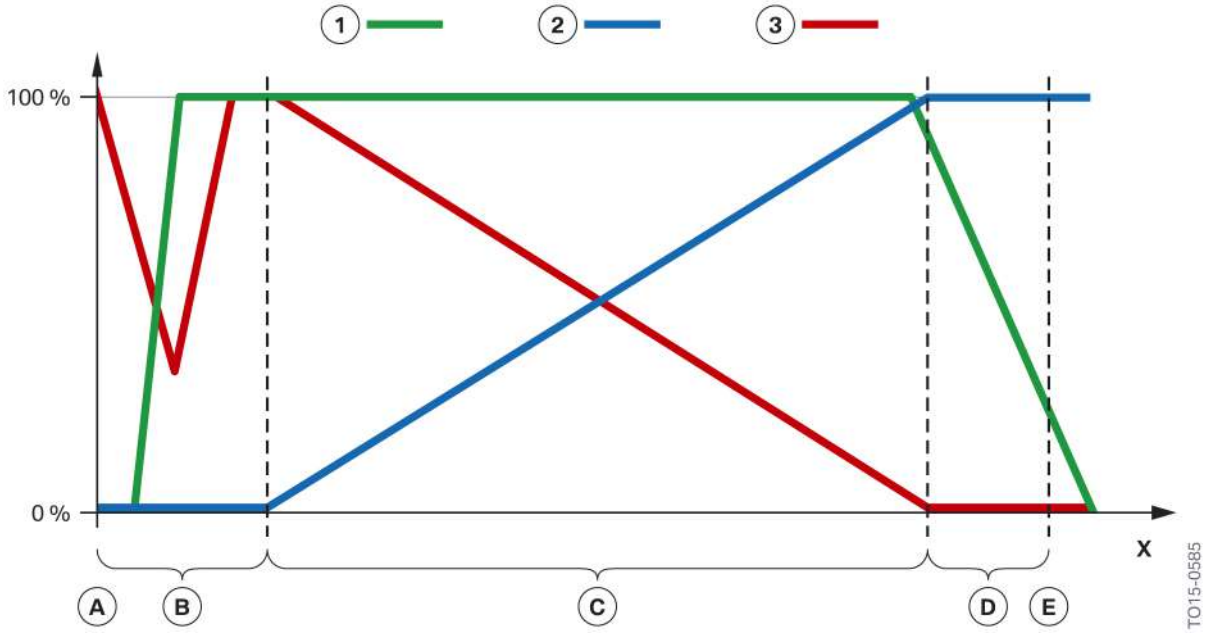
| Index | Explanation |
|---|---|
| A | Minor coolant circuit |
| B | Main coolant circuit |
| C | Heater circuit |

The coolant is directed via the radiator when the main coolant circuit is open.

If the minor coolant circuit is open, the coolant is directed from the crankcase directly into the heat management module using a short-circuit line.

When the heater circuit is open the coolant is directed via the heat exchanger for the heating system.

**Operating strategy**



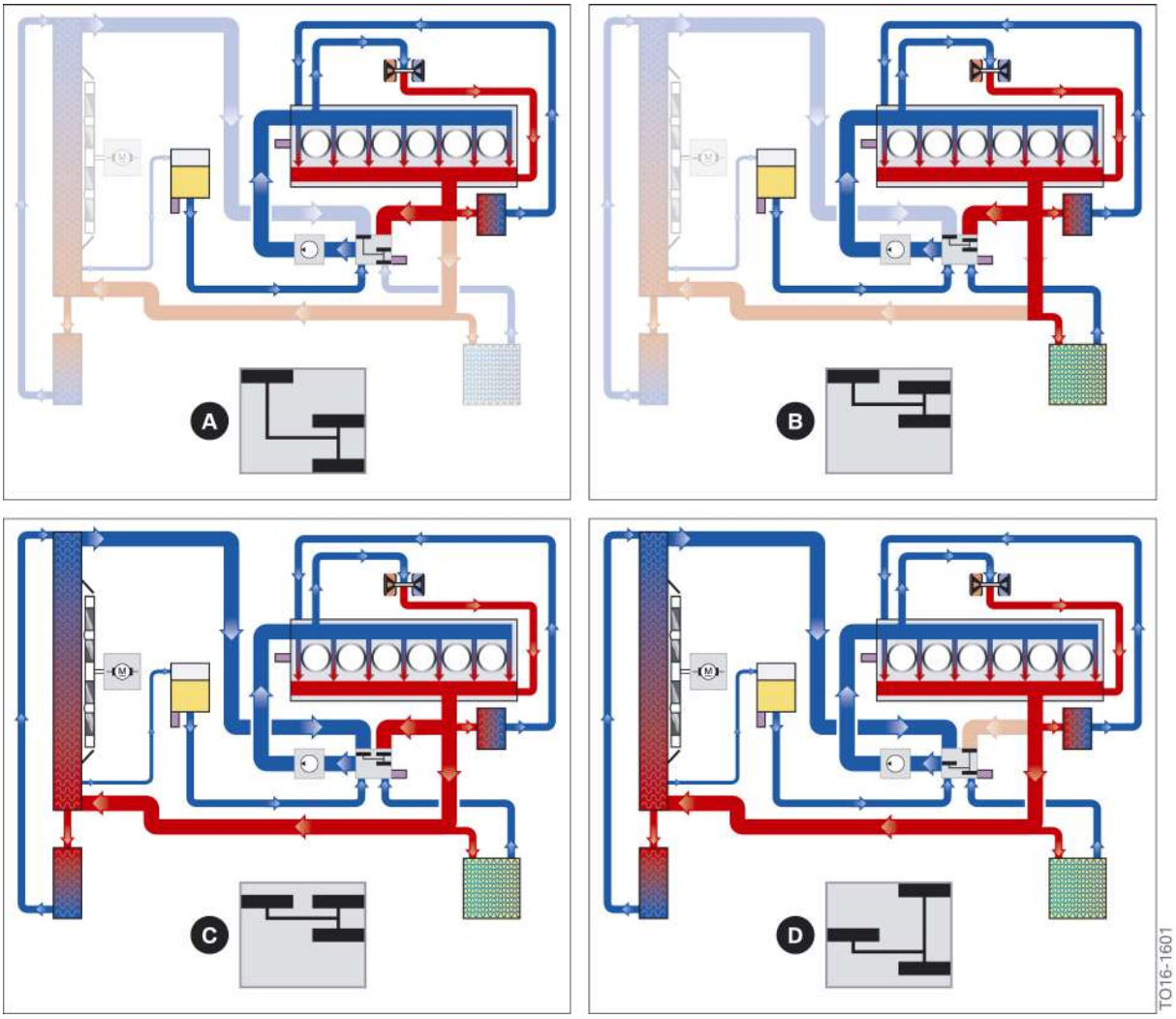

B58 engine, circuit diagram of the heat management module

| Index | Explanation |
|---|---|
| 0 % | Rotary valve closed |
| 100 % | Rotary valve open |
| A | Cold-start phase |
| B | Warm-up phase |
| C | Operating temperature |
| D | Transfer from normal operation to maximum cooling requirement |
| E | Maximum cooling requirement |
| X | Rotational angle of the rotary valve in degrees (increasing engine temperature) |
| 1 | Heater circuit |
| 2 | Main coolant circuit |
| 3 | Minor coolant circuit |

The diagram shows the rotary valve positions at increasing coolant temperature.

The openings on the rotary valve vary the cross-sections of the different coolant ducts depending on the rotational angle of the rotary valve. The following graphic schematically illustrates the phases from cold start to maximum cooling requirement.

# ENGINE MECHANICS – PETROL ENGINE.
# 6. COOLING.




B58 engine, heat management module switching modes

| Index | Explanation |
|---|---|
| A | Cold start |
| B | Warm-up phase |
| C | Operating temperature |
| D | Maximum cooling requirement |

**Cold-start phase figure A:** The short-circuit line is 100% open in the cold-start phase. The coolant connections towards the radiator and heating are closed.

**Warm-up phase figure B:** During the warm-up phase the connection to the heating is opened in addition to the short-circuit line. The radiator remains closed.

**Operating temperature figure C:** Figure C shows the control in normal conditions (operating temperature). The cross-sections to the corresponding connections are opened to a greater or lesser extent depending on the coolant temperature, so that coolant can flow through the minor and main coolant circuit and the heater circuit.

**Maximum cooling requirement figure D:** The radiator connection is opened by 100% and the short-circuit line is completely closed to provide maximum cooling at high dynamic loads and/or high ambient temperatures. In this process, the heat exchanger for the heating system is blocked by 90% to be able to provide the coolant radiator with even more volumetric flow.

The coolant connection to the expansion tank is not controlled. As a result, it is permanently open so that any coolant demand in the coolant circuit can be compensated for by the expansion tank at any time.



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